Lecture Notes on Information Theory and Coding

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Goals: The goal of this class is to establish an understanding of the instrinsic properties of transmission of information and the relation between coding and the foundamental limits of information transmission in the context of communication.

Course Outline:

- Entropy and Mutual information (Measure of information)
- Sourse coding
- Channel capacity
- The Gaussian channel
- Coding for a noisy channel (Block coding principles)
- Rate distortion theory

基本内容可以概括为：

IT $\begin{cases} \text{通信的基本性能限} \\ \text{逼近性能限的方法 – Coding: source coding, channel coding, network coding} \end{cases}$

Textbook: T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd edition, New York, Wiley, 2006. (中译本：信息论基础，机械工业出版社，2008；第一版影印本：清华大学出版社，2003).

References:

(1) C. E. Shannon, "A mathematical theory of communication", *Bell Syst. Tech. J.*, vol. 27, 1948.

(2) R. G. Gallager, *Information Theory and Reliable Communication*, Wiley, 1968.

(3) R. J. McEliece, *The Theory of Information and coding*, 1977. (This is a very readable small book)

(4) Raymond W. Yeung, *Information Theory and Network Coding*, Springer, 2008. (中译本：高等教育出版社，2011)

(5) J. Massey, *Digital Information Theory*, Course notes, ETH.

(6) M. Medard, *Transmission of Information*, Course notes, MIT.

(7) 王育民，李晖，*信息论与编码理论*，第二版，高等教育出版社，2013.

(8) 仇佩亮，*信息论与编码*，高等教育出版社，2003.

(9) 付祖芸，*信息论*，电子工业出版社.

(10) R. G. Gallager, "Claude E. Shannon: A retrospective on his life, work, and impact," *IEEE Trans. Inform. Theory*, vol.47, no.7, pp. 2681-2695, Nov. 2001.

Chapter 1

Introduction

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. — Shannon

In 1948, Claude E. Shannon published a landmark paper entitled "A Mathematical Theory of Communication". This paper laid the groundwork of a entirely new scientific discipline, "Information Theory".

Information theory studies the transmission, processing and utilization of information.

1.1 Relationship of information theory and communication theory

Information theory answers two fundamental questions in communication theory:

1. What is the ultimate data compression? H
2. What is the ultimate transmission rate? C

Information theory also suggests means of achieving these ultimate limits of communication. (e.g. random coding, ML decoding)

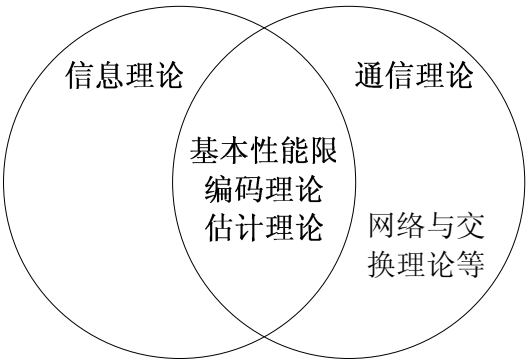


Figure 1.1: 信息理论和通信理论的关系示意图. Shannon理论主要研究基本限(fundamental limits)

通信系统传输的是信号；信号是消息的载体；消息中的不确定成分是信息。

- **狭义信息论(Shannon Theory)**
  Shannon在前人工作的基础上，用概率统计的方法研究通信系统。揭示了通信系统中传送的对象是信息；系统设计的中心问题是在干扰噪声中如何有效而可靠地传送信息。指出可以用编码方法实现这一目标；并在理论上证明了通信系统可达到的最佳性能限。
- 一般信息论：除Shannon理论外，还包括最佳接收理论（信号检测、估计与调制理论），噪声理论等。
- 广义信息论

信息论是通信与信息系统的基础理论，是现代通信发展的动力和源泉：

I have often remarked that the transistor and information theory, two Bell Laboratories breakthroughs within months of each other, have launched and powered the vehicle of modern digital communications. Solid state electronics provided the engine while information theory gave us the steering wheel with which to guide it. — Viterbi, IT News Lett., 1998.

- 信源编码定理→ 数据压缩技术→ 无线通信系统从1G变革到2G
- 信道编码定理→ 差错控制编码（Turbo,LDPC）→ 3G
- 数据处理定理→ 软判决译码
- 高斯噪声是最坏的加性噪声+ 多用户信息论→ CDMA、多用户检测
- MIMO容量理论→ 空时编码、预编码→ LTE、4G
- 多用户信息论→ 协作通信、网络编码→ 新一代无线系统

The recent work on the information-theoretic aspects of communication concentrated on: 1) Network information theory, and 2) MIMO systems.

1.2 What is information? (Measure of information)

For Shannon theory, information is what we receive when uncertainty is reduced.

How to measure:

- Amount of information should fulfill $I \geq 0$
- Amount of information should depend on probability $P(x)$
- For independent events: $P(X,Y) = P(X)P(Y) \rightarrow I = I(X) + I(Y)$

It should has the form of $\log \frac{1}{P_X(x)}$. (Self-information of the event $X = x$)

1.3 Applications

- Data compression: voice coder, MPEG, LZ algorithm.
- Modem

- Deep space communication (and coding was called a “marriage made in heaven”)
- CDMA, MIMO, 4G
- Physical layer security (Information-theoretic securiy)
- Quantum communication
- Stock market

1.4 Historical notes

- Sampling theorem: 1928 by Nyquist
- Hartley’s measure of information (1928)

$$I(X) = \log L,$$

  L=number of possible values of X.

- Information theory: 1948 by Shannon
  Investigate how to achieve the efficient and reliable communication
- Why using “entropy”?
  Shannon 与V. Neuman 讨论时, V. Neuman 建议用“熵”.

  1. 你的不确定函数在统计力学中已经被称为熵(entropy).
  2. 没有人知道熵到底是什么，所以有争论时你就永远立于不败之地.

- 在Shannon 1948年的原文中，既没有使用“mutual information”也没有用一个特殊符号来记它，而总是使用不确定性之差。术语“mutual information”及符号$I(X;Y)$是后来由Fano引入的.
- Shannon was born in Michigan, 1916. In 1936, he received B.S. degree in both electrical engineering and mathematics from the Univ. of Michigan. Received his M.S. and Ph.D. degree from MIT. In 1941, he joined Bell Lab. In 1958, he accepted a permanent appointment at MIT. 随后买了大房子, 房子里有很多玩具.
- Shannon的硕士论文是关于布尔代数与交换的，他基于此研究工作发表的第一篇论文won the 1940 Alfred Noble prize for the best paper in engineering published by an author under 30. It is widely recognized today as the foundation of the switching field and as one of the most important Master’s theses ever written. His Ph.D. dissertation, “An Algebra for Theoretical Genetics,” was completed in 1940. This thesis was never published.
- In 1961, Shannon published a pioneering paper ”Two-way Communication Channels”, which established the foundation for the discipline now known as ”multi-user information theory”; and later N. Abramson published his paper ”The Aloha System - Another Alternative for Computer Communications” in 1970 which introduced the concept of multiple access using a shared common channel. The information theoretic approach to multiaccess communication began in 1972 with a coding theorem developed by Ahlswede and Liao. In 1972, T. Cover published a paper “Broadcast channels”.

- 1995年，Gallager提出“Combining queueing theory with information theory for multiaccess”; 1998年，Ephremidus发表了论文“Information theory and communication networks: An unconsummated union”; 2000年，R. Alshwede, N. Cai, S.-Y. R. Li, and R. W. Yeung发表了著名论文“Network information flow”,提出了网络编码(Network coding)的思想; 2000年，P. Gupta和P. R. Kumar发表了论文“The Capacity of wireless networks”，提出了传送容量(Transport capacity)的概念；2003之后，研究正向着大规模网络的信息理论发展（Towards an IT of large networks）。

1.5 A model of digital communication systems

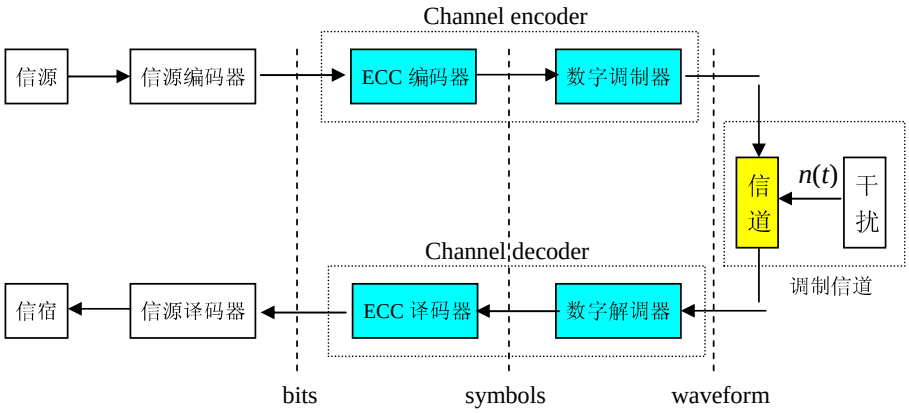


Figure 1.2: 数字通信系统示意图

- source: discrete/continuous; memoryless/with memory
- encoder: convert the messages into the signal which is suitable for transmission over physical channels.
- channel: wireless/cable, disk.
- interference.
- sink: destination.

1.6 Review of Probability

Bayes rule:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

For a discrete random variable (r.v.),

- Probability Mass Function (PMF)

  $$P_X(x) = P(X = x)$$

  denotes the Prob. of the event that X takes on the value x.

For a continuous r.v.,

- Cumulative Distribution Function (CDF)

$$F_X(x) = P(X \leq x)$$

- Probability Density Function (PDF)

$$p_X(x) = \frac{\mathrm{d}}{\mathrm{dx}} F_X(x)$$

Chapter 2

Entropy & Mutual Information (Shannon's measure of information)

This chapter introduces basic concepts and definitions required for the discussion later. Mainly include: Entropy, Mutual information(互信息), and relative entropy(相对熵).

2.1 Entropy

Let X be a discrete variable with alphabet $\mathcal{X}$ and PMF $P_X(x) = \Pr\{X = x\}, x \in \mathcal{X}$. For convenience, we will often write simply $P(x)$ for $P_X(x)$.

Definition 2.1.1. *The entropy of a discrete r.v. X is defined as*

$$\begin{aligned} H(X) &= \sum_{x\in\mathcal{X}} P(x) \log_b \frac{1}{P(x)} \\ &= -\sum_{x\in\mathcal{X}} P(x) \log_b P(x) \end{aligned} \tag{2.1}$$

When $b = 2$, the unit is called the *bit* (binary digit); when $b = e$, the unit is called the *nat* (natural unit). (Conversion is easy: $\log_b x = \log_b a \log_a x \Rightarrow H_b(x) = (\log_b a) H_a(x)$). Unless otherwise specified, we will take all logarithms to base 2, hence all entropies will be measured in bits.

In the above definition, we use the convention that $0 \log 0 = 0$. Note that equivalently, many books adopt the convention that the summation is taken over the corresponding support set. The support set of $P(X)$, denoted by $\mathcal{S}_X$, is the set of all $x \in \mathcal{X}$ such that $P(x) > 0$; i.e., $\mathcal{S}_X = \text{supp}(P_X) = \{x : P(x) > 0\}$.

The entropy $H(X)$ is also called the uncertainty of X, meaning that it is a measure of the randomness of X.

Note that the entropy $H(X)$ depends on the probabilities of different outcomes of X, but not on the names of the outcomes. For example,

$\mathcal{X} = \{Green, Blue, Red\}$	$\mathcal{Y} = \{Sunday, Monday, Friday\}$
$P(X) : 0.2, 0.3, 0.5$	$P(Y) : 0.2, 0.3, 0.5$

$$H(X) = H(Y)$$

Remark: The entropy of X can also be interpreted as the expected value of $\log \frac{1}{P(X)}$ (i.e., the average uncertainty):

$$H(X) = \mathbb{E}\left[\log \frac{1}{P(X)}\right] \tag{2.2}$$

where we define $\mathbb{E}[F(x)] = \sum_{x\in\mathcal{X}} P_X(x)F(x)$. Recall that $I(x) = \log \frac{1}{P_X(x)}$ is the self-information of the event $X = x$, so $H(X) = \mathbb{E}[I(x)]$ is also referred to as 平均自信息量。

A immediate consequence of the definition is that $H(X) \geq 0$.

Example 2.1.1. *Let*

$$X = \begin{cases} 1 & \textit{with probability } p \\ 0 & \textit{with probability } 1-p \end{cases}$$

Then

$$H(X) = -p\log p - (1-p)\log(1-p) \tag{2.3}$$

Equation (2.3) is often called the *binary entropy function*, and denoted by $H_2(p)$. Its graph is shown in Fig. 2.1. We can see that $H(X) = 1$ bit when $p = \frac{1}{2}$.

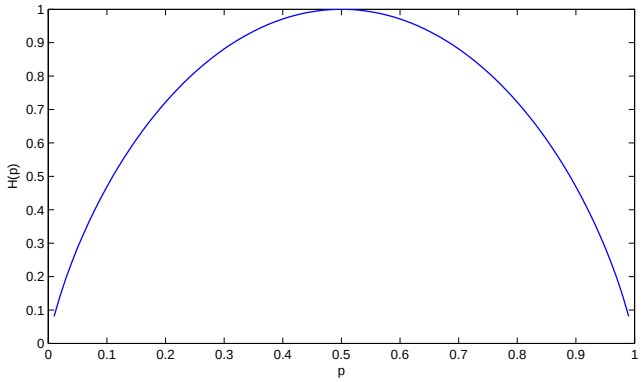


Figure 2.1: 二元随机变量的熵函数

Example 2.1.2. *Let*

$$X = \begin{cases} a & \textit{with probability } 1/2 \\ b & 1/4 \\ c & 1/8 \\ d & 1/8 \end{cases}$$

Then $H(X) = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{4}\log\frac{1}{4} - \frac{1}{8}\log\frac{1}{8} - \frac{1}{8}\log\frac{1}{8} = \frac{7}{4}$ *bits.*

On the other hand, if X takes on values in $\mathcal{X} = \{a, b, c, d\}$ with equal probability, then we have $H(X) = -\frac{1}{4}\log\frac{1}{4} \times 4 = 2$ bits=$\log|\mathcal{X}|$.

We can see that the uniform distribution over the range $\mathcal{X}$ is the maximum entropy distribution over this range. (In other words, the entropy of X is maximized when its values are equally likely.)

2.2 Joint entropy and conditional entropy

We now extend the definition of the entropy of a single random variable to a pair of random variables.

(X, Y) can be considered to be a single vector-valued random variable.

Definition 2.2.1. *The joint entropy $H(XY)$ of a pair of discrete random variables (X, Y) with a joint distribution $P(x, y)$ is defined as*

$$\begin{aligned} H(XY) &\triangleq \mathbb{E}[-\log P(X,Y)] \\ &= -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P(x,y)\log P(x,y) \\ &= -\sum_{(x,y)\in\mathcal{S}_{XY}} P(x,y)\log P(x,y) \end{aligned} \tag{2.4}$$

Definition 2.2.2. *The conditional entropy of the discrete random variable X, given that the event $Y = y$ occurs, is defined as*

$$\begin{aligned} H(X|Y=y) &= -\sum_{x\in\mathcal{X}} P(x|y)\log P(x|y) \\ &= \mathbb{E}\left[-\log P(X|Y)|Y=y\right] \end{aligned} \tag{2.5}$$

Definition 2.2.3. *If $(X, Y) \sim P(x, y)$, then the conditional entropy of the discrete random variable X, given the discrete random variable Y, is defined as*

$$\begin{aligned} H(X|Y) &= \sum_{y\in\mathcal{Y}} P(y)H(X|Y=y) \\ &= -\sum_{y\in\mathcal{Y}} P(y)\sum_{x\in\mathcal{X}} P(x|y)\log P(x|y) \\ &= -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P(x,y)\log P(x|y) \\ &= \mathbb{E}[-\log P(X|Y)] \end{aligned} \tag{2.6}$$

Notice that $H(Y|X) \neq H(X|Y)$.

Theorem 2.2.1. $H(XY) = H(X) + H(Y|X) = H(Y) + H(X|Y)$

Proof.

$$\begin{aligned} H(XY) &= -\sum_x\sum_y P(x,y)\log P(x,y) \\ &= -\sum_x\sum_y P(x,y)\left[\log P(x) + \log P(y|x)\right] \\ &= -\sum_x P(x)\log P(x) - \sum_x\sum_y P(x,y)\log P(y|x) \\ &= H(X) + H(Y|X) \end{aligned} \tag{2.7}$$

□

Corollary 2.2.2. $H(XY|Z) = H(X|Z) + H(Y|XZ)$

We now generalize the above theorem to a more general case.

Theorem 2.2.3 (Chain rule for entropy)**.** *Let $X_1, X_2, \ldots, X_N$ be discrete random variables drawn according to $P(x_1, x_2, \ldots, x_N)$. Then*

$$H(X_1, X_2, \ldots, X_N) = \sum_{n=1}^{N} H(X_n|X_1, X_2, \ldots, X_{n-1})$$

Proof.

$$\begin{aligned} H(X_1, X_2, \cdots, X_N) &= \mathbb{E}[-\log P(X_1, X_2, \cdots, X_N)] \\ &= \mathbb{E}\left[-\log \prod_{n=1}^{N} P(X_n|X_1, \cdots, X_{n-1})\right] \\ &= \sum_{n=1}^{N} \mathbb{E}[-\log P(X_n|X_1, \cdots, X_{n-1})] \\ &= \sum_{n=1}^{N} H(X_n|X_1, \cdots, X_{n-1}) \end{aligned} \tag{2.8}$$

□

If X_n are independent of each other, then

$$H(X_1, X_2, \cdots, X_N) = \sum_{n=1}^{N} H(X_n). \tag{2.9}$$

Similarly, we have

$$H(X_1, X_2, \cdots, X_N|Y) = \sum_{n=1}^{N} H(X_n|X_1, \cdots, X_{n-1}, Y).$$

2.3 Properties of the entropy function

Let X be a discrete random variable with alphabet $\mathcal{X} = \{x_k, k = 1, 2, \cdots, K\}$. Denote the pmf of X by $p_k = Pr(X = x_k), x_k \in \mathcal{X}$. Then the entropy $H(X)$ of X can be written as

$$H(\mathbf{p}) = -\sum_{k=1}^{K} p_k \log p_k$$

where $\mathbf{p} = (p_1, p_2, \cdots, p_K)$ is a K-vector of probabilities.

Property 2.3.1. $H(\mathbf{p}) \geq 0$ *(Non-negativity of entropy)*

Proof. Since $0 \leq p_k \leq 1$, we have $\log p_k \leq 0$. Hence, $H(\mathbf{p}) \geq 0$. □

Definition 2.3.1. *A function $f(x)$ is said to be convex-$\bigcup$ over an interval (a, b) if for every $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$,*

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

A function f is said to be strictly convex-$\bigcup$ if equality holds only when $\lambda = 0$ or $\lambda = 1$.

凸区域：若对于区域D中任意两点$\bar{\alpha}$ 和$\bar{\beta}$, $\bar{\alpha} \in D$, $\bar{\beta} \in D$, 有

$$\lambda\bar{\alpha} + (1-\lambda)\bar{\beta} \in D, \quad 0 \leq \forall\lambda \leq 1$$

则称D是凸区域。

凸函数：若定义在凸区域D上的函数$f(x)$满足

$$f(\lambda\bar{\alpha} + (1-\alpha)\bar{\beta}) \leq \lambda f(\bar{\alpha}) + (1-\lambda)f(\bar{\beta}), \qquad \forall\bar{\alpha}, \bar{\beta} \in D, 0 \leq \lambda \leq 1.$$

则称函数$f(x)$为凸-$\bigcap$ 函数.

Theorem 2.3.1. *If the function f has a second derivative which is non-negative (resp. positive) everywhere ($f''(x) \geq 0$), then the function is convex-$\bigcup$ (resp. strictly convex).*

Theorem 2.3.2 (Jensen's inequality)**.** *If f is a convex-$\bigcup$ function and X is a random variable, then*

$$\mathbb{E}[f(x)] \geq f(\mathbb{E}[X])$$

Proof. For a two mass point distribution, the inequality becomes

$$p_1 f(x_1) + p_2 f(x_2) \geq f(p_1 x_1 + p_2 x_2)$$

which follows directly from the definition of convex function.

Suppose that the theorem is true for distributions with $k-1$ mass points. Then writing $p_i' = \frac{p_i}{1-p_k}$ for $i = 1, 2, \ldots, k-1$, we have

$$\begin{aligned}
\sum_{i=1}^{k} p_i f(x_i) &= p_k f(x_k) + (1-p_k)\sum_{i=1}^{k-1} p_i' f(x_i) \\
&\geq p_k f(x_k) + (1-p_k) f(\sum_{i=1}^{k-1} p_i' x_i) \\
&\geq f\left(p_k x_k + (1-p_k)\sum_{i=1}^{k-1} p_i' x_i\right) \\
&= f(\sum_{i=1}^{k} p_i x_i) \qquad (2.10)
\end{aligned}$$

□

Property 2.3.2. *$H(\mathbf{p})$ is the convex-$\bigcap$ function of $\mathbf{p}$.*

IT-inequality: For a positive real number r,

$$\log r \leq (r-1)\log e \qquad (2.11)$$

with equality if and only if (iff) $r = 1$.

Proof. Let $f(r) = \ln r - (r-1)$. Then we have

$$f'(r) = \frac{1}{r} - 1 \text{ and } f''(r) = -\frac{1}{r^2} < 0.$$

We can see that $f(r)$ is convex-$\bigcap$ in the interval $r > 0$. Moreover, the maximize value is zero, which is achieved with $r = 1$. Therefore,

$$\ln r \leq r - 1$$

with equality iff $r = 1$. Equation (2.11) follows immediately. □

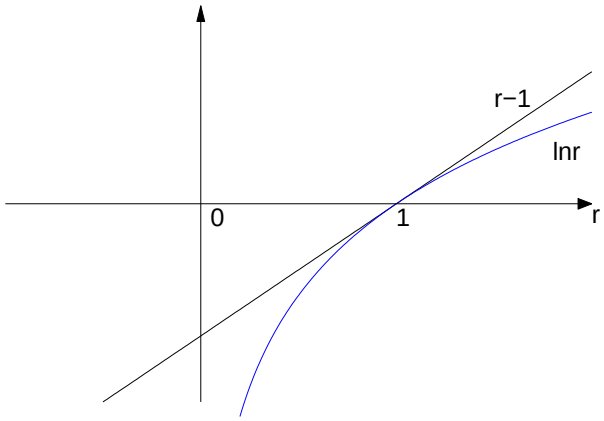


Figure 2.2: IT-inequality曲线

Theorem 2.3.3. *If the discrete r.v. X has K possible values, then $H(X) \leq \log K$, with equality iff $P(x) = \frac{1}{K}$ for all $x \in \mathcal{X}$.*

Proof.

$$\begin{aligned}
H(X) - \log K &= -\sum_{x \in \mathcal{S}_x} P(x) \log P(x) - \log K \\
&= \sum_x P(x) \left[\log \frac{1}{P(x)} - \log K \right] \\
&= \sum_x P(x) \log \frac{1}{KP(x)} \\
\text{(by IT-inequality)} \quad &\leq \sum_x P(x) \left[\frac{1}{KP(x)} - 1 \right] \log e \qquad (2.12) \\
&= \left[\sum_x \frac{1}{K} - \sum_x P(x) \right] \log e \\
&\leq \left[\sum_{x \in \mathcal{X}} \frac{1}{K} - 1 \right] \log e \\
&= (1-1) \log e = 0
\end{aligned}$$

where equality holds in (2.12) iff $KP(x) = 1$ for all $x \in \mathcal{S}_x$. □

Theorem 2.3.4 (Conditioning reduces entropy)**.** *For any two discrete r.v.'s X and Y,*

$$H(X|Y) \leq H(X)$$

with equality iff X and Y are independent.

(We can also use the relationship $I(X;Y) = H(X) - H(X|Y) \geq 0$ to obtain $H(X) \geq H(X|Y)$)

Proof.

$$\begin{aligned}
H(X|Y) - H(X) &= -\sum_{(x,y)\in\mathcal{S}_x y} P(x,y)\log P(x|y) + \sum_{x\in\mathcal{S}_x} P(x)\log P(x) \\
&= \sum_{x,y} P(x,y)\log\frac{P(x)}{P(x|y)} \\
&= \sum_{x,y} P(x,y)\log\frac{P(x)P(y)}{P(x,y)} \\
\text{(by IT-inequality)}\quad &\leq \sum_{x,y} P(x,y)\left(\frac{P(x)P(y)}{P(x,y)} - 1\right)\log e \\
&= \left(\sum_{x,y} P(x)P(y) - \sum_{x,y} P(x,y)\right)\log e \\
&\leq \left(\sum_{x\in\mathcal{X},y\in\mathcal{Y}} P(x)P(y) - 1\right)\log e \\
&= (1-1)\log e \\
&= 0
\end{aligned}$$

□

Note that, however $H(X|Y=y)$ may exceed $H(X)$.

Theorem 2.3.5 (Independence bound on entropy)**.** *Let $X_1,\ldots,X_N$ be drawn according to $P(X_1,X_2,\ldots,X_N)$. Then*

$$H(X_1,X_2,\ldots,X_N) \leq \sum_{n=1}^{N} H(X_n)$$

with equality iff the X_n are independent.

Proof. By the chain rule for entropies,

$$\begin{aligned}
H(X_1,X_2,\ldots,X_N) &= \sum_{n=1}^{N} H(X_n|X_1,\ldots,X_n-1) \\
&\leq \sum_{n=1}^{N} H(X_n) \qquad (2.13)
\end{aligned}$$

□

2.4 Relative entropy and mutual information

2.4.1 Relative entropy

The relative entropy is a measure of the "distance" between two distributions.

Definition 2.4.1. *The relative entropy (or Kullback Leiber distance) between two pmf $P(x)$ and $Q(x)$ is defined as*

$$D(P||Q) = \sum_{x\in\mathcal{X}} P(x)\log\frac{P(x)}{Q(x)} = \mathbb{E}_p\left[\log\frac{P(x)}{Q(x)}\right]$$

It is sometimes called the information divergence/cross-entropy/Kullback entropy between $P(x)$ and $Q(x)$.

- In general, $D(P||Q) \neq D(Q||P)$, so that relative entropy does not have the symmetry required for a true "distance" measure.
- The divergence inequality: $D(P||Q) \geq 0$
  with equality iff $P(x) = Q(x)$ for all $x \in \mathcal{X}$.

Proof. Let $\mathcal{S}_X = \{x : P(x) > 0\}$ be the support set of $P(x)$. Then

$$\begin{aligned} -D(P||Q) &= \sum_{x \in \mathcal{S}_X} P(x) \log \frac{Q(x)}{P(x)} \qquad (2.14) \\ (by\ IT - inequality) \quad &\leq \sum_x P(x) \left[\frac{Q(x)}{P(x)} - 1 \right] \log e \\ &= \left[\sum_x Q(x) - \sum_x P(x) \right] \log e \\ &\leq \left[\sum_{x \in \mathcal{X}} Q(x) - \sum_{x \in \mathcal{X}} P(x) \right] \log e \\ &= 0 \end{aligned}$$

□

- Conditional relative entropy: For joint PMFs $P(x, y)$ and $Q(x, y)$, the conditional relative entropy

$$\begin{aligned} D(P(y|x)||Q(y|x)) &= \sum_x P(x) \sum_y P(y|x) \log \frac{P(y|x)}{Q(y|x)} \\ &= \mathbb{E}_{P(x,y)} \left[\log \frac{P(y|x)}{Q(y|x)} \right] \qquad (2.15) \end{aligned}$$

- Chain rule for relative entropy:

$$D(P(x, y)||Q(x, y)) = D(P(x)||Q(x)) + D(P(y|x)||Q(y|x))$$

- Example: Entropy written as relative entropy
  Suppose that X take values in $\mathcal{X}$ with $|\mathcal{X}| = K$. Let U have the uniform distribution over $\mathcal{X}$. Then

$$\begin{aligned} D(P_X||P_U) &= \sum_x P_X(x) \log \frac{P_X(x)}{P_U(x)} \\ &= \sum_x P_X(x) \log \frac{P_X(x)}{1/K} \\ &= \log K - H(X) \end{aligned}$$

  It follows that $H(X) \leq \log |\mathcal{X}|$.

2.4.2 Mutual information

Mutual information is a measure of the amount of information that one r.v. contains about another r.v. It is the reduction in the uncertainty of one r.v. due to the knowledge of the other.

Definition 2.4.2. *Consider two random variables X and Y with a joint pmf $P(x,y)$ and marginal pmf $P(x)$ and $P(y)$. The (average) mutual information $I(X;Y)$ between X and Y is the relative entropy between $P(x,y)$ and $P(x)P(y)$, i.e.,*

$$\begin{aligned} I(X;Y) &= D(P(x,y)||P(x)P(y)) \\ &= \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P(x,y)\log\frac{P(x,y)}{P(x)P(y)} \\ &= \mathbb{E}_{p(x,y)}\left[\log\frac{P(x,y)}{P(x)P(y)}\right] \\ &= \sum_{x}\sum_{y} P(x,y)\log\frac{P(x|y)}{P(x)} \end{aligned} \tag{2.16}$$

Properties of mutual information:

1. Non-negativity of mutual information: $I(X;Y)\geq 0$
   with equality iff X and Y are independent.

   Proof. $I(X;Y) = D(P(x,y)||P(x)P(y)) \geq 0$, with equality iff $P(x,y)=P(x)P(y)$. □

2. Symmetry of mutual information: $I(X;Y) = I(Y;X)$

3. $$\begin{aligned} I(X;Y) &= \sum_{x}\sum_{y} P(x,y)\log\frac{P(x|y)}{P(x)} \\ &= -\sum_{x,y} P(x,y)\log P(x) - \left(-\sum_{x,y} P(x,y)\log P(x|y)\right) \\ &= H(X) - H(X|Y) \end{aligned} \tag{2.17}$$

   By symmetry, it also follows that $I(X;Y) = H(Y) - H(Y|X)$. Since $H(XY) = H(X) + H(Y|X)$, we have

   $$I(X;Y) = H(X) + H(Y) - H(XY)$$

4. $I(X;X) = H(X) - H(X|X) = H(X)$. Hence, entropy is sometimes referred to as *average self-information*.

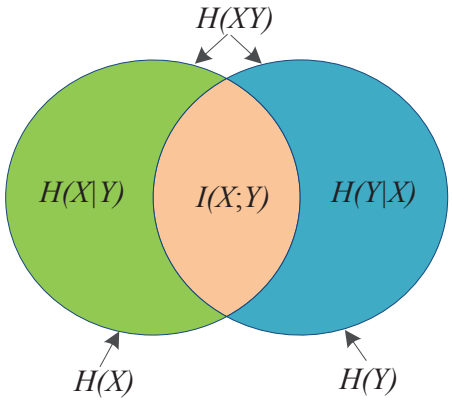


Figure 2.3: mutual information and conditional entropy

2.4.3 Conditional mutual information

Definition 2.4.3. *The conditional mutual information between the random variables X and Y, given the r.v. Z, is defined by*

$$\begin{aligned} I(X;Y|Z) &= H(X|Z) - H(X|YZ) \\ &= \sum_x \sum_y \sum_z P(x,y,z) \log \frac{P(x,y|z)}{P(x|z)P(y|z)} \\ &= \mathbb{E}_{p(x,y,z)} \left[\log \frac{P(x|yz)}{P(x|z)} \right] \end{aligned} \tag{2.18}$$

It follows from the definition that

$$I(X;Y|Z) = I(Y;X|Z) \geq 0$$

with equality iff conditional on each Z, X and Y are statistically independent; i.e., $P(x,y|z) = P(x|z)P(y|z)$ for each element in the joint sample space for which $P(z) > 0$.

We can visualize the situation in which $I(X;Y|Z) = 0$ as a pair of channels in cascade as shown in Fig. 2.4. We assume that the output of the 2nd channel depends statistically only on the input to the 2nd channel,i.e. $p(y|z) = p(y|z,x)$, all x, y, z with $p(z,x) > 0$.

Mutiplying both sides by $P(x|z)$, we obtain $P(x,y|z) = P(x|z)P(y|z)$, so that $I(X;Y|Z) = 0$.

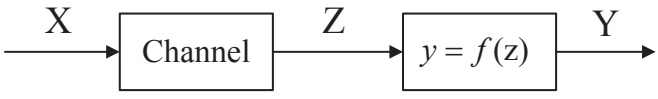


Figure 2.4: Channels in cascade

- **An important property:**

$$\begin{aligned} I(X;YZ) &= I(YZ;X) \\ &= I(X;Y) + I(X;Z|Y) \\ &= I(X;Z) + I(X;Y|Z) \end{aligned} \tag{2.19}$$

Proof.

$$\begin{aligned} I(X;YZ) &= H(YZ) - H(YZ|X) \\ &= H(Y) + H(Z|Y) - [H(Y|X) + H(Z|XY)] \\ &= [H(Y) - H(Y|X)] + [H(Z|Y) - H(Z|XY)] \\ &= I(X;Y) + I(X;Z|Y) \end{aligned} \tag{2.20}$$

□

Theorem 2.4.1 (Chain rule for mutual information)**.**

$$I(X_1, X_2, \ldots, X_N; Y) = \sum_{n=1}^{N} I(X_n; Y|X_1, \ldots, X_{n-1})$$

Proof.

$$\begin{aligned} I(X_1, X_2, \ldots, X_N; Y) &= H(X_1, X_2, \ldots, X_N) - H(X_1, X_2, \ldots, X_N|Y) \\ &= \sum_{n=1}^{N} H(X_n|X_1, X_2, \ldots, X_{n-1}) - \sum_{n=1}^{N} H(X_n|X_1, X_2, \ldots, X_{n-1}, Y) \\ &= \sum_{n} I(X_n; Y|X_1, \ldots, X_{n-1}) \end{aligned} \tag{2.21}$$

□

2.4.4 Data processing inequality

The data processing inequality can be used to show that no clever manipulation of the data can improve the inferences that can be made from the data.

Definition 2.4.4. *Random variables X, Z, Y are said to form a Markov Chain in that order (denoted by $X \to Z \to Y$) if the conditional distribution of Y depends only on Z and is conditionally independent of X. Specifically,*

$$X \to Z \to Y \quad \text{if} P(x, z, y) = P(x)P(z|x)P(y|z)$$

Theorem 2.4.2 (Data processing inequality)**.** *If $X \to Z \to Y$, then*

$$I(X; Z) \geq I(X; Y)$$

Proof. By the chain rule, we obtain

$$\begin{aligned} I(X;YZ) &= I(X;Y) + I(X;Z|Y) \\ &= I(X;Z) + I(X;Y|Z) \end{aligned}$$

Since X and Y are independent given Z, we have $I(X;Y|Z) = 0$. Thus,

$$I(X;Z) = I(X;Y) + I(X;Z|Y) \tag{2.22}$$

From the non-negativity of mutual information, $I(X;Z|Y) \geq 0$. Thus, (2.22) implies that

$$I(X;Z) \geq I(X;Y)$$

□

From the symmetry, it follows that

$$I(Y;Z) \geq I(X;Y)$$

This theorem demonstrates that no processing of Z, deterministic or random, can increase the information that Z contains about X.

Corollary 2.4.3. *In particular, if $Y = f(Z)$, we have $I(X;Z) \geq I(X;f(Z))$.*

Proof. $X \to Z \to f(Z)$ forms a Markov Chain. □

Corollary 2.4.4. *If $X \to Z \to Y$, then $I(X;Z|Y) \leq I(X;Z)$.*

Proof. Using the fact $I(X;Y) \geq 0$, the corollary follows immediately from (2.22). □

Expressing the data processing inequality in terms of entropies, we have

$$\begin{aligned} &H(X) - H(X|Z) \geq H(X) - H(X|Y) \\ &\Rightarrow H(X|Z) \leq H(X|Y) \end{aligned} \tag{2.23}$$

The average uncertainty $H(X|Z)$ about the input of a channel given the output is called *the equivocation on the channel*, and thus the above inequality yields the intuitively satisfying result that this uncertainty or equivocation can never decrease as we go further from the input on a sequence of cascaded channels.

Example 2.4.1. *Suppose that the random vector $[X,Y,Z]$ is equally likely to take on values in $\{[000],[010],[100],[101]\}$. Then*

$$H(X) = H_2\left(\frac{1}{2}\right) = 1\text{bit}$$

Note that $P_{Y|X}(0|1) = 1$ so that

$$H(Y|X=1) = 0$$

Similarly, $P_{Y|X}(0|0) = \frac{1}{2}$, so that

$$H(Y|X=0) = H_2\left(\frac{1}{2}\right) = 1\text{bit}$$

Thus, $H(Y|X) = \frac{1}{2} \times 1 = \frac{1}{2}$ bit.

$$\begin{aligned} H(Z|XY) &= \sum_{x,y} P(x,y) H(Z|X=x, Y=y) \\ &= \frac{1}{4}(0) + \frac{1}{4}(0) + \frac{1}{2}(1) \\ &= \frac{1}{2} \text{ bit} \end{aligned}$$

and $H(XYZ) = H(X) + H(Y|X) + H(Z|XY) = 1 + \frac{1}{2} + \frac{1}{2} = 2$ bits.

Because $P_Y(1) = \frac{1}{4}, P_Y(0) = \frac{3}{4}$, we have

$$H(Y) = H_2\left(\frac{1}{4}\right) = 0.811 \quad bits$$

we see that $H(Y|X) = \frac{1}{2} < H(Y)$. However, $H(Y|X=0) = 1 > H(Y)$.

Furthermore, $I(X;Y) = H(Y) - H(Y|X) = 0.811 - 0.5 = 0.311$ bits
In words, the first component of the random vector $[X,Y,Z]$ gives 0.311 bits of information about the 2nd component, and vice versa.

2.5 Sufficient Statistics

Suppose we have a family of pmfs $\{f_\theta(x)\}$ indexed by θ, and let X be a sample from a distribution in this family. Let $T(X)$ be any statistic (like the sample mean or variance). Then

$$\theta \to X \to T(X), \text{ and } I(\theta; T(X)) \leq I(\theta; X)$$

A statistic $T(X)$ is called sufficient for θ if it contains all the information in X about θ.

Definition 2.5.1. *A function $T(X)$ is said to be a sufficient statistic relative to the family $\{f_\theta(x)\}$ if X is independent of θ given $T(X)$; i.e., $\theta \to X \to T(X)$ forms a Markov Chain.*

It means that $I(\theta; X) = I(\theta; T(X))$ for all distributions on θ.

Example 2.5.1. *Let $X_1, \ldots, X_n, X_i \in \{0, 1\}$, be an i.i.d sequence of coin tosses of a coin with unknown parameter $\theta = Pr(X_i = 1)$. Given n, the number of 1's is a sufficient statistics for θ. Here, $T(X_1, \ldots, X_n) = \sum_{i=1}^n X_i$.*

Example 2.5.2. *If $X \sim N(0, 1)$, i.e., if*

$$f_\theta(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{(x-\theta)^2}{2}} = N(\theta, 1)$$

and $X_1, \ldots, X_n$ are drawn independently according to this distribution, then $\bar{X}_n = T(X_1, \ldots, X_n) = \frac{1}{n}\sum_{i=1}^n X_i$ is a sufficient statistic for θ.

2.6 Fano's inequality

Suppose that we wish to estimate a r.v. X based on a correlated r.v. Y. Fano's inequality relates the probability of error in estimating X to its conditional entropy $H(X|Y)$.

Theorem 2.6.1. *For any r.v.'s X and Y, we try to guess X by $\hat{X} = g(Y)$. The error probability $P_e = P(\hat{X} \neq X)$ satisfies*

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y) \tag{2.24}$$

Proof. Applying chain rule on $H(X, E|\hat{X})$, where error r.v.

$$E = \begin{cases} 1, & \text{if } \hat{X} \neq X \\ 0, & \text{if } \hat{X} = X \end{cases}$$

we have

$$\begin{aligned}
H(E, X|\hat{X}) &= H(X|\hat{X}) + H(E|X\hat{X}) = H(X|\hat{X}) \\
H(E, X|\hat{X}) &= H(E|\hat{X}) + H(X|E\hat{X}) \\
&\leq H(E) + H(X|E\hat{X}) \\
&= H(P_e) + P(E=1)H(X|\hat{X}, E=1) + P(E=0)H(X|\hat{X}, E=0) \\
&= H(P_e) + P_e H(X|\hat{X}, E=1) + (1-P_e)H(X|\hat{X}, E=0) \\
&= H(P_e) + P_e H(X|\hat{X}, E=1) + 0 \\
&\leq H(P_e) + P_e \log(|\mathcal{X}| - 1) \quad X\text{取除}\hat{X}\text{之外的其它}|\mathcal{X}| - 1\text{个值} \\
&\leq 1 + P_e \log|\mathcal{X}| \qquad (2.25)
\end{aligned}$$

Thus we can obtain

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}).$$

By the data processing theorem, we have $I(X;\hat{X}) \leq I(X;Y) \Rightarrow H(X|\hat{X}) \geq H(X|Y)$. Then (2.24) follows. □

物理解释：观察到Y（即已知Y）的条件下，对X还存在的不确定性可分为两部分：

- 判断猜测结果是否正确，其不确定性为$H(P_e)$；
- 如果判决是错的（with probability P_e），则这时X可能取除$\hat{X}$之外的其它$|\mathcal{X}| - 1$个值。为了确定是哪一个，所需的信息量$\leq \log(|\mathcal{X}| - 1)$.

2.7 Convex functions (互信息的凸性)

2.7.1 Concavity of entropy

Theorem 2.7.1. *Entropy $H(X)$ is concave in $P(x)$.*

That is, if X_1, X_2 are r.v.'s defined on $\mathcal{X}$ with distribution $P_1(x)$ and $P_2(x)$, respectively. For any $\theta \in [0,1]$, consider a r.v. X with

$$P_X(x) = \theta P_1(x) + (1-\theta)P_2(x) \quad \forall x$$

then

$$H(X) \geq \theta H(X_1) + (1-\theta)H(X_2)$$

Proof. Let Z be a binary r.v. with $P(Z=0) = \theta$. Let

$$X = \begin{cases} X_1 & \text{if } Z = 0 \\ X_2 & \text{if } Z = 1 \end{cases}$$

Then

$$\begin{aligned} H(X) &\geq H(X|Z) \\ &= \theta H(X|Z=0) + (1-\theta)H(X|Z=1) \\ &= \theta H(X_1) + (1-\theta)H(X_2) \end{aligned} \tag{2.26}$$

or

$$H(\theta P_1 + (1-\theta)P_2) \geq \theta H(P_1) + (1-\theta)H(P_2)$$

□

2.7.2 Concavity of mutual information

Theorem 2.7.2. *For a fixed transition probability $P(y|x)$, $I(X;Y)$ is a concave (convex-$\cap$) function of $P(x)$.*

Proof. Construct X_1, X_2, X, and Z as above. Consider

$$\begin{aligned} I(XZ;Y) &= I(X;Y) + I(Z;Y|X) \\ &= I(Z;Y) + I(X;Y|Z) \end{aligned} \tag{2.27}$$

Conditioned on X, r.v.'s Y and Z are independent, i.e., $P(y|x,z) = P(y|x)$. Using $I(Y;Z|X) = 0$, we have

$$\begin{aligned} I(X;Y) &\geq I(X;Y|Z) \\ &= \theta I(X;Y|Z=0) + (1-\theta) I(X;Y|Z=1) && (2.28) \\ &= \theta I(X_1;Y) + (1-\theta) I(X_2;Y) && (2.29) \end{aligned}$$

□

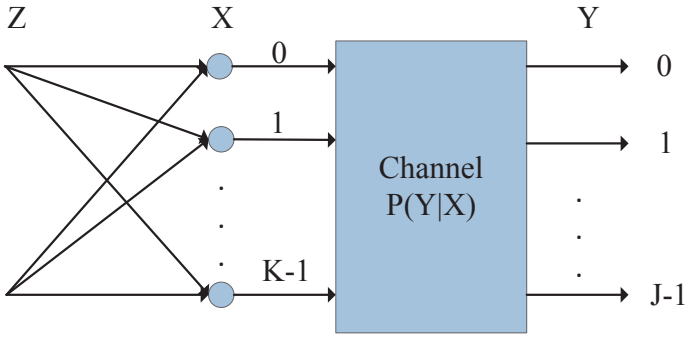


Figure 2.5: 给定信道下互信息的凸性

Theorem 2.7.3. *For a fixed input distribution $P(x)$, $I(X;Y)$ is convex-$\cup$ in $P(y|x)$.*

Proof. Consider a r.v. X and two channels with $P_1(y|x)$ and $P_2(y|x)$. When feed with X, the outputs of the two channels are denoted by Y_1 and Y_2.

Now let one channel be chosen randomly according to a binary r.v. Z that is independent of X, and denote the output by Y.

$$\begin{aligned} I(X;YZ) &= I(X;Z) + I(X;Y|Z) \\ &= I(X;Y) + I(X;Z|Y) \end{aligned} \tag{2.30}$$

Thus, $I(X;Y) < I(X;Y|Z) = \theta I(X;Y_1) + (1-\theta) I(X;Y_2)$. □

2.7.3 互信息的凸性—另一证明方法

Theorem 2.7.4. *Let $(X,Y) \sim P(x,y) = P(x)P(y|x)$. The mutual information $I(X;Y)$ is a convex-$\cap$ function of $P(x)$ for fixed $P(y|x)$ and a convex-$\cup$ function of $P(y|x)$ for fixed $P(x)$.*

Proof.

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum_x P(x) H(Y|X=x) \end{aligned} \tag{2.31}$$

$$\left\{\begin{array}{l}\text{if } P(y|x) \text{ is fixed, then } P(y) \text{ is a linear function of } P(x)\\ H(Y) \text{ is a convex-}\cap \text{ function of } P(y)\end{array}\right.$$

$$\left.\begin{array}{l}\Rightarrow\ H(Y) \text{ is a convex-}\cap \text{ function of } P(x)\\ \text{The 2nd term in (2.31) is a linear function of } P(x)\end{array}\right\}\Rightarrow$$

The difference is a convex-$\cap$ function of $P(x)$.

To prove the 2nd part, we fix $P(x)$ and consider two different conditional distributions $P_1(y|x)$ and $P_2(y|x)$.

Let

$$P_\lambda(y|x) = \lambda P_1(y|x) + (1-\lambda)P_2(y|x)$$

Then

$$P_\lambda(x,y) = \lambda P_1(x,y) + (1-\lambda)P_2(x,y)$$

and

$$P_\lambda(y) = \lambda P_1(y) + (1-\lambda)P_2(y)$$

where $P_1(x,y) = P_1(y|x)P(x)$ and $P_2(x,y) = P_2(y|x)P(x)$.

If we let $Q_\lambda(x,y) = P(x)P_\lambda(y)$, then we have

$$\begin{aligned} Q_\lambda(x,y) &= P(x)[\lambda P_1(y) + (1-\lambda)P_2(y)] \\ &= \lambda Q_1(x,y) + (1-\lambda)Q_2(x,y) \end{aligned} \tag{2.32}$$

Since $I(X;Y) = D(P_\lambda||Q_\lambda)$ and $D(P||Q)$ is a convex function of (P,Q). it follows that $I(X;Y)$ is a convex-$\cup$ function of $P(y|x)$. □

2.8 Differential Entropy

Differential entropy is the entropy of a continuous random variable.

Let X be a continuous random variable with cumulative distribution function $F(x) = \Pr\{X \le x\}$. The probability density function of X (if it exists) is given by

$$p(x) = \frac{\mathrm{d}F(x)}{\mathrm{d}x} \qquad\qquad x \in (-\infty, +\infty)$$

The set of x where $p(x) > 0$ is called the support set of X.

Definition 2.8.1. *The differential entropy $h(X)$ (or $H_c(X)$) of a continuous random variable X with a density $p(x)$ is defined by*

$$h(X) = \int_S p(x)\log\frac{1}{p(x)}\mathrm{d}x = -\int_S p(x)\log p(x)\mathrm{d}x \tag{2.33}$$

where S is the support set of the random variable.

- Motivation for the definition: Think of quantizing the X axis into intervals of length $\triangle x$, then the quantized outcomes form a discrete ensemble with

$$P_r(x_i < x \le x_i + \triangle x) \approx p(x_i)\triangle x.$$

The self-information of an X is given by $\log \frac{1}{p(x_i)\triangle x}$.

The entropy of the quantized r.v. is $H_{\triangle x}(X) = -\sum_{i=-\infty}^{+\infty} p(x_i)\triangle x \log p(x_i)\triangle x$. As $\triangle x \to 0$, we have

$$\begin{aligned}\lim_{\triangle x\to 0} H_{\triangle x}(x) &= -\int_{-\infty}^{\infty} p(x)\log p(x)\mathrm{d}x - \lim_{\triangle x\to 0}\log\triangle x\int_{-\infty}^{\infty} p(x)\mathrm{d}x \\ &= -\int_{-\infty}^{\infty} p(x)\log p(x)\mathrm{d}x - \lim_{\triangle x\to 0}\log\triangle x \end{aligned} \tag{2.34}$$

It is seen that the 2nd term in the RHS of (2.35) approaches ∞ as $\triangle x \to 0$. For the purpose of calculation, the first term is often used to define the entropy of a continuous ensemble.

Example 2.8.1 (Uniform distribution)**.** *Consider a random variable X distributed uniformly over (a, b). Its differential entropy is*

$$h(x) = -\int_a^b \frac{1}{b-a}\log\frac{1}{b-a}\mathrm{d}x = \log(b-a)$$

Notice that the differential entropy are not necessarily positive, not necessarily finite.

Example 2.8.2 (Gaussian distribution)**.** *$X \sim \mathcal{N}(0, \sigma^2)$. The p.d.f of X is*

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{x^2}{2\sigma^2}\right)$$

The differential entropy is then given by

$$\begin{aligned} h(x) &= -\int_{-\infty}^{\infty} p(x)\left[-\ln\sqrt{2\pi}\sigma - \frac{x^2}{2\sigma^2}\right]\mathrm{d}x \\ &= \frac{1}{2}\ln 2\pi\sigma^2 + \frac{1}{2\sigma^2}\int_{-\infty}^{\infty} p(x)x^2\mathrm{d}x \\ &= \frac{1}{2}\ln 2\pi\sigma^2 + \frac{1}{2} \\ &= \frac{1}{2}\ln 2\pi\sigma^2 + \frac{1}{2}\ln \mathrm{e} \\ &= \frac{1}{2}\ln 2\pi\mathrm{e}\sigma^2 \qquad \text{nats} \\ &= \frac{1}{2}\log_2 2\pi\mathrm{e}\sigma^2 \qquad \text{bits} \\ &(\log_2\Phi = \ln\Phi\cdot\log_2\mathrm{e}) \end{aligned} \tag{2.35}$$

From (2.35), the variance of X can be expressed as

$$\sigma^2 = \frac{1}{2\pi\mathrm{e}}\mathrm{e}^{2h(X)}$$

which is sometimes called the "熵功率".

2.8.1 Joint and conditional differential entropy

Definition 2.8.2. *The differential entropy of a set $\{X_1, X_2, \ldots, X_n\}$ of random variables with density $p(x_1, x_2, \ldots, x_n)$ is defined as*

$$h(X_1, X_2, \ldots, X_n) = -\int p(x_1, x_2, \ldots, x_n)\log p(x_1, x_2, \ldots, x_n)\mathrm{d}x_1\mathrm{d}x_2\ldots\mathrm{d}x_n$$

Definition 2.8.3. *If X, Y have a joint pdf $p(x, y)$, then the conditional differential entropy is*

$$h(X|Y) = -\iint p(x,y)\log p(x|y)\mathrm{d}x\mathrm{d}y$$

Since $p(x, y) = p(x|y)p(y)$, we can obtain

$$\begin{aligned} h(XY) &= h(Y) + h(X|Y) \\ &= h(X) + h(Y|X) \end{aligned} \tag{2.36}$$

Theorem 2.8.1. *Let $X_1, X_2, \ldots, X_n$ have a multivariate normal distribution with mean $\mathbf{m}$ and covariance matrix $\mathbf{K}$. Then*

$$h(X_1, X_2, \ldots, X_n) = \frac{1}{2}\log\left((2\pi\mathrm{e})^n|\mathbf{K}|\right) \qquad \textit{bits} \tag{2.37}$$

Proof. The pdf of $X_1, X_2, \ldots, X_n$ is

$$p(\mathbf{X}) = \frac{1}{(\sqrt{2\pi})^n|\mathbf{K}|^{\frac{1}{2}}}\mathrm{e}^{-\frac{1}{2}(\mathbf{x}-\mathbf{m})^T\mathbf{K}^{-1}(\mathbf{x}-\mathbf{m})}$$

Then

$$\begin{aligned} h(X_1, X_2, \ldots, X_n) &= -\int p(\mathbf{x})\left[-\frac{1}{2}(\mathbf{x}-\mathbf{m})^T\mathbf{K}^{-1}(\mathbf{x}-\mathbf{m}) - \ln(\sqrt{2\pi})^n|\mathbf{K}|^{\frac{1}{2}}\right]\mathrm{d}\mathbf{x} \\ &= \frac{1}{2}\mathbb{E}\left[\sum_i\sum_j (x_i - m_i)(\mathbf{K}^{-1})_{ij}(x_j - m_j)\right] + \frac{1}{2}\ln(2\pi)^n|\mathbf{K}| \\ &= \frac{1}{2}\sum_i\sum_j \mathbb{E}\left[(x_i - m_i)(x_j - m_j)\right](\mathbf{K}^{-1})_{ij} + \frac{1}{2}\ln(2\pi)^n|\mathbf{K}| \\ &= \frac{1}{2}\sum_j\sum_i \mathbf{K}_{ji}(\mathbf{K}^{-1})_{ij} + \frac{1}{2}\ln(2\pi)^n|\mathbf{K}| \\ &= \frac{1}{2}\sum_j (\mathbf{K}\mathbf{K}^{-1})_{jj} + \frac{1}{2}\ln(2\pi)^n|\mathbf{K}| \\ &= \frac{1}{2}\sum_j \mathbf{I}_{jj} + \frac{1}{2}\ln(2\pi)^n|\mathbf{K}| \\ &= \frac{n}{2} + \frac{1}{2}\ln(2\pi)^n|\mathbf{K}| \\ &= \frac{1}{2}\ln(2\pi\mathrm{e})^n|\mathbf{K}| \qquad \text{nats} \\ &= \frac{1}{2}\log_2\left((2\pi\mathrm{e})^n|\mathbf{K}|\right) \qquad \text{bits} \end{aligned}$$

($\mathbf{K}$是对称矩阵: $\mathbf{K}_{ij} = \mathbf{K}_{ji}$) □

2.8.2 Relative entropy and mutual information

Definition 2.8.4. *The relative entropy $D(p||q)$ between two densities p and q is defined by*

$$D(p||q) = \int p\log\frac{p}{q}$$

Note that $D(p||q)$ is finite only if the support set of p is contained in the support set of q.

Definition 2.8.5. *The mutual information $I(X;Y)$ between two random variables with joint density $p(x,y)$ is defined as*

$$I(X;Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}\mathrm{d}x\mathrm{d}y \tag{2.38}$$

It is clear that

$$\begin{aligned} I(X;Y) &= h(X) - h(X|Y) \\ &= h(Y) - h(Y|X) = h(X) + h(Y) - h(XY) \end{aligned} \tag{2.39}$$

and $I(X;Y) = D\left(p(x,y)||p(x)p(y)\right)$.

- The (average) conditional mutual information is given by

$$I(X;Y|Z) = \iiint p(x,y,z)\frac{p(x,y|z)}{p(x|z)p(y|z)}\mathrm{d}x\mathrm{d}y\mathrm{d}z$$

2.8.3 Properties

(1) $D(p(x)||q(x)) \geq 0$ with equality if $p(x) = q(x)$ almost everywhere (a.e.).

Proof. Let S be the support set of p. Then

$$-D(p||q) = \int_S p\log\frac{q}{p}dx \leq \int_S p\left(\frac{q}{p} - 1\right)\log\mathrm{e}dx = \int_S q(x)dx - 1 \leq 0$$

□

Corollary 2.8.2. *$I(X;Y) \geq 0$ with equality iff X and Y are independent.*

(2) $h(X|Y) \leq h(X)$ with equality iff X and Y are independent.

(3) Chain rule for differential entropy

$$h(X_1, X_2, \ldots, X_N) = \sum_{n=1}^{N} h(X_n|X_1, X_2, \ldots, X_{n-1})$$

Proof. Follows directly from the definitions. □

(4) $h(X + c) = h(X)$
$h(aX) = h(X) + \log|a|$

Proof. Let $Y = aX$. Then $p_Y(y) = \frac{1}{|a|}p_X\left(\frac{y}{a}\right)$, and $p_Y(y) = \frac{p_X(x)}{|f'(x)|}$,$x = \frac{y}{a}$.

$$\begin{aligned} h(aX) &= -\int p_Y(y)\log p_Y(y)\mathrm{d}y \\ &= -\int \frac{1}{|a|}p_X\left(\frac{y}{a}\right)\log\frac{1}{|a|}p_X\left(\frac{y}{a}\right)\mathrm{d}y \\ &= -\left[\int \frac{1}{|a|}p_X\left(\frac{y}{a}\right)\log p_X\left(\frac{y}{a}\right) - \log|a|\int\frac{1}{|a|}p_X\left(\frac{y}{a}\right)\mathrm{d}y\right] \\ &= -\int p_X(x)\log p_X(x) + \log|a| \\ &= h(X) + \log|a| \end{aligned}$$

□

Similarly, let $\mathbf{X}$ be a random vector with joint pdf $f(\mathbf{x})$, and $\mathbf{c}$ be a column vector in $\mathbb{R}^n$. Then $h(\mathbf{X}+\mathbf{c}) = h(\mathbf{X})$.

Let $\mathbf{A}$ be a nonsingular $n \times n$ matrix. Then we have $h(\mathbf{AX}) = h(\mathbf{X}) + \log|det(\mathbf{A})|$, where $|det(\mathbf{A})|$ is the absolute value of the determinant.

- In general, if $Y = f(X)$, where f is a one-to-one correspondence between X and Y, then

$$h(Y) = h(X) + \mathsf{E}\left[\log\left|\frac{\mathrm{d}y}{\mathrm{d}x}\right|\right]$$

(5) **Data processing theorem:** If X is the input to a channel, Z the output, and Y a transformation of Z (i.e., $Y = f(Z)$), then

$$I(X;Z) \geq I(X;Y).$$

Suppose that Y is a reversible transformation of Z, so that $z = f^{-1}(y)$. Then $I(X;Y) \geq I(X;Z)$. Combining these two equations, we have $I(X;Y) = I(X;Z)$.

This implies that the average mutual information between two ensembles is invariant to any reversible transformation of one of the outcomes.

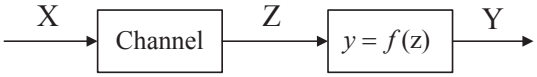


Figure 2.6: 信息处理定理

(6)

$$I(X;YZ) = I(X;Y) + I(X;Z|Y) \tag{2.40}$$
$$= I(X;Z) + I(X;Y|Z) \tag{2.41}$$

(7) The Gaussian distribution has the largest differential entropy of any distribution with a specified mean and variance. In other words, if the random variable X has mean m and variance $\sigma^2 < \infty$ then

$$h(X) \leq \frac{1}{2}\log 2\pi\mathrm{e}\sigma^2$$

Proof. Suppose that $q(x)$ is the density of a $N(m,\sigma^2)$ variable X. Then

$$\log q(x) = \log\left(\frac{1}{\sqrt{2\pi}\sigma}\mathrm{e}^{-\frac{(x-m)^2}{2\sigma^2}}\right) = -\log\sqrt{2\pi}\sigma - \frac{(x-m)^2}{2\sigma^2}$$

Let $p(x)$ be any density of a random variable with mean m_p and variance σ_p^2. We have

$$\begin{aligned}
-\int_{-\infty}^{\infty} p(x)\log q(x)\mathrm{d}x &= \log\sqrt{2\pi}\sigma + \frac{1}{2\sigma^2}\int_{-\infty}^{\infty} p(x)(x-m)^2\mathrm{d}x \\
&= \log\sqrt{2\pi}\sigma + \frac{1}{2\sigma^2}\int_{-\infty}^{\infty} p(x)\left[(x-m_p)+(m_p-m)\right]^2\mathrm{d}x \\
&= \log\sqrt{2\pi}\sigma + \frac{1}{2\sigma^2}\left[\sigma_p^2 + (m_p-m)^2\right]
\end{aligned}$$

Thus, if $p(x)$ and $p'(x)$ have the same mean and the same variance, then

$$-\int_{-\infty}^{\infty} p(x)\log q(x)\mathrm{d}x = -\int_{-\infty}^{\infty} p'(x)\log q(x)\mathrm{d}x$$

With this equation, we obtain

$$\begin{aligned}
0 &\le D(p||q) \\
&= \int_{-\infty}^{\infty} p(x)\log p(x)dx - \int_{-\infty}^{\infty} p(x)\log q(x)dx \\
&= -h(p) - \int_{-\infty}^{\infty} q(x)\log q(x)dx \\
&= -h(p) + h(q)
\end{aligned}$$

□

- **A summery:**

$$\begin{cases} \text{given } \sigma^2, & \text{the Gaussian 分布 has 最大熵} \\ \text{given } h(X), & \text{the Gaussian 分布 has the smallest } \sigma^2\text{(average power)} \end{cases}$$

The above argument can be extended to a random vector.

Theorem 2.8.3. *Let the random vector $\mathbf{X} \in \mathcal{R}^n$ has zero mean and covariance $\mathbf{K} = \mathrm{E}\left[\mathbf{X}\mathbf{X}^T\right]$ i.e., $K_{ij} = \mathrm{E}\left[X_i X_j\right], 1 \le i, j \le n$. Then*

$$h(\mathbf{X}) \le \frac{1}{2}\log(2\pi\mathrm{e})^n|\mathbf{K}|$$

with equality iff $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$.

(8) If the random variable X takes on values in the interval $[-M, M]$, i.e., $x \in [-M, M]$, then $h(X) \le \log(2M)$, with equality iff X has a uniform distribution over $[-M, M]$.

Proof. See [textbook, pp.30] ——Method 1
Method 2: Let $p(x)$ be the uniform density over $[a, b]$ satisfying $\int_a^b p(x)\mathrm{d}x = 1$. Let $q(x)$ be an arbitrary density with $\int_a^b q(x)\mathrm{d}x = 1$. Then

$$\begin{aligned}
h\left[x, q(x)\right] - h\left[x, p(x)\right] &= -\int_a^b q(x)\log q(x)\mathrm{d}x + \int_a^b p(x)\log p(x)\mathrm{d}x \\
&= -\int_a^b q(x)\log q(x)\mathrm{d}x + \int_a^b p(x)\log\left(\frac{1}{b-a}\right)\mathrm{d}x \\
&= -\int_a^b q(x)\log q(x)\mathrm{d}x + \left[\log\left(\frac{1}{b-a}\right)\int_a^b p(x)\mathrm{d}x\right] \\
&= -\int_a^b q(x)\log q(x)\mathrm{d}x + \left[\log\left(\frac{1}{b-a}\right)\int_a^b q(x)\mathrm{d}x\right] \\
&= -\int_a^b q(x)\log q(x)\mathrm{d}x + \int_a^b q(x)\log p(x)\mathrm{d}x \\
&= \int_a^b q(x)\log\frac{p(x)}{q(x)}\mathrm{d}x \le \log\left[\int_a^b q(x)\frac{p(x)}{q(x)}\mathrm{d}x\right] = 0
\end{aligned}$$

其中“≤” by Jensen's inequality. □

Chapter 3

The Asymptotic Equipartition Property (渐进等同分割性)

3.1 预备知识：Convergence of random variables

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i \to \mathbb{E}[X] \qquad \text{for i.i.d. r.v.'s}$$

3.1.1 Type of convergence

- Almost sure convergence (also called convergence with probability 1):

  $$P\left(\lim_{n\to\infty} Y_n(w) = Y(w)\right) = 1$$

  write $Y_n \xrightarrow[a.s.]{} Y$

- Mean-square convergence:

  $$\lim_{n\to\infty} \mathbb{E}[|Y_n - Y|^2] = 0$$

- Convergence in probability: $\forall \varepsilon > 0$

  $$\lim_{n\to\infty} P\left(|Y_n(w) - Y(w)| > \varepsilon\right) = 0$$

- Convergence in distribution: The CDF $F_n(y) = \Pr\{Y_n \le y\}$ satisfy:

  $$\lim_{n\to\infty} F_n(y) \to F_Y(y)$$

  at all y for which F is continuous.

3.1.2 Weak law of large numbers (WLLN)

Assume that $X_1, X_2, \ldots, X_n$ are i.i.d. r.v.'s having finite mean μ and variance σ^2. And

$$S_n = \frac{X_1 + \cdots + X_n}{n}$$

Weak LLN:

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n}X_i-\mu\right|\geq\varepsilon\right)\leq\frac{\sigma^2}{n\varepsilon^2}\triangleq\delta$$

3.2 Typical sequence

考虑一个长度 $n=20$ 的二元序列（信源输出）with

$$P_X(0)=3/4 \text{ and } P_X(1)=1/4$$

下面给出3个序列，其中一个是信源实际输出的，猜测哪一个是"real" 输出序列：

(1) 全 1 序列：$1,1,\ldots,1,1$

(2) 10101000000000110001

(3) 全 0 序列：$0,0,\ldots,0,0$

直观是（2）。

- 大数定律says roughly that if some event has probability p, then 在多次独立试验中该事件出现的次数（比例）will be close to p with high probability. 序列（2）中"1"出现的次数比例是$6/20=0.3$，该序列更符合WLLN。
- 1948年，Shannon spoke of "typical long sequences", but gave no precise definition of such sequences. 此后许多人给出了定义，但这些定义是方便的for some purpose, for另一目的就不方便。所以，讨论基于典型序列的arguments时，我们应首先明确选择哪种典型序列定义。
- ε-letter-typical:
$$(1-\varepsilon)P_X(a)\leq\frac{N_a(x^n)}{n}\leq(1+\varepsilon)P_X(a)$$
- Entropy-typical sequences（弱典型）

3.3 Weakly typical sequences

Theorem 3.3.1 (AEP)**.** *If $X_1,X_2,\ldots,X_n$ are i.i.d. $\sim P(x)$, then*

$$-\frac{1}{n}\log P(x_1,x_2,\cdots,x_n)\to H(X)\qquad \textit{in Probability}$$

Proof. Create r.v. $Y_i=\log P(X_i)$. Then apply the WLLN to Y:

$$\begin{aligned}-\frac{1}{n}\log P(x_1,x_2,\ldots,x_n)&=-\frac{1}{n}\sum_{i=1}^{n}\log P(X_i)\\&=-\frac{1}{n}\sum_{i=1}^{n}Y_i\to-\mathbb{E}[Y]\quad\text{in Prob.}\\&=H(X)\end{aligned}\tag{3.1}$$

i.e., for all $\varepsilon > 0$,

$$\lim_{n\to\infty} \Pr\left(\left|-\frac{1}{n}\log P(x_1, x_2, \cdots, x_n) - H(X)\right| \le \varepsilon\right) = 1$$

□

Definition 3.3.1. *The typical set T_ε with respect to $P(x)$ is defined as*

$$T_\varepsilon = \left\{\mathbf{x} \triangleq (x_1, x_2, \cdots, x_n) \in \mathcal{X}^n : \left|-\frac{1}{n}\log P(\mathbf{x}) - H(X)\right| \le \varepsilon\right\}$$

Thus, T_ε is the set of sequence $(x_1, x_2, \cdots, x_n) \in \mathcal{X}^n$ for which the sample average of the log pmf is within ε of its mean $H(X)$.

It can be rewritten as

$$T_\varepsilon = \left\{\mathbf{x} \in \mathcal{X}^n : 2^{-n[H(X)+\varepsilon]} \le P(\mathbf{x}) \le 2^{-n[H(X)-\varepsilon]}\right\}$$

which implies that the sequences in T_ε are approximately equiprobable.

- As a consequence of AEP, we have

$$P(\mathbf{x} \in T_\varepsilon) > 1 - \delta \to 1 \qquad \text{(高概率集合)}$$

Theorem 3.3.2 (size of the typical set)**.** *The number of elements, $|T_\varepsilon|$, in the typical set T_ε satisfies*

$$(1-\varepsilon)2^{n[H(X)-\varepsilon]} \le |T_\varepsilon| \le 2^{n[H(X)+\varepsilon]}$$

Proof. Since $P(\mathbf{x}) \ge 2^{-n[H(X)+\varepsilon]}$ for each $\mathbf{x} \in T_\varepsilon$, we have

$$1 = \sum_{\mathbf{x}\in\mathcal{X}^n} P(\mathbf{x}) \ge \sum_{\mathbf{x}\in T_\varepsilon} P(\mathbf{x}) \ge \sum_{T_\varepsilon} 2^{-n[H(X)+\varepsilon]} = |T_\varepsilon| 2^{-n[H(X)+\varepsilon]}$$

This implies that $|T_\varepsilon| \le 2^{n[H(X)+\varepsilon]}$.

Conversely, since $\Pr\{T_\varepsilon\} > (1 - \frac{\sigma^2}{n\varepsilon}) = 1 - \delta = 1 - \varepsilon$, we have

$$1 - \varepsilon < \sum_{\mathbf{x}\in T_\varepsilon} P(\mathbf{x}) \le \sum_{T_\varepsilon} 2^{-n[H(X)-\varepsilon]} = |T_\varepsilon| 2^{-n[H(X)-\varepsilon]}$$

which implies $|T_\varepsilon| > (1-\varepsilon)2^{n[H(X)-\varepsilon]}$. □

Compare to $|\mathcal{X}^n| = 2^{n\log|\mathcal{X}|}$: Let

$$\alpha \triangleq \frac{|T_\varepsilon|}{|\mathcal{X}^n|} \le 2^{-n[\log|\mathcal{X}|-H(X)-\varepsilon]} \xrightarrow{\text{as n}\uparrow} 0.$$

即典型序列的数目远比非典型序列少.

- **Summary**:

We conclude that for large n, the typical set T_ε has aggregate probability approximately 1 and contains approximately $2^{n[H(X)]}$ elements, each of which has probaility approximately $2^{-nH(X)}$. That is, asymptotically for very large n, the r.v. X^n resembles an equiprobable source with alphabet size $2^{nH(X)}$.

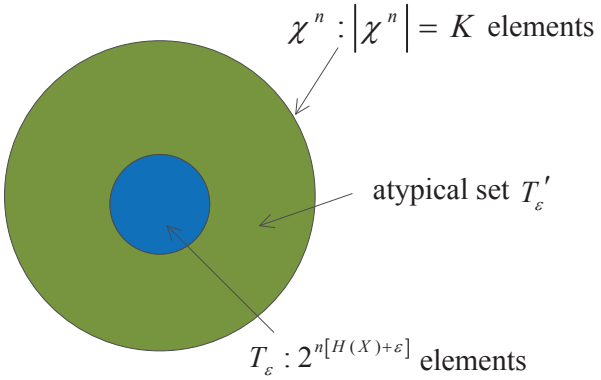


Figure 3.1: 典型序列集示意图

Example 3.3.1. *Consider binary r.v.'s X_i, i.i.d. with $P(X=0)=p$ and $P(X=1)=1-p$.*

A "typical" sequence of length n has roughly np 0's and $n(1-p)$ 1's, the probability for that to happen is

$$p^{np}(1-p)^{n(1-p)} = 2^{n[p\log p+(1-p)\log(1-p)]} = 2^{-nH(X)}$$

How many "typical" sequences are there?

$$\begin{aligned}\binom{n}{np} &= \frac{n!}{(np)!(n(1-p))!} \approx \frac{n^n e^{-n}}{(np)^{np}e^{-np}(n(1-p))^{n(1-p)}e^{-n(1-p)}} \\ &= \frac{1}{p^{np}(1-p)^{n(1-p)}} = 2^{nH(X)}\end{aligned} \tag{3.2}$$

(Note: Stirling formula: $n! \approx n^n e^{-n}\sqrt{2\pi n}$)

3.4 AEP for continuous random variables

Definition 3.4.1. *Let $\mathbf{X}=(X_1,X_2,\ldots,X_n)$ be a sequence of r.v.'s drawn iid according to the density $p(x)$. Then the typical set with respect to $p(x)$ is defined as*

$$T_\varepsilon = \left\{\mathbf{x}\in S^n : \left|-\frac{1}{n}\log p(\mathbf{x}) - h(X)\right| \le \varepsilon\right\}$$

where $p(\mathbf{x}) = \prod_{i=1}^n p(x_i)$.

The size of the typical set T_ε is the number of sequences in T_ε in the discrete case, whereas in the continuous case $|T_\varepsilon|$ is the volume of T_ε.

Definition 3.4.2. *The volume Vol(A) of a set $A\subset\mathbb{R}^n$ is defined as*

$$\mathrm{Vol}(A) = \int_A dx_1 dx_2\cdots dx_n$$

Theorem 3.4.1. *The typical set T_ε has the following properties:*

(1) $\Pr\{T_\varepsilon\} > 1-\varepsilon$ *for n sufficiently large.*

(2) $\mathrm{Vol}(T_\varepsilon) \leqslant 2^{n[h(X)+\varepsilon]}$ *for all n.*

(3) $\mathrm{Vol}(T_\varepsilon) \geqslant (1-\varepsilon)2^{n[h(X)-\varepsilon]}$ *for n sufficiently large.*

3.5 Using the typical set for data compression

Motivated by the AEP, we divided all sequence in $\mathcal{X}^n$ into two sets: T_ε and T_ε^c (its complement).

3.5.1 Source encoding method

- We order all elements in the T_ε and T_ε^c according to some order.
- Then we can represent each sequence in T_ε by giving the index of length $\leq n[H(X)+\varepsilon]+1$ bits (correction of 1 bit because of integrality)
- We prefix all these sequences by a $0 \Rightarrow$ total length $\leq n[H(X)+\varepsilon]+2$
- Similarly, we can index each sequence not in T_ε by using no more than $n\log|\mathcal{X}|+1$ bits. Prefix these indices by 1.
- Thus, we have a code for all the sequences in $\mathcal{X}^n$.

3.5.2 The average length of codewords

Let $l(\mathbf{x})$ = length of the binary codeword corresponding to $\mathbf{x}$. Then the expected length of the codeword is

$$\begin{aligned}
\mathbb{E}[l(\mathbf{x})] &= \sum_{\mathbf{x}\in T_\varepsilon} P(\mathbf{x})l(\mathbf{x}) + \sum_{\mathbf{x}\in T_\varepsilon^c} P(\mathbf{x})l(\mathbf{x}) \\
&\leq \sum_{T_\varepsilon} P(\mathbf{x})[n(H(X)+\varepsilon)+2] + \sum_{T_\varepsilon^c} P(\mathbf{x})[n\log|\mathcal{X}|+2] \\
&\leq (1-\varepsilon)[n(H(X)+\varepsilon)+2] + \varepsilon(n\log|\mathcal{X}|+2) \\
&= n(H(x)+\varepsilon)+2-\varepsilon[n(H(X)+\varepsilon)+2]+\varepsilon n\log|\mathcal{X}|+2\varepsilon \\
&\leq n(H(X)+\varepsilon)+\varepsilon n\log|\mathcal{X}|+2 \\
&= n(H(X)+\varepsilon'). \qquad (3.3)
\end{aligned}$$

where $\varepsilon' = \varepsilon + \varepsilon\log|\mathcal{X}| + \dfrac{2}{n}$.

So $\mathbb{E}\left[\dfrac{1}{n}l(\mathbf{x})\right] \leq H(X)+\varepsilon$.

Chapter 4

Entropy Rates of a Stochastic Process

4.1 Stochastic processes

- A stochastic process is an indexed sequence of random variables $X_1, X_2, \ldots$, a map from Ω to $\mathcal{X}^\infty$.
- A stochastic process is characterized by the joint PMF:

$$P_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n)$$

- The entropy of a stochastic process

$$H(X_1, X_2, \ldots) = H(X_1) + H(X_2|X_1) + \cdots + H(X_i|X_1 \ldots X_{i-1}) + \ldots$$

- Difficulties:

$$\begin{cases} \text{Sum to infinity.} \\ \text{All terms are different in general.} \end{cases}$$

4.2 Entropy rate

- The entropy rate of a stochastic process $\{X_i\}$ is defined by

$$H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X_n)$$

if it exists.

4.3 Entropy rate of stationary processes

- Chain rule:

$$\frac{1}{n} H(X_1, X_2, \ldots, X_n) = \frac{1}{n} \sum_{i=1}^{n} H(X_i|X_1 \ldots X_{i-1})$$

- For a stationary process,

$$\begin{aligned} H(X_{n+1}|X_1^n) &\leq H(X_{n+1}|X_2^n) \\ &= H(X_n|X_1^{n-1}). \end{aligned} \tag{4.1}$$

  Therefor, the sequence $H(X_n|X_1^{n-1})$ is non-increasing and non-negative, so limit exists.

- **Theorem 4.3.1.** *For a stationary process, the entropy rate*

$$\lim_{n\to\infty} \frac{1}{n} H(X_1^n) = \lim_{n\to\infty} H(X_n|X_1^{n-1})$$

- Markov Chain: A discrete stochastic process is a Markov Chain if

$$P_{X_n|X_0\ldots X_{n-1}}(x_n|x_0,\ldots,x_{n-1}) = P_{X_n|X_{n-1}}(x_n|x_{n-1})$$

  for $n = 1, 2, \ldots$, and all $(x_0, \ldots, x_n) \in \mathcal{X}^{n+1}$

- Denote $p_{ij} = P(X_{n+1} = j|X_n = i)$

- The entropy rate of Markov Chain:

$$\lim_{n\to\infty} \frac{1}{n} H(X_1^n) = \lim_{n\to\infty} \frac{1}{n} H(X_n|X_{n-1}) = -\sum_{i,j} \pi_i p_{ij} \log p_{ij}$$

- Significance of the entropy rate of a stochastic process arises from the AEP for a stationary ergodic process:

$$-\frac{1}{n} \log P(X_1, X_2, \ldots, X_n) \to H(\mathcal{X}) \text{ with probability 1.}$$

Chapter 5

Coding for Discrete Sources

5.1 Introduction

Three important classes of sources:

- Discrete sources (离散时间、离散信号取值集合)
  - The output of a discrete source is a sequence of letters from a given discrete alphabet $\mathcal{X}$.
  - A discrete alphabet is either a finite set of letters or a countably infinite set of letters.
- Analog sources (also called continuous-time, continuous-amplitude sources)
  - The output is an analog waveform.
- Discrete-time continuous-amplitude sources
  The output is a sequence of values which could be real number or multi-dimensional real numbers.
- 其它分类:

$$\left\{\begin{array}{l}\text{无记忆源}\\ \text{有记忆源}\end{array}\right. \qquad \left\{\begin{array}{l}\text{高斯源}\\ \text{Markov源}\end{array}\right. \qquad \left\{\begin{array}{l}\text{平稳源}\\ \text{各态历经源}\end{array}\right.$$

- *Discrete memoryless source* (DMS): DMS is a device whose output is a semi-infinite i.i.d. sequence of random variables $X_1, X_2, \ldots$, drawn from the finite set $\mathcal{X}$.
- An important parameter of a source is the rate R_s [source letters/s].

5.2 Coding a single random variable

Definition 5.2.1. *A source code $\mathcal{C}$ for a random variable X is a mapping from $\mathcal{X}$ to $\mathcal{D}^*$, the set of finite length strings of symbols from a D-ary alphabet.*

The same definition applies for sequence of r.v.'s, X^n.

- x (or x^n) – source symbol (string) $\in \mathcal{X}$.
  $\mathcal{D}$ – set of coded symbols.
  $C(x)$ – codeword corresponding to x.
  $\ell(x)$ – length of $C(x)$.

- For example, $\mathcal{X} = \{red, blue\}, C(red) = 00, C(blue) = 11$ is a source code with alphabet $\mathcal{D} = \{0, 1\}$.

- Without loss of generality, we will assume that $\mathcal{D} = \{0, 1, \ldots, D-1\}$.

Definition 5.2.2. *The expected length, $L(\mathcal{C})$, of a code is given by*

$$L(\mathcal{C}) = \sum_{x \in \mathcal{X}} P_X(x)\ell(x) = \mathbb{E}[\ell(x)]$$

- **Goal**: For a given source, find a code to minimize the expected length (per source symbol).

5.3 Fixed-length source codes (等长编码)

- Convert each source letter individually into a fixed-length block of ℓ D-ary symbols.

- The number of letters in the source alphabet, $K = |\mathcal{X}|$, satisfies $K \leq D^\ell$, then a different D-ary sequence of length ℓ may be assigned to each letter $x \in \mathcal{X}$. The resulting code is uniquely decodable.

- For example, for $\mathcal{X} = \{a, b, c, d, e, f, g\}$, $K = 7$, $\mathcal{D} = \{0, 1\}$, there exists an invertible mapping for $\mathcal{X}$ to binary 3-tuples:

$$a \to 000, b \to 001, \ldots, g \to 110$$

X	$C(x)$
a	000
b	001
⋮	⋮

Figure 5.1: $\ell = 3$的信源编码

- We can see that this coding method requires $\ell = \lceil \log_2 |\mathcal{X}| \rceil$ bits to encode each source letter.

- If we want to encode blocks of n source symbols at a time, the resulting source alphabet is the n-fold Cartesian product $\mathcal{X}^n = \mathcal{X} \times \mathcal{X} \times \cdots \times \mathcal{X}$, which has size $|\mathcal{X}^n| = K^n$.

- Using fixed-length source coding, we can encode each block of n source symbols into $\ell = \lceil \log_2 K^n \rceil$ bits. The rate $R = \frac{\ell}{n}$ of coded bits required per source symbol is then

$$R = \frac{\lceil \log_2 K^n \rceil}{n} \geq \frac{n \log_2 K}{n} = \log_2 K$$
$$R = \frac{\lceil \log_2 K^n \rceil}{n} < \frac{n \log_2 K + 1}{n} = \log_2 K + \frac{1}{n}$$

  If n is sufficiently large, then $R \to \log_2 K$.

5.3.1 等长编码定理

Goal: Minimize the average rate $R = \mathbb{E}[\ell(x^n)]/n$

- 令 M 为待编码消息序列个数（或码字总数，codebook size），则对于等长编码，有 $R = \frac{1}{n} \log_2 M$
- Encoder "compresses" x^n into an index $w \in \{1, 2, \cdots, 2^{nR}\}$. That is, the encoder sends nR bits for every source sequence x^n.
- 若采用 D 元等长编码，码长为 L，则 D 元码字个数 $M = D^L$，$R = \frac{L}{n} \log_2 D$

编码方法：Data compression by AEP

- Since $|T_\varepsilon| \leq 2^{n(H+\varepsilon)}$, we use $n(H+\varepsilon)+1$ bits to index all sequence in T_ε.
- Use an extra bit to indicate T_ε.
- We have shown that

$$\mathbb{E}[\ell(x^n)] = \sum_{x^n} P(x^n)\ell(x^n) \leq n[H(X) + \varepsilon'] \Rightarrow R = \frac{1}{n}\mathbb{E}[\ell(x^n)] \leq H(X) + \varepsilon'.$$

另一方面，by Fano inequality,

$$\begin{aligned} nR &\geq H(\hat{X}^n) \quad (\because \text{最多有 } 2^{nR} \text{ 个不同序列 } \hat{x}^n) \\ &\geq I(X^n; \hat{X}^n) \\ &= H(X^n) - H(X^n|\hat{X}^n) \\ &= nH(X) - H(X^n|\hat{X}^n) \\ &\geq n\left[H(X) - \frac{H_2(P_e)}{n} - P_e \log_2 |\mathcal{X}|\right] \end{aligned} \tag{5.1}$$

(Fano 不等式：$H_2(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X^n|\hat{X}^n)$)

$\because$ 要求$P_e \to 0$ with n,

$\therefore R \geq H(X)$ $(\Longrightarrow \frac{L}{n} \geq \frac{H(X)}{\log_2 D})$.

Theorem 5.3.1 (Fixed-to-fixed source coding theorem, 无扰编码定理)**.** *若 $R > H(X)$，则 R 是可达的；若 $R < H(X)$，则 R 不可达。*

- 编码效率：$\eta = \dfrac{H(X)}{R} \leq 1$.
- Example: 见中文教材的例3.2.3。

5.4 Variable-length source codes (不等长编码)

- A variable-length source code maps each source letter $x \in \mathcal{X}$ to a codeword $C(x)$ of length $\ell(x)$.
  For example,
  $$\mathcal{X} = \{a, b, c\}, \mathcal{D} = \{0, 1\}$$
  $$C(a) = 0, C(b) = 10, C(c) = 11$$

- The major property that we usually require for any variable-length code is that of *unique decodability*. This means that the input sequence of source letters can be reconstructed unambiguously from the encoded symbol sequence.

- Clearly, unique decodability requires that $C(x) \neq C(x')$ for $x \neq x'$.

- Definition: The extension of a code $\mathcal{C}$ is the code for finite strings of $\mathcal{X}$ given by the concatenation of the individual codewords: $C(x_1, x_2, \ldots, x_n) = C(x_1)C(x_2)\ldots C(x_n)$.
  For example,
  $$\left.\begin{array}{l} C(x_1) = 00 \\ C(x_2) = 11 \end{array}\right\} \Rightarrow C(x_1x_2) = 0011$$

  – A code is called non-singular if
  $$x_i \neq x_j \Rightarrow C(x_i) \neq C(x_j)$$

  – A code is called uniquely decodable if its extension is non-singular.

  For example, $C(a) = 0, C(b) = 10, C(c) = 11$ is prefix-free and uniquely decodable. However, the code $\mathcal{C}'$ defined by
  $$C'(a) = 0, C'(b) = 1, C'(c) = 01$$
  is not uniquely decodable.

5.4.1 Prefix-free codes

Checking whether a code is uniquely decodable can be quite complicated. However, there is a good class of uniquely decodable codes called *prefix-free codes*.

Definition 5.4.1. *A code is said to be prefix-free if no codeword is a prefix of any other codeword.*

Advantages $\left\{\begin{array}{l} \text{easy to check whether a code is prefix-free and therefore uniquely decodable.} \\ \text{can be decoded with no delay (instantaneous code).} \end{array}\right.$

- Any fixed-length code is prefix-free.

- Classes of codes:

X	Uniquely decodable, But not prefix-free	Prefix-free
a	10	0
b	00	10
c	11	110
d	110	111

Figure 5.2: 唯一可译码的分类及编码方式

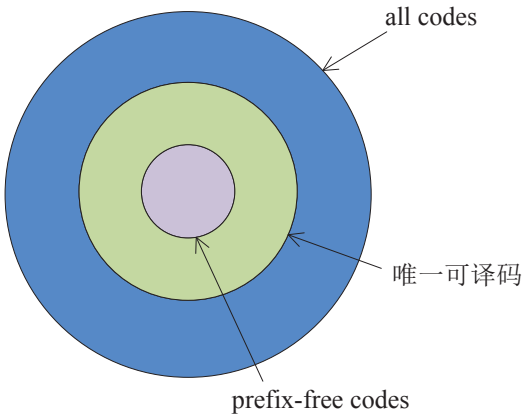


Figure 5.3: 码的分类

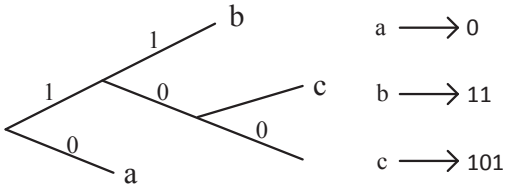


Figure 5.4: The binary code tree

5.4.2 Code tree

- The digits in the codewords are represented as the labels on the branches of a rooted tree.
- For prefix-free codes, each codeword corresponds to a leaf node.
- A prefix-free code will be called full if no new codeword can be added without destroying the prefix-free property.

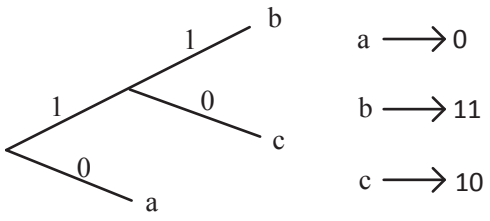


Figure 5.5: full tree

- D-ary tree: A D-ary tree is a finite rooted tree such that D branches stem outward from each (intermediate) node. D branches are labeled with the D different D-ary letters.

- Why the prefix-free condition guarantees unique decodability?

5.5 Kraft Inequality

The Kraft inequality tells us whether it it possible to construct a prefix-free code for a given source alphabet $\mathcal{X}$ with a given set of codeword length $\{\ell(x)\}$.

Theorem 5.5.1 (Kraft Inequality)**.** *There exists a D-ary prefix-free code with codeword lengths $\ell_1, \ell_2, \ldots, \ell_k$ if and only if*

$$\sum_{i=1}^{k} D^{-\ell_i} \leq 1 \tag{5.2}$$

Every full prefix-free code satisfies (5.1) with equality.

Proof. First assume that $\mathcal{C}$ is a prefix-free code with codeword lengths $\{\ell_1, \ell_2, \ldots, \ell_k\}$. Let $\ell_{max} = \max_i \ell_i$. Consider constructing a D-ary tree for the code $\mathcal{C}$ by pruning the full D-ary tree of length ℓ_{max} at all nodes corresponding to codewords:

a) A codeword at depth ℓ_i has $D^{\ell_{max}-\ell_i}$ descendants (leaves) at depth ℓ_{max}. [Each of these descendant sets must be disjoint]

b) Begin with $i = 1$, we find the node X_i corresponding to a codeword.

c) We prune the tree to make this node a leaf at depth ℓ_i.

d) By this process, we delete $D^{\ell_{max}-\ell_i}$ leaves from the tree. None of these leaves could have previously been deleted because of the prefix-free condition.

e) But there are only $D^{\ell_{max}}$ leaves that can be deleted. So we have

$$\sum_i D^{\ell_{max}-\ell_i} \leq D^{\ell_{max}}$$

or

$$\sum_i D^{-\ell_i} \leq 1$$

Next, conversely, suppose that we are given the set of codeword lengths $\ell_1, \ell_2, \ldots, \ell_k$ for which (5.1) is satisfied. □

- Without loss of generality, assume that we have ordered these lengths so that $\ell_1 \leq \ell_2 \leq \cdots \leq \ell_k$.
- Consider the following algorithm:
  a) Start with the full D-ary tree of length ℓ_{max}, and $i \leftarrow 1$.
  b) Choose x_i as any surviving node at depth ℓ_i (not yet used as a codeword), and remove its descendants from the tree. Stop if there is no such surviving node.
  c) If $i = k$, stop. Otherwise $i = i + 1$ and goto b).

We now show that we can indeed choose x_i in step b) for all $i < k$. Suppose that $x_1, x_2, \ldots, x_{i-1}$ has been chosen. The number of surviving leaves at depth ℓ_{max} not stemming from any codeword is

$$D^{\ell_{max}} - \left(\sum_{j=1}^{i-1} D^{\ell_{max}-\ell_j} \right) = D^{\ell_{max}} \left(1 - \sum_{j=1}^{i-1} D^{-\ell_j} \right) > 0$$

with condition (3.1). There must be (unused) surviving nodes at depth $\ell_i < \ell_{max}$. Since $\ell_1 \leq \cdots \leq \ell_{i-1} \leq \ell_i$, no already chosen codeword can stem outward from such a surviving node and hence this surviving node may be chosen as x_i.

Example:
Construct a binary prefix-free code with lengths $\ell_1 = \ell_2 = \ell_3 = 2, \ell_4 = 3$ and $\ell_5 = 4$. Since $\sum_{i=1}^{5} 2^{-\ell_i} = \frac{15}{16} < 1$, such a prefix-free code exists.

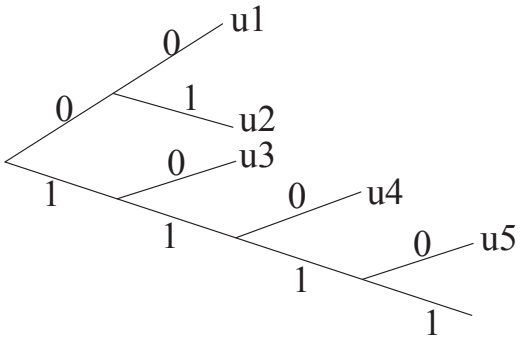


Figure 5.6: a binary prefix-free code tree

- Note: Just because a code has lengths that satisfy (5.1), it does not follow that the code is prefix-free, or even uniquely decodable.
- The same theorem holds for uniquely decodable codes. (这意味着对某一码字长度集合，存在唯一可译码和无前缀码。So why use any code other than a prefix-free code?)

Theorem 5.5.2 (Extended Kraft Inequality)**.** *For any prefix-free code over an alphabet of size D, the codeword length satisfy*

$$\sum_{i=1}^{\infty} D^{-\ell_i} \leq 1$$

Conversely, for any given set of codeword lengths that satisfy the inequality, we can construct a prefix-free code with these lengths.

Proof. Consider a codeword $y_1 y_2 \dots y_{\ell_i}$, where $y_j \in \{0, 1, \dots, D-1\} \triangleq \mathcal{D}$. Let $0.y_1 y_2 \dots y_{\ell_i} = \sum_{j=1}^{\ell_i} y_i D^{-j} \in [0, 1]$. This codeword corresponds to an interval

$$(0.y_1 y_2 \dots y_{\ell_i}, 0.y_1 y_2 \dots y_{\ell_i} + \frac{1}{D^{\ell_i}})$$

Prefix-free code implies the intervals are disjoint. Hence the sum of their lengths ≤ 1. □

5.6 Optimal codes

5.6.1 Problem formulation and Shannon codes

Let $\mathcal{X} = \{a_1, a_2, \dots, a_k\}$ be the source alphabet, and $p_i = P_X(X = a_i) > 0$. Suppose that we encode each source symbol into prefix-free codeword. Denote by $C(a_i)$ the codeword for a_i and by ℓ_i the length of $C(a_i)$.

- Optimal code is defined as code with smallest possible $L(\mathcal{C})$ with respect to P_X.
- We now consider the problem of finding the prefix code with minimum expected length: $L = \sum_{i=1}^{k} p_i \ell_i = \sum_{x \in \mathcal{X}} P(x)\ell(x)$.
- Mathematically, this is a standard optimization problem:

$$\text{Minimize} \quad L = \sum_i p_i \ell_i$$

$$\text{subject to} \quad \sum_i D^{-\ell_i} \leq 1$$

  and $\ell_i = \ell(x)$ are integers.
- We first ignore the integer constraint on ℓ_i. With real variables, we may assume that $\sum_i D^{-\ell_i} = 1$.
- Using a Lagrange multiplier λ, we want to minimize

$$J = \sum_i p_i \ell_i + \lambda(\sum_i D^{-\ell_i} - 1)$$

Setting $\frac{\partial J}{\partial \ell_i} = 0$, we obtain

$$\frac{\partial J}{\partial \ell_i} = p_i - \lambda D^{-\ell_i} \ln D = 0 \qquad ((a^x)' = (a^x) \ln a)$$

equivalently, $D^{-\ell_i} = \dfrac{p_i}{\lambda \ln D}$.

Since $\sum_i p_i = 1$ and $\sum_i D^{-\ell_i} = 1$, we have $\lambda = \frac{1}{\ln D}$, and hence

$$p_i = D^{-\ell_i}.$$

This yields optimal lengths $\ell_i^* = -\log_D p_i$.

The expected codeword length

$$\begin{aligned} L^*(\ell_i \text{ non-integer}) &= -\sum_i p_i \log_D p_i \\ &= -\sum_i p_i \log_2 p_i / \log_2 D \\ &= H(X)/\log_2 D \end{aligned}$$

With the constraint that ℓ_i are integers, we have the following theorem:

Theorem 5.6.1 (Entropy bounds for prefix-free codes)**.** *Let $L_{\min}$ be the minimum expected codeword length over all D-ary prefix-free codes. Then*

$$\frac{H(X)}{\log_2 D} \le L_{\min} < \frac{H(X)}{\log_2 D} + 1$$

Proof. ① Let $\ell_1, \ldots, \ell_k$ be the codeword lengths of an arbitrary prefix-free code.

$$\begin{aligned} \frac{H(X)}{\log D} - L &= \frac{1}{\log D} \sum_i p_i \log \frac{1}{p_i} - \sum_i p_i \ell_i \\ &= \frac{1}{\log D} \sum_i p_i \log \frac{1}{p_i} + \sum_i p_i \log_D D^{-\ell_i} \\ &= \frac{1}{\log D} \left[\sum_i p_i \log \frac{D^{-\ell_i}}{p_i} \right] \\ \text{(by IT inequality)} \quad &\le \frac{1}{\log D} \left[\log e \sum_i p_i \left(\frac{D^{-\ell_i}}{p_i} - 1 \right) \right] \qquad (*) \\ &= \frac{\log e}{\log D} \left(\sum_i D^{-\ell_i} - \sum_i p_i \right) \\ \text{(by Kraft inequality)} \quad &\le 0 \end{aligned} \tag{5.3}$$

(*) is satisfied with equality iff $\frac{D^{-\ell_i}}{p_i} = 1$; i.e., $p_i = D^{-\ell_i}$.

② We now show that there exists a prefix-free code with $L(\mathcal{C}) < \frac{H(X)}{\log D} + 1$.
Let us choose the codeword length to be $\ell_i = \lceil -\log_D p_i \rceil$. Then

$$\begin{aligned} &\underbrace{-\log_D p_i \le \ell_i}_{\Downarrow} < -\log_D p_i + 1 \\ &D^{-\ell_i} \le p_i \\ &\textstyle\sum D^{-\ell_i} \le \sum p_i = 1 \Rightarrow \text{Kraft inequality is satisfied} \end{aligned} \qquad (**)$$

Thus, a prefix-free code exists with the above length.
From the RHS of (**),

$$L = \sum_i p_i \ell_i < \sum p_i(-\log_D p_i + 1) = \frac{H(X)}{\log_2 D} + 1$$

③ Therefore, $\frac{H(X)}{\log_2 D} \le L_{\min} \le L < \frac{H(X)}{\log_2 D} + 1$.

□

Summary [Shannon code]:

- Ideal codeword length $\ell_i = -\log_D p_i$.
  This is optimal when $-\log_D p_i$ is an integer for any i.
- For general distribution, set
  $$\ell_i = \lceil -\log_D p_i \rceil$$
- Bounds for the codeword length:
  $$-\log_D p_i \le \ell_i < -\log_D p_i + 1$$
  Average codeword length
  $$H_D(X) \le L < H_D(X) + 1$$
- Example: $P_X(x) = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{4}, \frac{1}{12}\}$.
  Then
  $$\begin{aligned} H(X) &= 1.8554 \\ \ell_i &= \lceil -\log_D p_i \rceil = (2,2,2,4) \\ \mathbb{E}[\ell(x)] &= \frac{13}{6} = 2.1667 \end{aligned}$$

  Comparing to the obvious codeword length assignment (2,2,2,2,), loss 0.1667 bit per source symbol.

5.6.2 Improvement

Coding over multiple i.i.d. source symbols: View $(X_1, X_2, \ldots, X_n)$ as one super-symbol from $\mathcal{X}^n$. Apply the bounds derived above,

$$H(X_1 \ldots X_n) \le \mathbb{E}[\ell(X_1^n)] < H(X_1 \ldots X_n) + 1$$

Since $X_1, \ldots, X_n$ are i.i.d, $H(X_1^n) = \sum_i H(X_i) = nH(X)$, we have

$$H(X) \le \frac{1}{n}\mathbb{E}[\ell(X_1^n)] < H(X) + \frac{1}{n}$$

Theorem 5.6.2 (Prefix-free source coding theorem)**.** *For any DMS with entropy $H(X)$, there exists a D-ary prefix-free coding of source blocks of length n such that the expected codeword length per source symbol L_n satisfies*

$$\frac{H(X)}{\log D} \le L_n < \frac{H(X)}{\log D} + \frac{1}{n}$$

From this theorem, $H(X)$ is the minimum expected codeword length per source symbol required to describe the source.

Usually, $R = L_n \log_2 D$ [bits/source symbol] is called the rate of a prefix-free code.

5.6.3 Unknown distribution

If assign the codeword length as

$$\ell_i = \lceil -\log q(x) \rceil$$

and the true distribution of X is $P_X(i) = p_i$, then

$$H(p) + D(p||q) \le \mathbb{E}_p[\ell(X)] < H(p) + D(p||q) + 1$$

Proof.

$$\begin{aligned}
\mathbb{E}_p[\ell(X)] &= \sum_x p(x) \lceil \log \frac{1}{q(x)} \rceil \\
&< \sum_x p(x) \left[\log \frac{1}{q(x)} + 1 \right] \\
&= \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \frac{1}{p(x)} \right) + 1 \\
&= \sum_x p(x) \log \frac{p(x)}{q(x)} + \sum_x p(x) \log \frac{1}{p(x)} + 1 \\
&= D(p||q) + H(p) + 1
\end{aligned}$$

□

- Penalty of $D(p||q)$ bits per source symbol due to the wrong distribution.
- Discussion
  - For any n, any code over i.i.d sequence X_1^n, $\frac{1}{n}\mathbb{E}[\ell(X_1^n)] \ge H(X)$.
  - We can achieve this when $n \to \infty$, AEP code, Shannon code.

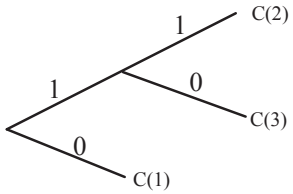
$$\lim_{n\to\infty} \frac{1}{n} \mathbb{E}[\ell(X_1^n)] = H(X)$$

  - True or False: for finite n,
    - Shannon code is "optimal"?
    - A code with codeword length $\ell_i = -\log P_X(a_i), \forall i$, is optimal
    - Any prefix code must satisfy $\ell_i \ge -\log p_i, \forall i$
    - The optimal code must satisfy $\ell_i \le \lceil -\log p_i \rceil$, $\forall i$

5.7 Huffman codes

The optimal prefix code (in the sense of minimal L) for a given distribution can be constructed by a simple algorithm discovered by Huffman in 1950 (as a term paper in Fano's IT class at MIT).

- Huffman's trick, in today's jargon, was to "think outside the box". He ignored the Kraft inequality, and looked at the binary code tree to establish properties that an optimal prefix-free code should have.



P	X	c(x)	ℓ (x)
0.6	1	0	1
0.3	2	11	2
0.1	3	10	2

Figure 5.7: a simple optimal code

- **Example:**
  A simple optimal code

Lemma 5.7.1. *Optimal codes have the property that if $\ell_i > \ell_j$, then $p_i \leq p_j$.*

Proof. Let $\mathcal{C}$ be the optimal code. Consider a code $\mathcal{C}'$, with the codewords i and j of $\mathcal{C}$ interchanged. Then

$$\begin{aligned} L(\mathcal{C}') - L(\mathcal{C}) &= \sum_k p_k \ell'_k - \sum_k p_k \ell_k \\ &= [p_i \ell_j + p_j \ell_i] - [p_i \ell_i + p_j \ell_j] \\ &= (p_j - p_i)(\ell_i - \ell_j) \end{aligned}$$

Note that $\ell_i - \ell_j > 0$, and since $\mathcal{C}$ is optimal, $L(\mathcal{C}') - L(\mathcal{C}) \geq 0$. Hence we must have $p_j \geq p_i$. □

Lemma 5.7.2. *Optimal prefix-free codes have the property that the two longest codewords have the same length.*

Proof. Otherwise, one can delete the last bit of the longer one, preserving the prefix-free property and achieving lower codeword length. □

Lemma 5.7.3. *The two longest codewords differ only in the last bit and correspond to the two least likely symbols.*

Proof. By Lemma(5.7.1), the longest codewords must belong to the least probable source symbols.
If there is a maximal length codeword without a sibling, then we can delete the last bit of codeword and still satisfy the prefix-free property. This reduces the average codeword length. □

- The Huffman algorithm chooses an optimal code tree by starting at leaves for the least likely symbols and working in.

- **Example 1**:

  Optimal length:

  $$\begin{aligned} -\log p_i &= \{1.32, 2.32, 2.74, 2.74, 3.32\} \\ &\approx \text{the length } \{1, 3, 3, 3, 3\} \text{ of the optimal code} \end{aligned}$$

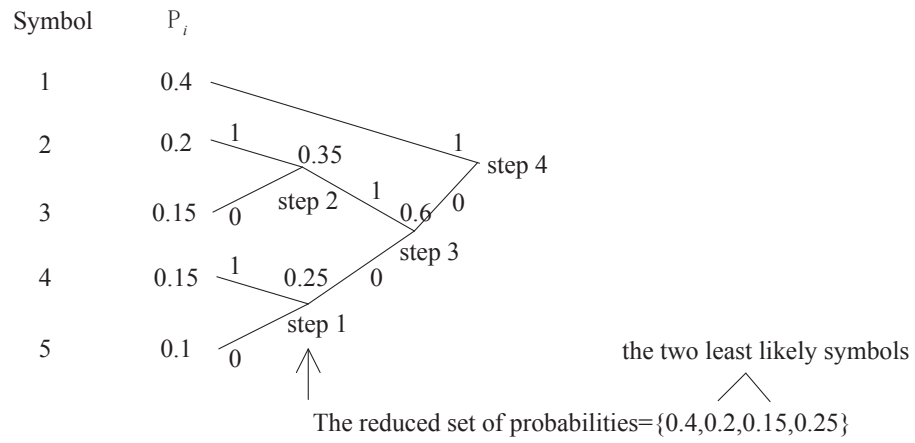


Figure 5.8: Huffman tree

$$\begin{cases} H(X) = 2.15 \text{ bits/symbol} \\ L(\mathcal{C}) = 2.2 \text{ bits/symbol} \end{cases}$$

- **Example 2**:

Symbol	p_i	codeword
1	0.35	11
2	0.2	01
3	0.2	00
4	0.15	101
5	0.1	100

- Let X be a r.v. over $\mathcal{X}$ with $\mathbf{p}$, and X' be a r.v. with reduced set of probabilities $\mathbf{p}'$. Let L' be the expected codeword length of any code for X'. Then

$$L = L' + (p_k + p_{k-1}) \qquad (\because L = \sum_{i=1}^{k-2} p_i \ell_i + p_{k-1}(\ell'_{k-1} + 1) + p_k(\ell'_{k-1} + 1))$$

  $L_{min} = L'_{min} + p_k + p_{k-1} \Rightarrow$ finding the optimal reduced code yields the optimal final code.

- Note that there are many optimal codes. The Huffman algorithm produce one such optimal code.

- D-ary Huffman code: $|\mathcal{X}| = (D-1)\theta + D, \quad K = |\mathcal{X}|$

$$\begin{aligned} \text{第一次合并的消息个数} &= [(K-2) \bmod (D-1)] + 2 \\ &= R_{D-1}(K-2) + 2 \end{aligned}$$

5.8 Shannon-Fano Coding

- Order the K probabilities in decreasing order, and divide source symbols into D sets of almost equal probability. (将源符号近似等概地划分为 D 个子集)
  — Choose m such that $\left|\sum_{i=1}^{m} p_i - \sum_{i=m+1}^{K} p_i\right|$ is minimized. (for binary codes)

- Assign 0 for the first bit of the upper set and 1 for the lower set.

- Example: $P(X) = \{1/3, 1/9, 1/9, 1/9, 1/9, 1/9, 1/27, 1/27, 1/27\}$

5.9 Shannon-Fano-Elias Coding

- Modified CDF:
$$\bar{F}(x) = \sum_{a<x} P(a) + \frac{1}{2}P(x)$$
It is the midpoint of the step corresponding to x.

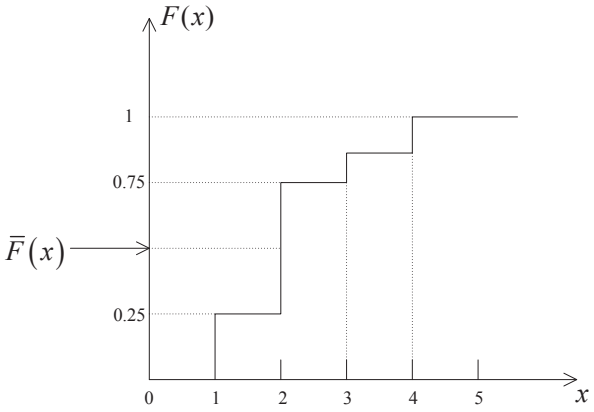


Figure 5.9: $P(x) = \{0.25, 0.5, 0.125, 0.125\}$

- Using the value of $\bar{F}(x)$ as a code for x: the first $\ell(x)$ bits of $\bar{F}(x)$
$\bar{F}(x)$ in binary $\rightarrow$ codeword. Then

$$\begin{aligned}\bar{F}(x) - \lfloor \bar{F}(x) \rfloor_{\ell(x)} &< \frac{1}{2^{\ell(x)}} \\ &< \frac{P(x)}{2} \quad (\because \ell(x) = \lceil \log_2 \frac{1}{P(x)} \rceil + 1 > \log_2 \frac{1}{P(x)} + 1) \\ &= \bar{F}(x) - F(x-1)\end{aligned}$$

That is, $\lfloor \bar{F}(x) \rfloor_{\ell(x)}$ lies within the step corresponding to x. Thus $\ell(x)$ bits suffice to describe x.

- Each codeword $z_1 z_2 \ldots z_\ell$ 是将$\bar{F}(x)$截取前$\ell(x)$位得到的，因此它不是代表一个点，it is considered to represent the interval $[0.z_1 z_2 \ldots z_\ell, 0.z_1 z_2 \ldots z_\ell + \frac{1}{2^\ell}]$. 因为对应的区间互不相交$\rightarrow$ the code is prefix-free.

- The expected length of this code is
$$L = \sum_x P(x)\ell(x) = \sum_x P(x) \left[\lceil \log \frac{1}{P(x)} \rceil + 1 \right] < \sum_x \left[\log \frac{1}{P(x)} + 2 \right] = H(X) + 2$$

- Example:

x	$P(x)$	$\bar{F}(x)$	$\bar{F}(x)$ in binary	$\ell(x)=\lceil\log\frac{1}{P(x)}+1\rceil$	$\mathcal{C}(x)$
1	0.25	0.125	0.001	3	001
2	0.5	0.5	0.10	2	10
3	0.125	0.8125	0.1101	4	1101
4	0.125	0.9375	0.1111	4	1111

$$\begin{cases} L(\mathcal{C}) = 2.75 \text{ bits} \\ H(X) = 1.75 \text{ bits} \end{cases}$$

code efficiency $\eta \triangleq \dfrac{H(X)}{R} = \dfrac{H(X)}{L} \leq 1$.

5.10 Arithmetic Coding

- Calculate the pmf $P(x^n)$ and the CDF $F(x^n)$ for sequence x^n.
- Using Elias coding, we can use a number in $[(F(x^n)-P(x^n)), F(x^n)]$ as the code for x^n.
- $\{F(a_i)\}$将区间$[0,1]$分为了K个子区间，每个区间的长度为$P(a_i)$.
- **Example:** 对二进制序列0100进行算术编码。假定$P_X(0) = P_0 = \frac{3}{4}, P_X(1) = P_1 = \frac{1}{4}$

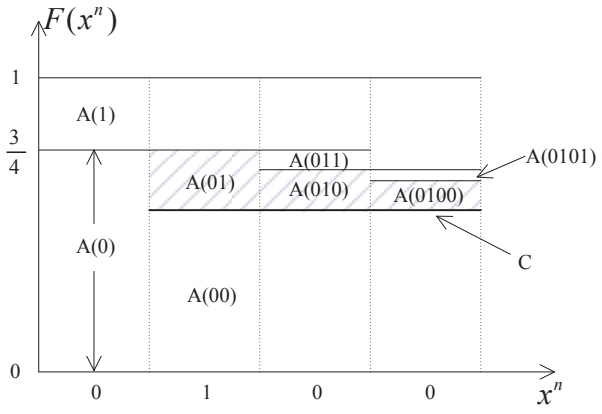


Figure 5.10: 算术编码

  - 输入区间$A=[0,1]$
  - 输入第一位“0”后，区间A按概率分割为两个区间：$A(0) = A \cdot P_0, A(1) = [P_0, 1] = AP_1$（长度）并取$A(0)$为下次编码区间，即新的$A + A(0)$。
  - 输入第二位“1”后，区间$A = A(0)$分割为：

$$A(00) = A \cdot P_0, A(01) = AP_1$$

    并取$A = A(01)$为下次分割区间。
  - 可以看出，A区间在不断变小，编码区间底线C要么不变，要么上升。
  - 编码过程：

① 初始化：$C = 0, A = 1$
② 输入高概率符号“0”，$C = 0$,$A = AP_0 = \frac{3}{4}$
③ 输入低概率符号“1”，$C = C + AP_0 = \frac{9}{16}$,$A = AP_1 = \frac{3}{16}$
④ 输入高概率符号“0”，$C = \frac{9}{16}$,$A = \frac{3}{16} \times \frac{3}{4} = \frac{9}{64}$
⑤ 输入低概率符号“0”，

$$C = \frac{9}{16} = 0.1001 \qquad A = \frac{9}{64} \times \frac{3}{4} = \frac{27}{256} = (\frac{16}{256}) + \frac{11}{256} = 0.00011011 \quad \text{(binary)}$$

最后码字区间为：$0.1001 \leq \mathcal{C}(x) < 0.10101011$
在该区间中任取一点并适当截断，得到码字: 101

- 算术编码中的码字长度：Cover书中取：$\lceil \log \frac{1}{P(x^n)} \rceil + 1$ bits
  西电书中取小数点后 l 位：$l = \lceil \log \frac{1}{P(x^n)} \rceil$
- Example: $\mathcal{X} = \{a_1, a_2, a_3, a_4\}, P(a_1) = 1/2, P(a_2) = 1/4, P(a_3) = 1/8, P(a_4) = 1/8$. $x^n = a_2a_1a_1a_3a_4a_1a_2a_1$.

5.11 Lempel-Ziv Universal Source Coding

已知信源的概率分布，我们能用Huffman算法构造最佳码。对于未知概率分布的源，it is desirable to have a *one-pass* (or online) algorithm to compress the data that “learns” the probability distribution of the data and uses it to compress the incoming symbols.

- Universally optimal: asymptotic compression rate approached the entropy rate of the source for any stationary ergodic source.
- Lempel-Ziv (LZ) algorithms do not require prior knowledge of the source statistics.
- LZ77: sliding window LZ algorithm, it uses string-matching on a sliding window. (它的实现晚一点，典型实现：MS Windows)
- LZ78: tree-structured LZ algorithm, it uses an adaptive dictionary. (典型实现：UNIX的compress)

The key idea of LZ algorithm is to parse the string into phrases and to replace phrases by pointers to where the string has occurred in the past.

5.11.1 LZ78算法

- Parse a string into phrase, where each phrase is the shortest phrase not seen earlier, This algorithm can be viewed as building a dictionary in the form of a tree, where the nodes correspond to phrases seen so far.
  (分段规则：尽可能取最短的符号串，并保证每段都不相同)
- Example: The string is *a b b a b b a b b b a a b a b a a*
  We parse it as: $a, b, ba, bb, ab, bba, aba, baa$，这就构成分段字典
- Code this phrases by giving the location of the prefix (经常用段号表示) and the value of the last symbol.
  ∴ 码字= 段号+ the last symbol的编码。(单字符的码字段号为0)

- Example: $\mathcal{X} = \{a_1, a_2, a_3, a_4\}$. 源序列为$x^n = a_1\ a_2\ a_1\ a_3\ a_2\ a_4\ a_2\ a_4\ a_3\ a_1\ a_1\ a_4$.
  - 可分段为：$a_1, a_2, a_1a_3, a_2a_4, a_2a_4a_3, a_1a_1, a_4$
  - 编码表：

段号	phrases	codeword	
1	a_1	$\underbrace{000}_{\text{段号}}\vdots\ \underbrace{00}_{a_1\text{的编码}}$	
2	a_2	$000\vdots 01$	
3	a_1a_3	$001\vdots\ \underbrace{10}_{a_3}$	
4	a_2a_4	$010\vdots 11$	
5	$a_2a_4a_3$	$100\vdots 10$	$\longrightarrow a_2a_4$段的段号4
6	a_1a_1	$001\vdots 00$	
7	a_4	$000\vdots 11$	

5.11.2 LZ77算法（Ref: Gallager'08）

- The algorithm keeps the w most recently encoded source symbols in memory. This is called a sliding window of size w.
- Suppose the source symbols x_1^P have been encoded. The encoder looks for the longest match (长为n) between the not-yet-encoded n-string x_{P+1}^{P+n} and a stored string x_{P+1-u}^{P+n-u} starting in the window of length w.
- With LZ77, this string of n symbols is encoded simply by encoding the integers n and u.
  - 初始化: Encode the first w symbols in a fixed-length code without compression, using $\lceil \log |\mathcal{X}| \rceil$ bits per symbol.
  - Set $P = w$
  - Find the largest $n \geq 2$ such that $x_{P+1}^{P+n} = x_{P+1-u}^{P+n-u}$ for some u in the range $1 \leq u \leq w$
  - Encode the integer n into a codeword from the unary-binary code.
  - If $n > 1$, encode $u \leq w$ using a fixed-length code of length $\log w$ bits.
  - Set P to $P + n$.
- If a match is not found in the window，the next character is sent uncompressed. To distinguish between these two cases, a flag bit is needed.
  Phrase types: (flag f, 匹配位置 u（反向计数）, 匹配长度 n) or (f, c (未压缩字符))
- **Example:**

$$\overbrace{a\ c\ d\ b\ c\ d\ a\ c\ b\ \underbrace{a\ b\ a}_{match}\ c\ d\ b\ c}^{w}\ \overbrace{a\ b\ a}^{n=3}\ b\ d\ c\ a\ \cdots$$

$$u=7$$

- Note that the target of a (u, n) pair could extend beyond the window, so that it overlaps with the new phrase.
- **Example:** $w = 4$, the string is abbabbabbbaababa
  Phrase表示：

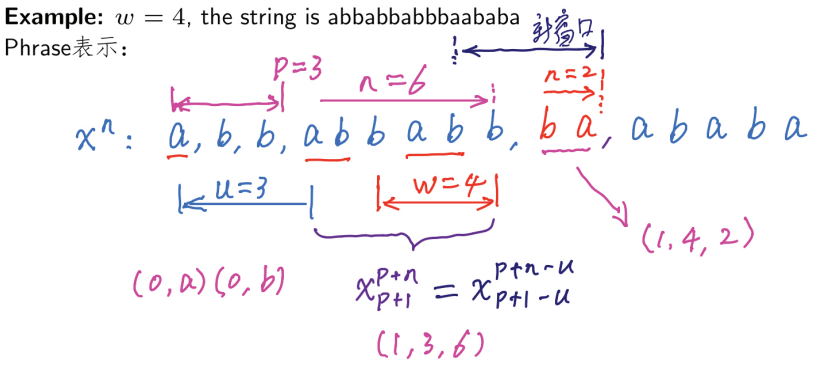


Figure 5.11: LZ77算法举例

Chapter 6

Channel Capacity

6.1 Categories of channels (channel models)

- $\begin{cases} \text{discrete channel: discrete-valued inputs and outputs} \\ \text{continuous channel: continuous-valued inputs and outputs} \\ \text{semi-continuous channel: discrete inputs and continuous outputs} \\ \text{waveform channel.} \end{cases}$
- $\begin{cases} \text{single-user channel(point-to-point)} \\ \text{multiuser channel: multiaccess channel; broadcast channel} \end{cases}$
- $\begin{cases} \text{memoryless channel} \\ \text{channel with memory} \end{cases}$
- $\begin{cases} \text{time-invariant channel: 恒参信道} \\ \text{time-varying channel: 随参信道} \end{cases}$
- physical channel: 卫星，光纤，电缆

6.2 Discrete memoryless channels (DMCs) and capacity

A DMC is specified by 3 quantities:

1) the input alphabet $\mathcal{X} = \{0, 1, \ldots, K-1\}$ or $\mathcal{A}_x = \{a_1, a_2, \ldots, a_K\}$.

2) the output alphabet $\mathcal{Y} = \{0, 1, \ldots, J-1\}$ or $\mathcal{A}_y = \{b_1, b_2, \ldots, b_J\}$.

3) the conditional probability distribution $P_{Y|X}(y|x)$ over $\mathcal{Y}$ for each $x \in \mathcal{X}$:

$$P(y_N|x_1 \ldots x_{N-1}, x_N, y_1 \ldots y_{N-1}) = P(y_N|x_N) \tag{6.1}$$

(平稳信道，时不变： $P_{Y_n|X_n}$不依赖于 n)

That is, the probability distribution of the output depends only on the input at that time and is conditionally independent of previous channel inputs and outputs.

Example 6.2.1. *BSC*

$$P(y|x) = \begin{cases} 1-p, & \text{if } y = x \\ p, & \text{if } y \neq x \end{cases}$$

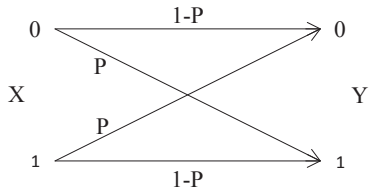


Figure 6.1: BSC

Example 6.2.2. *BEC (Binary erasure channel)*

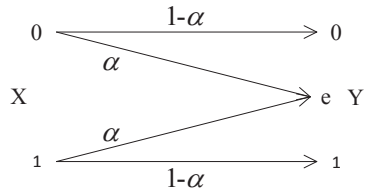


Figure 6.2: BEC

Theorem 6.2.1. *When a DMC is used without feedback,*

$$P_N(y^N|x^N) = P_N(y_1 \dots y_N|x_1 \dots x_N) = \prod_{n=1}^{N} P(y_n|x_n) \tag{6.2}$$

Proof.

$$\begin{aligned} P(x_1 \dots x_N, y_1 \dots y_N) &= \prod_{n=1}^{N} P(x_n|x_1 \dots x_{n-1}, y_1 \dots y_{n-1}) P(y_n|x_1 \dots x_{n-1}, x_n, y_1 \dots y_{n-1}) \\ &\qquad \text{(无反馈)} \qquad\qquad \text{(定义)} \\ &= \prod_{n=1}^{N} P(x_n|x_1 \dots x_{n-1}) P_{Y|X}(y_n|x_n) \\ &= \prod_{n=1}^{N} P(x_n|x_1 \dots x_{n-1}) \prod_{i=1}^{N} P_{Y|X}(y_i|x_i) \\ &= P(x_1 \dots x_N) \prod_{i=1}^{N} P_{Y|X}(y_i|x_i) \end{aligned}$$

Dividing $P(x_1 \dots x_N)$, we obtain (6.2). □

Definition 6.2.1 (Information channel capacity)**.** *The capacity of a DMC is defined as*

$$C \triangleq \max_{P(x)} I(X;Y) = \max_{P(x)} [H(Y) - H(Y|X)]$$

where the maximum is taken over all possible input distribution $P(x)$.

- Operational interpretation: The capacity is the maximum rate of information that can be transferred over the channel reliably. 更正式地，有下列定义：

Definition 6.2.2 (Operational channel capacity)**.** *The capacity is the supremum of all achievable rates.*

- 可达: 存在$(\lceil 2^{nR}, n\rceil)$码序列，使得最大（或平均）错误概率$P_e \to 0$ as $n \to \infty$.
- ε-可达：$P_e \le \varepsilon$.

Theorem 6.2.2 (Noisy channel coding theorem)**.** *All rates below* $C = \max_{P(x)} I(X;Y)$ *are achievable.*

- 它建立了*Operational channel capacity = Information channel capacity* for DMCs，即 C 就是容量。
- 数学意义：解决了容量的计算问题 (at least in principle)。
- 证明：见后面的信道编码定理。

Our goal now is to find $P(x)$ to maximum $I(X;Y)$, and later use this distribution to achieve the maximum communication rate.

Example 6.2.3. *Consider a BSC channel.*

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum_x P(x)H(Y|X=x) \\ &= H(Y) - \sum_x P(x)H_2(p) \\ &= H(Y) - H_2(p) \\ &\le 1 - H_2(p) \end{aligned}$$

$$\therefore C = 1 - H_2(p) \quad \textit{bits/symbol}$$

The optimal input distribution: $P_X(0) = P_X(1) = \frac{1}{2}$

Example 6.2.4. *BEC*

- *方法一：Let E be a indicator variable that is 1 if there is an error and is 0 otherwise. Then*

$$\begin{aligned} C &= \max_{P(x)}[H(Y) - H(Y|X)] \\ &= \max_{P(x)}[H(Y,E) - H(Y|X)] \\ &= \max_{P(x)}\left[H(E) + H(Y|E) - \sum_x P(x)H(Y|X=x)\right] \\ &= \max_{P(x)}[H(E) + H(Y|E) - H(\alpha)] \end{aligned}$$

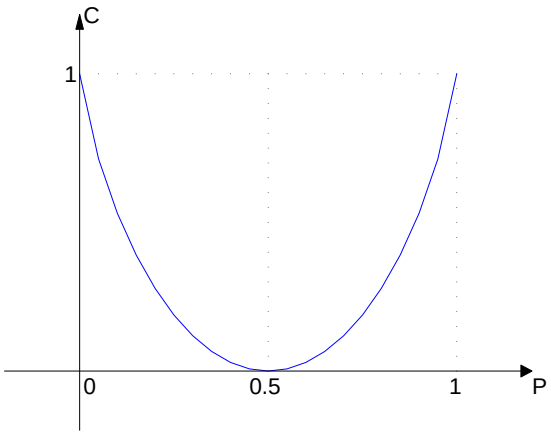


Figure 6.3: BSC capacity

$\because H(E) = H(\alpha)$

$$\begin{aligned} H(Y|E) &= P(E=0)H(Y|E=0) + P(E=1)\underbrace{H(Y|E=1)}_{=0} \\ &= H(X) - \alpha H(X) \\ &= (1-\alpha)H(X) \end{aligned}$$

$\therefore C = \max_{P(x)}[(1-\alpha)H(X)] = 1-\alpha$
where capacity is achieved by $P_X(0) = P_X(1) = \frac{1}{2}$.

- *方法二：令* $\pi = P(X=0)$ *表示* $X=0$ *的概率，则*

$$\begin{aligned} H(Y) &= H(\pi(1-\alpha), \alpha, (1-\pi)(1-\alpha)) \\ &= -\pi(1-\alpha)\log\pi(1-\alpha) - \alpha\log\alpha - (1-\pi)(1-\alpha)\log[(1-\pi)(1-\alpha)] \\ &= -(1-\alpha)[-\pi\log\pi - (1-\pi)\log(1-\pi)] - \alpha\log\pi - (1-\alpha)\log(1-\alpha) \\ &= H(\alpha) + (1-\alpha)H(X) \end{aligned}$$

- *方法三：*

$$\begin{aligned} I(X;Y) &= H(X) - H(X|Y) \\ &= H(X) - P(Y=0)H(X|Y=0) - P(Y=1)H(X|Y=1) - P(Y=e)H(X|Y=e) \\ &= H(X) - \alpha H(X|Y=e) \end{aligned}$$

Thus, $C = 1-\alpha$.

We can also consider the capacity for n uses of the channel,

$$C^{(n)} = \frac{1}{n}\max_{P(x^n)} I(X^n;Y^n)$$

Theorem 6.2.3. *Let* $P(x^n)$ *be an arbitrary joint probability distribution on sequence of* n *inputs for a DMC. Then*

$$I(X^n;Y^n) \le \sum_{i=1}^{n} I(X_i;Y_i) \tag{6.3}$$

$$I(X^n;Y^n) \le nC \tag{6.4}$$

Proof.

$$\begin{aligned}
I(X^n;Y^n) &= H(Y^n) - H(Y^n|X^n) \\
&= \sum_{i=1}^{n} H(Y_i|Y_1 \dots Y_{i-1}) - \sum_{i=1}^{n} H(Y_i|Y_1 \dots Y_{i-1}, X^n) \\
\text{(by memoryless assumption)} &= \sum_{i=1}^{n} H(Y_i|Y_1 \dots Y_{i-1}) - \sum_{i=1}^{n} H(Y_i|X_i) \\
&\le \sum_{i=1}^{n} H(Y_i) - \sum_{i=1}^{n} H(Y_i|X_i) \qquad (6.5) \\
&= \sum_{i=1}^{n} I(X_i;Y_i) \\
\text{(From the definition of capacity)} &\le nC
\end{aligned}$$

□

Alternative one:

$$\begin{aligned}
H(Y^n|X^n) &= \sum_{x,y} P(x^n, y^n) \log \frac{1}{P(y^n|x^n)} \\
&= \mathbb{E}\left[\sum_{n} \log \frac{1}{P(y_i|x_i)}\right] \\
&= \sum_{n} \mathbb{E}\left[\log \frac{1}{P(y_i|x_i)}\right]
\end{aligned}$$

Equality in (6.5) is achieved when the output letters are statistically independent. If the input letters are independent; i.e., $P(x^n) = \prod_{i=1}^{n} P(x_i)$, then

$$\begin{aligned}
P(y^n) &= \sum_{x^n} \prod_{i=1}^{n} P(x_i)P(y_i|x_i) \qquad (\because P(y^n) = \sum_{x^n} P(x^n)P(y^n|x^n)) \\
&= \prod_{i=1}^{n} \sum_{x_i} P(x_i)P(y_i|x_i) \\
&= \prod_{i=1}^{n} P(y_i) \rightarrow y_i\text{统计独立}
\end{aligned}$$

Note: It should not be concluded from this theorem that statistical dependence between inputs is sometimes to be avoided.

进一步讨论：

- Does "式(6.5)取等号"（输出独立→ 输入独立）imply no coding is needed to achieve capacity?
  No. 这只是不考虑 P_e 的"information capacity"的优化结果。*Coding is always needed to achieve capacity.*
- Reliable communication cannot be achieved at exactly the rate C and when the channel outputs are exactly independent.
- For a rate strictly less than C, no matter how close, the coding theorem guarantees that reliable communication is possible.

- Equality in (6.5) is achieved if the inputs are independent over time. Thus, optimize $P(x^n)$ to maximize $I(X^n;Y^n)$ $\longrightarrow$在单符号上优化 $P(x_i)$ 使 $I(X_i;Y_i)$ 最大。即

$$C^{(n)} = \frac{1}{n}\sum_i \max_{P(x_i)} I(X_i;Y_i) = \max_{P(x_1)} I(X_1;Y_1)$$

- For a memoryless channel, we can focus on maximizing the mutual information of one channel use. This does not mean we can communicate reliably with just one channel use.

- Pick codewords randomly according to i.i.d input distribution （随机编码）is totally different from sending uncoded symbols.

6.3 Symmetric channels

- Let us consider the channel with transition probability matrix

$$\mathbf{T} = [P(y|x)] = \begin{bmatrix} P(y_1|x_1) & P(y_2|x_1) & \cdots & P(y_J|x_1) \\ P(y_1|x_2) & P(y_2|x_2) & \cdots & P(y_J|x_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(y_1|x_K) & P(y_2|x_K) & \cdots & P(y_J|x_K) \end{bmatrix}, size = |\mathcal{X}| \times |\mathcal{Y}|$$

For example,

$$\mathbf{T} = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix}$$

$$\begin{cases} \text{each row of } \mathbf{T} \text{ corresponds to an input of the channel} \\ \text{each column of } \mathbf{T} \text{ corresponds to an output of the channel} \end{cases}$$

In this channel, each row is a permutation of each other row (关于输入对称)and so is each column (关于输出对称). Such a channel is said to be *symmetric*.

Definition 6.3.1. *A DMC is symmetric iff all the rows of* $\mathbf{T}$ *are permutations of each other, and the columns are permutations of each other.*

- Denote a row of $\mathbf{T}$ by $\mathbf{r} = \left[r_1, \cdots, r_{|\mathcal{Y}|}\right]$. Then

$$H(\mathbf{r}) = -\sum_i r_i \log r_i$$

- **Optimal input distribution:**

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \mathbb{E}_X\left[H(Y|X=x)\right] \\ &= H(Y) - H(\mathbf{r}) \\ &\le \log|\mathcal{Y}| - H(\mathbf{r}) \end{aligned}$$

The equality holds only if Y is uniformly distributed.

– Let X be uniformly distributed; i.e., $P(x) = \frac{1}{|\mathcal{X}|}$ for all $x \in \mathcal{X}$.

$$P_Y(y) = \sum_{x \in \mathcal{X}} P(x) P(y|x) = \frac{1}{|\mathcal{X}|} \sum_x P(y|x) = \frac{c}{|\mathcal{X}|}$$

其中 $c = \sum$ entries in one column.

Therefore, the uniform input distribution is optimal. The capacity of the above channel is $C = \log 3 - H(0.3, 0.2, 0.5)$.

- **Weakly symmetric channel:**
Every row of $\mathbf{T}$ is a permutation of every other row, and all the column sums $\sum_x P(y|x)$ are equal.

– Example:

$$\mathbf{T} = \begin{bmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{bmatrix}$$

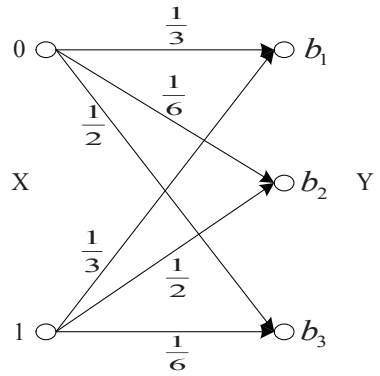


Figure 6.4: Example

– The capacity $C = \log |\mathcal{Y}| - H(\mathbf{r})$.

Example 6.3.1. *Consider a K-ary symmetric channel.*

$$P(y|x) = \begin{cases} 1-p, \text{if } y = x \\ \\ \frac{p}{K-1}, \text{otherwise} \end{cases}$$

The channel capacity is given by

$$\begin{aligned} C &= \log K + P(0|0) \log P(0|0) + (|\mathcal{Y}| - 1) \cdot \frac{p}{K-1} \log \frac{p}{K-1} \\ &= \log K + (1-p) \log (1-p) + p \log \frac{p}{K-1} \\ &= \log K - H(p) - p \log (K-1) \end{aligned}$$

When $K = 2, C = 1 - H(p) \longrightarrow$ BSC.

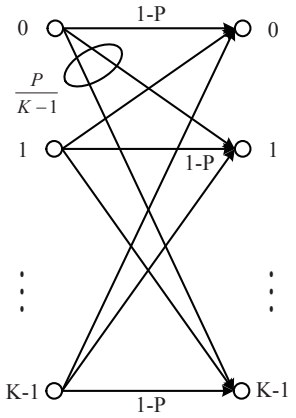


Figure 6.5: Example

Theorem 6.3.1. *令 $Q_k \triangleq P_X(X=k)$，则输入概率矢量 $\mathbf{P}_X=\{Q_0,Q_1,\ldots,Q_{K-1}\}$ 达到 DMC 信道 $P(j|k)$ 容量 C 的充要条件是*

$$\begin{aligned} &I(X=k;Y)=C, \quad \textit{for all } k \textit{ with } Q_k>0\\ &I(X=k;Y)\le C, \quad \textit{for all } k \textit{ with } Q_k=0. \end{aligned}$$

其中 $I(X=k;Y)=\sum_j P(j|k)\log\frac{P(j|k)}{\sum_i Q_iP(j|i)}$.

说明：在上述条件下，有

$$\begin{aligned} I(X;Y)&=\sum_k P(X=k)I(X=k;Y)\\ &=\sum_k (Q_k\cdot C)=C\cdot\sum_k Q_k=C\\ &\qquad\qquad (Q_k\neq 0) \end{aligned}$$

Proof. 由定理2.6.1中的 K-T 条件可知，输入分布 Q_k 使 $I(X;Y)$ 最大化的充要条件是

$$\frac{\partial I(X;Y)}{\partial Q_k}=\lambda, \quad Q_k>0 \tag{6.6}$$

$$\frac{\partial I(X;Y)}{\partial Q_k}\le\lambda, \quad Q_k=0$$

其中 λ 为一常数.

记 $I(X;Y)=\sum_j\sum_m Q_mP(j|m)\log\frac{P(j|m)}{\sum_i Q_iP(j|i)}$，则

$$\begin{aligned} \frac{\partial I(X;Y)}{\partial Q_k}&=\sum_j\left[P(j|k)\log\frac{P(j|k)}{\sum_i Q_iP(j|i)}+\log e\sum_m Q_mP(j|m)\cdot\frac{\sum_i Q_iP(j|i)}{P(j|m)}\cdot\frac{-P(j|m)}{(\sum Q_iP(j|i))^2}\cdot P(j|k)\right]\\ &=\sum_j\left[P(j|k)\log\frac{P(j|k)}{\sum_i Q_iP(j|i)}-\log e\sum_m Q_mP(j|m)\ cdot\frac{P(j|k)}{\sum_i Q_iP(j|i)}\right]\\ &=\sum_j P(j|k)\log\frac{P(j|k)}{\sum_i Q_iP(j|i)}-\log e\sum_j P(j|k)\\ &=I(X=k;Y)-\log e \end{aligned}$$

$$\therefore \quad I(X=k;Y) = \frac{\partial I(X;Y)}{\partial Q_k} + \log e$$

由式 (6.6)：

$$I(X=k;Y) \begin{cases} = \lambda + \log e, & \text{for } Q_k > 0 \\ \leq \lambda + \log e, & \text{for } Q_k = 0 \end{cases}$$

记 $C = \lambda + \log e$，就可得到定理公式。 □

6.3.1 准对称信道

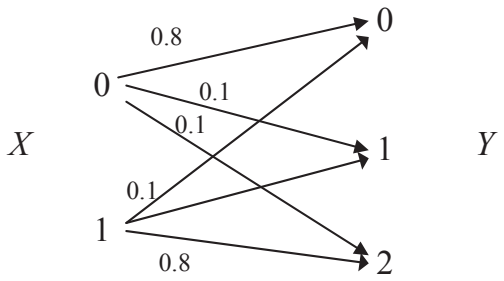
信道输出 Y 可划分为 n 个子集，而每个子集对应的信道转移概率矩阵 $\mathbf{P}$ 中的列组成的子阵是对称的。例:

$$\mathbf{P} = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.8 \end{bmatrix} \quad \longrightarrow \text{关于输入对称}$$

$$Y \Rightarrow \quad 0 \quad 1 \quad 2$$

Y 可划分为 2 个子集：$\{0,2\}$, $\{1\}$.

$$\begin{bmatrix} 0.8 & 0.1 \\ 0.1 & 0.8 \end{bmatrix} \quad \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}$$



Theorem 6.3.2. *准对称 DMC 信道的最佳输入分布为等概分布。*

$$C = I(X=k;Y)$$

Example 6.3.2. *考虑如下信道：*

$$\mathbf{P} = \begin{bmatrix} 1-P-Q & Q & P \\ P & Q & 1-P-Q \end{bmatrix}$$

$$\begin{aligned} C &= I(X=0;Y) \\ &= \sum_j P(j|X=0) \log \frac{P(j|X=0)}{\frac{1}{K}\sum_i P(j|i)} \\ &= (1-P-Q)\log\frac{1-P-Q}{\frac{1}{2}(1-Q)} + Q\log\frac{Q}{\frac{1}{2}\cdot 2Q} + P\log\frac{P}{\frac{1}{2}(1-Q)} \end{aligned}$$

作业：4.1, 4.3, 4.4, 4.6, 4.8, 4.12, 4.13, 4.14

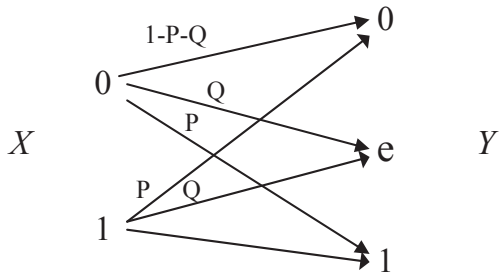


Figure 6.6: 二元对称删除信道

6.4 Computing the Channel Capacity

- In general, analytical solutions of the optimal input distribution that maximizes the mutual information is not always available. We need to complete the capacity numeracally.
- The difficulties: high dimensional nonlinear optimization.
- An iterative algorithm for computing channel capacity was suggested by S. Arimoto and Blahut in 1972.

6.4.1 Alternating optimization

- Consider the double supremum

$$\max_{u_1 \in A_1} \max_{u_2 \in A_2} f(u_1, u_2)$$

where A_1, A_2 are convex subset of $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}$, respectively, and f is a function defined on $A_1 \times A_2$.
Suppose f is bounded from above and is continuous and has continuous partial derivatives on $A_1 \times A_2$.

- Suppose now for all $u_1 \in A_1$, there exists a unique $C_2(u_1) \in A_2$ such that

$$f(u_1, C_2(u_1)) = \max_{u'_2 \in A_2} f\left(u_1, u'_2\right)$$

and for any $u_2 \in A_2$, there exists a unique $C_1(u_2)$ that

$$f(C_1(u_2), u_2) = \max_{u'_1 \in A_1} f\left(u'_1, u_2\right).$$

- Now we can find the maximum with an alternating algorithm.
  - Initialization: Pick any $u_1^{(0)}$.
  - Find $u_2^{(0)} = C_2\left(u_1^{(0)}\right)$.
  - For any $k \geq 1$, $u_1^{(k)}$ and $u_2^{(k)}$ are computed as

$$u_1^{(k)} = C_1\left(u_2^{(k-1)}\right) = \arg\max_{u'_1} f\left(u'_1, u_2^{(k-1)}\right) \quad u_2^{(k)} = C_2\left(u_1^{(k)}\right) = \arg\max_{u'_2} f\left(u_1^{(k)}, u'_2\right)$$

Since $f^{(k)} = f\left(u_1^{(k)}, u_2^{(k)}\right) \geq f\left(u_1^{(k)}, u_2^{(k-1)}\right) \geq f\left(u_1^{(k-1)}, u_2^{(k-1)}\right) = f^{(k-1)}$, i.e., the function value is non-decreasing in each step, the sequence must converge because f is bounded from above.

- Does it necessarily converge to the maximum?

6.4.2 The Blahut-Arimoto algorithm

Lemma 6.4.1. *Fix any input distribution P_X, and channel $P_{Y|X}$. Suppose $P_X(x) > 0$ for any $x \in \mathcal{X}$. Then*

$$\max_{Q_{X|Y}} \sum_x \sum_y \log \frac{Q(x|y)}{P(x)} = \sum_x \sum_y P(x) P(y|x) \log \frac{Q^*(x|y)}{P(x)}$$

where the maximization is taken over all Q such that

$$Q(x|y) = 0 \text{ iff } P(y|x) = 0 \tag{6.7}$$

and

$$Q^*(x|y) = \frac{P(x) P(y|x)}{\sum_{x'} P(x') P(y|x')} \tag{6.8}$$

i.e., the maxmizing Q is the true transition distribution from Y to X given the input $P(x)$ and the channel $P(y|x)$.

Proof. First check that Q^* satisfies the constranit.

$$\begin{aligned}
&\sum_x \sum_y P(x) P(y/x) \log \frac{Q^*(x/y)}{P(x)} - \sum_x \sum_y P(x) P(y/x) \log \frac{Q(x/y)}{P(x)} \\
&= \sum_x \sum_y P(x) P(y/x) \log \frac{Q^*(x/y)}{Q(x/y)} \\
&= \sum_x \sum_y P(y) Q^*(x/y) \log \frac{Q^*(x/y)}{Q(x/y)} \text{from (6.11)} \\
&= \sum_y P(y) D\left(Q^*(x/y) \,||Q(x/y)\right) \\
&\geq 0
\end{aligned}$$

□

- How to use this?

  Theorem 6.4.2. *For a DMC,*

$$\begin{aligned}
C &= \max_{P_X} I(X;Y) \\
&= \sup_{P(x)>0} \max_{Q(x|y)} \sum_{x,y} P(x) P(y|x) \log \frac{Q(x|y)}{P(x)}
\end{aligned}$$

- To obtain the 2nd equality above, we need $P(x) > 0$ strictly for any x, while the optimal input distribution might have for some x, $P^*(x) > 0$.
- The continuity of $I(X;Y)$ with respect to the input.
- Now we have a joint optimization of two variables: $P(x)$ and $Q(x|y)$ to apply the alternating algorithm, need to check:
  (1) The space of distributions is convex.
  (2) The function is bounded.

$$f(P,Q) \leq I_P(X;Y) \leq H(X) \leq \log|\mathcal{X}|.$$

where

$$f(P,Q) = \sum_x \sum_y P(x) P(y|x) \log \frac{Q(x|y)}{P(x)},$$

and

$$I_P(X;Y) = \sum_x \sum_y P(x) P(y|x) \log \frac{Q^*(x|y)}{P(x)}.$$

- Updating rules

(1) Arbitrarily choose a distribution $P(x) > 0$ strictly. Let it be $P^{(0)}$. Then given $P^{(k)}$.

$$\begin{aligned} Q^{(k)} &= \arg \max_{Q(x/y)} \sum_{x,y} P^{(k)}(x) P(y/x) \log \frac{Q(x/y)}{P(x)} \\ &= \frac{P^{(k)}(x) P(y/x)}{\sum_{x'} P^{(k)}(x') P(y/x')} \end{aligned}$$

(2) Given $Q^{(k)}$.

$$P^{(k+1)} = \arg \sup_{P>0} \sum_{x,y} P(x) P(y/x) \log \frac{Q^{(k)}(x/y)}{P(x)}$$

the maximization is taken over all the distributions on $\mathcal{X}$. The constraints are

$$\begin{cases} \sum_x P(x) = 1 \\ P(x) > 0 \text{ for all } x \in \mathcal{X} \end{cases}$$

- Using the method of Lagrange multipliers to find the best $P(x)$ by ignoring temporarily the positive constraints. Let

$$J = \sum_{x,y} P(x) P(y/x) \log \frac{Q(x/y)}{P(x)} - \lambda \sum_x P(x)$$

Set

$$\begin{aligned} 0 = \frac{\partial J}{\partial P(x)} &= \sum_y P(y/x) \log Q(x/y) - \sum_y P(y/x) [\log P(x) + 1] - \lambda \\ &= \sum_y P(y/x) \log Q(x/y) - \log P(x) - 1 - \lambda \end{aligned}$$

Therefore,

$$P(x) = e^{-(\lambda+1)} \prod_y Q(x/y)^{P(y/x)}$$

By considering $\sum_x P(x) = 1$, we can eliminate λ and obtain

$$P(x) = \frac{\prod_y Q(x/y)^{P(y/x)}}{\sum_{x'} \prod_y Q(x'/y)^{P(y/x')}}$$

The above product is over all y such that $P(y/x) > 0$, and $Q(x/y) > 0$ for all such y. This implies that both the numerator and the denominator above are positive, and therefore $P(x) > 0$. That is, the $P(x)$ obtained happen to satisfy the positive constraints.
So, define

$$P^{(k)}(x) = \frac{\prod_y Q^{(k-1)}(x/y)^{P(y/x)}}{\sum_{x'} \prod_y Q^{(k-1)}(x'/y)^{P(y/x')}}$$

- $f^{(k)} = f\left(P^{(k)}, Q^{(k)}\right) \to C$.
- Convergence of the alternating optimization algorithm:

  Lemma 6.4.3. *If f is concave, then $f\left(u_1^{(k)}, u_2^{(k)}\right) \to f^*$, where f^* is the desired maximum value.*

  - For a concave function, local maximum is global maximum.

- Convergence to the channel capacity:
  We only need to check

$$f(P, Q) = \sum_{x,y} P(x) P(y/x) \log \frac{Q(x/y)}{P(x)}$$

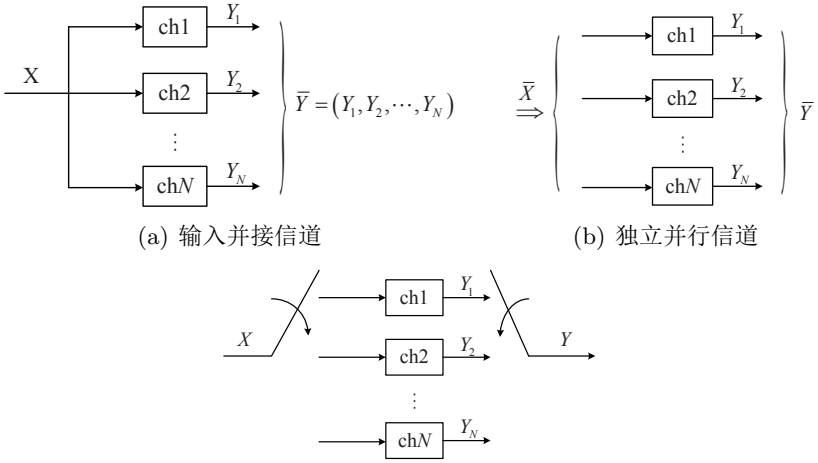
  is concave in P and Q.

6.5 Capacity of cascaded channels and parallel channels

6.5.1 Parallel Channels (积信道)

- The input of two component channels are independent; i.e., X_1 and X_2 are independent of each other.
- The output Y_1 and Y_2 of two component channels are independent.
- Two component channels are used simultaneously.

$$\mathcal{X} = \mathcal{X}_1 \times \mathcal{X}_2 = \left\{\left(a_k, a'_k\right)\right\} = \left\{\left(k, k'\right)\right\}$$
$$\mathcal{Y} = \mathcal{Y}_1 \times \mathcal{Y}_2 = \left\{\left(b_j, b'_j\right)\right\} = \left\{\left(j, j'\right)\right\}$$
$$P\left(jj'|kk'\right) = P(j|k)\, P\left(j'|k'\right)$$



(a) 输入并接信道 (b) 独立并行信道

(c) 和信道(并信道)

Figure 6.7: 信道模型

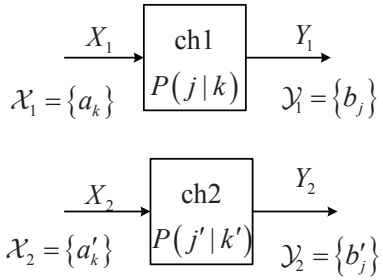


Figure 6.8: Parallel channel

Theorem 6.5.1. *The channel capacity of the set of N independent channels (in parallel) is given by*

$$C = \sum_{n=1}^{N} C_n$$

Proof. Since $P(y_1 \cdots y_N | x_1 \cdots x_N) = \prod_{n=1}^{N} P(y_n|x_n)$, we have

$$\begin{aligned} I(\mathbf{X};\mathbf{Y}) &= H(\mathbf{Y}) - H(\mathbf{Y}|\mathbf{X}) \\ &= H(\mathbf{Y}) - \sum_{n=1}^{N} H(Y_n|X_n) \\ &\leqslant \sum_{n=1}^{N} H(Y_n) - \sum_{n=1}^{N} H(Y_n|X_n) \\ &= \sum_{n=1}^{N} I(X_n;Y_n) \end{aligned}$$

Therefore,

$$\begin{aligned} C &= \max_{P(\mathbf{x})} I(\mathbf{X};\mathbf{Y}) \\ &\leqslant \max_{P(\mathbf{x})} \sum_{n=1}^{N} I(X_n;Y_n) \\ &= \sum_{n=1}^{N} C_n \end{aligned}$$

$\therefore\ C \leqslant \sum_{n=1}^{N} C_n$, 当输入独立时等号成立。 □

6.5.2 Union of channels (和信道)

- At each time interval, we randomly select one of N channels to use.
- The probability that the i-th channel is selected is assumed to be P_i, and $\sum_i P_i = 1$.
- For this type of channels, the input alphabet $\mathcal{X} = \bigcup_{i=1}^{N} \mathcal{X}_i$ and the output alphabet $\mathcal{Y} = \bigcup_{i=1}^{N} \mathcal{Y}_i$.
- Let $\mathcal{X}_i = \{1, 2, \cdots, K_i\}$, $\mathcal{Y}_i = \{1, 2, \cdots, J_i\}$.

  $$P(j_i|k_i) = P(Y_i = j_i | X_i = k_i).$$

  Then the channel transition matrix

  $$\mathbf{P} = \begin{bmatrix} \mathbf{P}_1 & & & \\ & \mathbf{P}_2 & & \\ & & \ddots & \\ & & & \mathbf{P}_N \end{bmatrix}$$

- 和信道的容量
  - 设信道 i 的（先验）输入概率分布为 Q_i，则有

    $$Q(x) = \begin{cases} P_1 Q_1(x), & \text{if } x \in \mathcal{X}_1 \\ P_2 Q_2(x), & \text{if } x \in \mathcal{X}_2 \end{cases}$$

  - 和信道的互信息为：

    $$\begin{aligned} I(X;Y) &= \sum_{x,y} Q(x)P(y|x) \log \frac{P(y|x)}{P(y)} \\ (\text{对于} x \in \mathcal{X}_1) \qquad &= \sum_{x_1,y_1} Q(x_1)P(y_1|x_1) \log \frac{P(y_1|x_1)}{P_1 P(y_1)} \\ (\text{对于} x \in \mathcal{X}_2) \qquad &+ \sum_{x_2,y_2} Q(x_2)P(y_2|x_2) \log \frac{P(y_2|x_2)}{P_2 P(y_2)} \end{aligned}$$

    其中 $P_1 P(y_1) = \underbrace{\sum_{x_1} P_1 Q_1(x) P(y_1|x_1)}_{y \in \mathcal{Y}_1 \text{时的} P(y)}$

由上式有 $I(X;Y) = P_1 I(X_1;Y_1) - P_1 \log P_1 + P_2 I(X_2;Y_2) - P_2 \log P_2$

$$\begin{aligned} C &= \max_{P_i, Q_i} I(X;Y) \\ &= \max_{P_1} \left\{ \max_{Q_1} P_1 I(X_1;Y_1) - P_1 \log P_1 + \max_{Q_2} P_2 I(X_2;Y_2) - P_2 \log P_2 \right\} \\ &= \max_{P_1} \{ P_1 C_1 + P_2 C_2 - P_1 \log P_1 - P_2 \log P_2 \} \end{aligned} \tag{6.9}$$

$$\begin{aligned} \text{令 } f(P_1) &= P_1 C_1 + P_2 C_2 - P_1 \log P_1 - P_2 \log P_2 \\ &= P_1 C_1 + (1 - P_1) C_2 - P_1 \log P_1 - (1 - P_1) \log (1 - P_1) \end{aligned}$$

$$\begin{aligned} \text{则 } \frac{df(P_1)}{dP_1} &= C_1 - C_2 - \log P_1 + \log (1 - P_1) \\ &= C_1 - \log P_1 - (C_2 - \log (1 - P_1)) = 0 \end{aligned}$$

$$\Longrightarrow C_1 - \log P_1 = C_2 - \log P_2 \triangleq \lambda$$

$$\therefore \begin{aligned} P_1 &= 2^{C_1 - \lambda} \\ P_2 &= 2^{C_2 - \lambda} \end{aligned}$$

$$P_1 + P_2 = 1 \Longrightarrow \begin{aligned} 2^{\lambda} &= 2^{C_1} + 2^{C_2} \\ \lambda &= \log_2 \left(2^{C_1} + 2^{C_2} \right) \end{aligned}$$

将P_1, P_2 代入式 (6.9)，有

$$\begin{aligned} C &= P_1 C_1 + P_2 C_2 - P_1 \log 2^{C_1 - \lambda} - P_2 \log 2^{C_2 - \lambda} \\ &= P_1 C_1 + P_2 C_2 - P_1 (C_1 - \lambda) - P_2 (C_2 - \lambda) \\ &= (P_1 + P_2) \lambda \\ &= \lambda \end{aligned}$$

$\therefore C = \log_2 \left(2^{C_1} + 2^{C_2} \right)$

一般地：$C = \log \left(\sum_{i=1}^{N} 2^{C_i} \right)$

6.5.3 Cascaded Channels (级联信道)

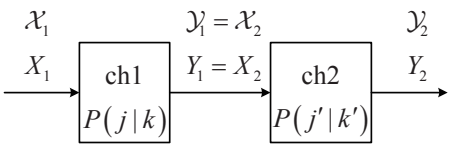


Figure 6.9: Cascaded channel

$$\begin{aligned} P\left(j'|k\right) &= \sum_{j} P\left(j, j'|k\right) \\ &= \sum_{j} P\left(j|k\right) P\left(j'|j, k\right) \\ &= \sum_{j} P\left(j|k\right) P\left(j'|j\right) \end{aligned}$$

Thus $\mathbf{P} = \mathbf{P}_1\mathbf{P}_2\cdots\mathbf{P}_N = \prod_{n=1}^{N}\mathbf{P}_n$.

$$\begin{cases} I(X_1;Y_2) \leqslant I(X_1;Y_1) \\ I(X_1;Y_2) \leqslant I(X_2;Y_2) \end{cases}$$

6.6 Maximum Likelihood Decoding

- Message W uniformly distributed in $\{1, 2, \ldots, M\}, M = 2^{nR}$.
- Encoder transmit codeword $x^n(m)$ if the incoming message is $W = m$.
- Receiver receives y^n, and find the most likely transmitted codeword.

$$\hat{W} = \arg\max_m P(y^n|x^n(m))$$

- Define $\mathcal{Y}_m = \{y^n : y^n \text{ decodes to } m\}$.
(将空间y^n按译码规则划分为不相交的判决区域)
- The probability of error conditioned on $W = m$ is transmitted:

$$P_{e,m} = \sum_{y^n \in \mathcal{Y}_m^c} P(y^n|x^n(m))$$

- The overall probability of block decoding error: $P_e = \sum_{m=1}^{M} P(W = m)P_{e,m}$.
- Bit error probability $P_b = \frac{1}{K}\sum_{i=1}^{K} P_{ei}$，其中$P_{ei} = P(\hat{U}_i \neq U_i)$ 是第 i 位出错的概率。这里，$K = nR$，$(U_1, U_2, \ldots, U_K)$ 是发送消息 W 的二进制表示。
- Decoding criterion:

  (1) The minimum error probability decoding rule:

$$P_e(y^n) = P(\hat{m} \neq m|y^n) = 1 - P(\hat{m} = m|y^n)$$

  (2) The MAP decoding rule:

$$\hat{m} = \arg\max_m P(x^n(m)|y^n)$$

  (3) The ML decoding rule: $\hat{m} = \arg\max_m P(y^n|x^n(m))$

$$\left(\because\ P(x^n(m)|y^n) = \frac{P(y^n|x^n(m))\,P(m)}{P(y^n)}\right)$$

  In the case of DMCs, this becomes

$$\hat{m} = \arg\max_m \prod_{i=1}^{n} P(y_i|x_{m,i})$$

  (4) 最小距离译码：

  AWGN: $P(y|x) \propto e^{-\frac{(y-x)^2}{2\sigma^2}}$

$$\hat{m} = \arg\min_m ||y^n - x^n(m)||^2$$

  BSC: $P(y^n|x^n) = p^{d_{\mathrm{H}}(y^n,x^n)}(1-p)^{n-d_{\mathrm{H}}(y^n,x^n)}$. For $p < \frac{1}{2}$，由单调函数有

$$P(y^n|x^n(m)) > P\left(y^n|x^n(m')\right) \quad \text{iff } d_{\mathrm{H}}(y^n, x^n(m)) < d_{\mathrm{H}}\left(y^n, x^n(m')\right)$$

6.7 Preview of channel coding theorem

- Reliable communication requires disjoint partitioning in the received signal space, corresponding to the different possible transmitted signals.
- For each (typical) input sequence X^n, there are approximately $2^{nH(Y|X)}$ possible Y^n sequences.
- There are in total $2^{nH(Y)}$ (typical) Y^n sequences.
  Divide this set into sets of size $2^{nH(Y|X)}$, each corresponding to one input X^n sequences.
- The number of disjoint sets is no more than $2^{nH(Y)-nH(Y|X)} = 2^{nI(X;Y)}$.
- Therefore we can send at most $2^{nI(X;Y)}$ different sequences that are distinguishable.
- The number of codewords $|\mathcal{C}| = M < 2^{nI(X;Y)} \Rightarrow$

$$\underbrace{\frac{1}{n}\log_2 M}_{\text{code rate } R} < I(X;Y)$$

  code rate R [bits/symbol], $R < C$.

6.8 Joint AEP

- $X^n = (X_1, \ldots, X_n) \in \mathcal{X}^n$, $Y^n = (Y_1, \ldots, Y_n) \in \mathcal{Y}^n$.

$$P(x^n, y^n) = \prod_{i=1}^{n} P(x_i, y_i)$$

Definition 6.8.1. *The set of the jointly typical sequences* $\{(x^n, y^n)\}$ *is*

$$\begin{aligned} T_\epsilon^{(n)} = \Big\{ & (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \\ & \left| -\frac{1}{n}\log P(x^n) - H(X) \right| < \epsilon, && \longrightarrow x^n \in T_\epsilon^{(n)}(X) \\ & \left| -\frac{1}{n}\log P(y^n) - H(Y) \right| < \epsilon, && \longrightarrow y^n \in T_\epsilon^{(n)}(Y) \\ & \left| -\frac{1}{n}\log P(x^n, y^n) - H(XY) \right| < \epsilon \Big\} && \longrightarrow (x^n, y^n) \in T_\epsilon^{(n)}(XY) \end{aligned}$$

- From the above conditions, we have

$$\begin{aligned} & -\epsilon < -\frac{1}{n}\log P(x^n, y^n) - H(XY) < \epsilon \\ \Longrightarrow \quad & 2^{-n[H(XY)+\epsilon]} < P(x^n, y^n) < 2^{-n[H(XY)-\epsilon]} \end{aligned}$$

Theorem 6.8.1. *Consider sequences* (X^n, Y^n) *drawn i.i.d. from* $P(x, y)$. *Then*

(1) $P\left((x^n, y^n) \in T_\epsilon^{(n)}\right) \to 1$ as $n \to \infty$

(2) $|T_\epsilon^{(n)}| \leq 2^{n[H(XY)+\epsilon]}$

Proof. (1)

$$\begin{aligned} P\left((x^n, y^n) \in T_\epsilon^{(n)}\right) &= 1 - P\left((x^n, y^n) \notin T_\epsilon^{(n)}\right) \\ &= 1 - P\left\{x^n \notin T_\epsilon^{(n)}(X) \cup y^n \notin T_\epsilon^{(n)}(Y) \cup (x^n, y^n) \notin T_\epsilon^{(n)}(XY)\right\} \end{aligned}$$

Since $P(\cup A_i) \leq \sum_i P(A_i)$,

$$\begin{aligned} P\left((x^n, y^n) \in T_\epsilon^{(n)}\right) \geq 1 - \Big\{ & P\left(x^n \notin T_\epsilon^{(n)}(X)\right) + P\left(y^n \notin T_\epsilon^{(n)}(Y)\right) \\ & + P\left(\left|-\frac{1}{2}\log P(x^n, y^n) - H(XY)\right| \geq \epsilon\right)\Big\} \end{aligned}$$

By the weak law of large numbers, given ϵ, there exists n_1 such that for all $n > n_1$,

$$P\left(x^n \notin T_\epsilon^{(n)}(X)\right) = \Pr\left(\left|-\frac{1}{n}\lg P(x^n) - H(X)\right| \geq \epsilon\right) \leq \frac{\sigma_w^2}{n\epsilon^2} = \frac{\epsilon}{3}$$

Similarly, there exist n_2 and n_3 such that for all $n > n_2$,

$$P\left(y^n \notin T_\epsilon^{(n)}(Y)\right) = \Pr\left(\left|-\frac{1}{n}\lg P(y^n) - H(Y)\right| \geq \epsilon\right) \leq \frac{\epsilon}{3}$$

and for all $n > n_3$,

$$P\left(\left|-\frac{1}{n}\lg P(x^n, y^n) - H(XY)\right| \geq \epsilon\right) \leq \frac{\epsilon}{3}$$

Choosing $n > \max\{n_1, n_2, n_3\}$, we have

$$P\left((x^n, y^n) \in T_\epsilon^{(n)}\right) \geq 1 - \left(\frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3}\right) = 1 - \epsilon$$

(2)

$$\begin{aligned} 1 = \sum_{(x^n, y^n) \in \mathcal{X}_n \times \mathcal{Y}_n} P(x^n, y^n) &\geq \sum_{(x^n, y^n) \in T_\epsilon^{(n)}} P(x^n, y^n) \\ &\geq \sum_{(x^n, y^n) \in T_\epsilon^{(n)}} 2^{-n[H(XY)+\epsilon]} \\ &= |T_\epsilon^{(n)}| \cdot 2^{-n[H(XY)+\epsilon]} \end{aligned}$$

so $|T_\epsilon^{(n)}| \leq 2^{n[H(XY)+\epsilon]}$.
Since $1 - \epsilon \leq \sum_{(x^n, y^n) \in T_\epsilon^{(n)}} P(x^n, y^n) = |T_\epsilon^{(n)}| \cdot 2^{-n[H(XY)+\epsilon]}$
$\therefore \quad |T_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n[H(XY)+\epsilon]}$

□

- Question:
  If we randomly pick a typical sequence $\widetilde{X}^n \in T_\epsilon^{(n)}(X)$, $\widetilde{Y}^n \in T_\epsilon^{(n)}(Y)$, are they joint typical?
  $$T_\epsilon^{(n)}(X) \times T_\epsilon^{(n)}(Y) = T_\epsilon^{(n)}(Y)$$
  $$2^{nH(X)} \cdot 2^{nH(Y)} \geq 2^{nH(XY)}$$

Theorem 6.8.2. *If* $(\widetilde{X}^n, \widetilde{Y}^n) \sim P(x^n)P(y^n)$*, then*

$$P\left((\widetilde{X}^n, \widetilde{Y}^n) \in T_\epsilon^{(n)}(XY)\right) \leq 2^{-n[I(X;Y)-3\epsilon]}$$

and for n *large enough,*

$$P\left((\widetilde{X}^n, \widetilde{Y}^n) \in T_\epsilon^{(n)}(XY)\right) \geq (1-\epsilon)2^{-n[I(X;Y)+3\epsilon]}$$

Proof.

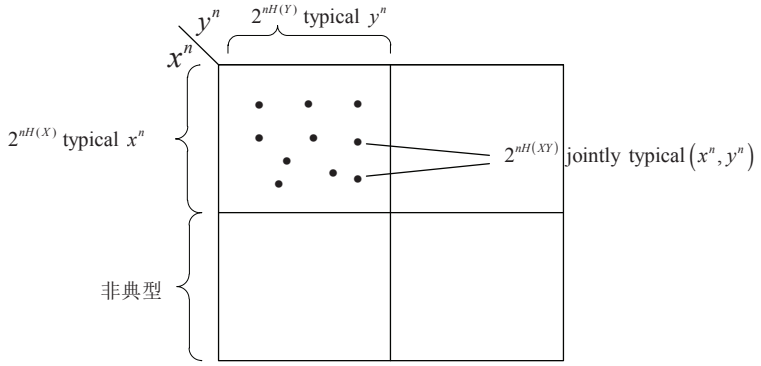
$$\begin{aligned} P\left((\widetilde{X}^n, \widetilde{Y}^n) \in T_\epsilon^{(n)}(XY)\right) &= \sum_{T_\epsilon^{(n)}(X,Y)} P(x^n)P(y^n) \\ &\leq 2^{n[H(XY)+\epsilon]} \cdot 2^{-n[H(X)-\epsilon]} \cdot 2^{-n[H(Y)-\epsilon]} \\ &= 2^{-n[I(X;Y)-3\epsilon]} \end{aligned}$$

$$\begin{aligned} P\left((\widetilde{X}^n, \widetilde{Y}^n) \in T_\epsilon^{(n)}(XY)\right) &= \sum_{T_\epsilon^{(n)}(X,Y)} P(x^n)P(y^n) \\ &\geq (1-\epsilon)2^{n[H(XY)-\epsilon]} \cdot 2^{-n[H(X)+\epsilon]} \cdot 2^{-n[H(Y)+\epsilon]} \\ &= (1-\epsilon)2^{-n[H(X)+H(Y)-H(XY)+3\epsilon]} \qquad = (1-\epsilon)2^{-n[I(X;Y)+3\epsilon]} \end{aligned}$$

□

- Fix a typical y^n, out of the $2^{nH(X)}$ typical x^n sequences, there are $\approx 2^{nH(X|Y)}$ sequences that are jointly typical with y^n. The probability that some randomly selected other $\tilde{x}^n$ (not the true input signal x^n) is jointly typical with y^n is $2^{nH(X|Y)}/2^{nH(X)} = 2^{-nI(X;Y)}$.

- Jointly typical sequences



- 上述定理说明，随着 $n \uparrow$，接收矢量几乎只能是与发送码字联合典型的序列，取其他序列的概率 $\to 0$。

  这样我们有译码方法：取与接收矢量联合典型的码字作为译码器的输出。（在条件 y^n 下 x^n 的典型序列集中）

Example 6.8.1. *Binary source and BSC*

- Consider a binary source x^n passes through a BSC with flipping probability p.
- Fix an arbitrary output sequence y^n, what x^n's are jointly typical?
- Write $w^n = y^n - x^n$, there should be approximately np 1's in w^n .
- How many of such x^n are there? $\doteq 2^{nH(p)}$
- Check: This set is much smaller than $|\mathcal{X}|^n = 2^n$.

6.9 The noisy-channel coding theorem

6.9.1 System model

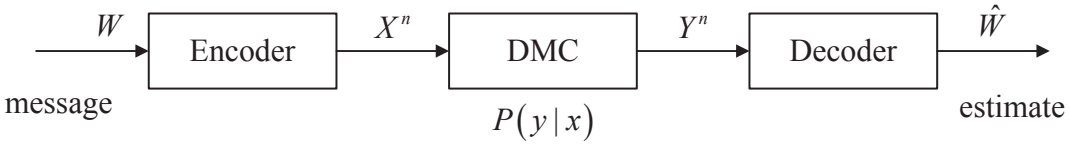


Figure 6.10: system model

- Denote the codebook $\mathcal{C} = \{x^n(w) = (x_1(w), x_2(w), \ldots, x_n(w)), 1 \leq w \leq M\}$
- A message $W = w \in \{1, 2, \ldots, M\}$ results in the transmitted codeword $x^n(w) \in \mathcal{C}$.
- Let $Y^n \backsim P(y^n|x^n)$ be the received sequence.
- On the basis of the received sequence Y^n, the decoder produces an integer $\hat{w}$. A decision error occurs if $\hat{w} \neq w$.
- The probability of block decoding error is

$$P_e = P(\hat{w} \neq w)$$

$$P_e(\mathbf{y}) = P(\hat{w} \neq w|\mathbf{y}) = 1 - P(\hat{w} = w|\mathbf{y})$$

- Decoding method: Jointly typical decoding, MAP, ML
- The code rate $R = \frac{1}{n}\log_2|\mathcal{C}| = \frac{k}{n}$ bits/channel use (bits per transmission)
- A rate R is said ro be achievable if there exists a sequence of $(\lceil nR \rceil, n)$ codes such that the maximal $P_e \to 0$ as $n \to \infty$ (for which the probability of error goes to 0 as $n \to \infty$).

6.9.2 Channel coding theorem

Theorem 6.9.1. *All rates below capacity C are achievable. Specifically, for every rate $R < C$ there exists a sequence of $(2^{nR}, n)$ codes with maximum $P_e \to 0$ as $n \to \infty$.*

Proof. (1) Construction of codebooks: *Random coding*

- Generate $M = 2^{nR}$ codewords, each with length n, i.i.d. from an input distribution $P(x)$.
  具体地说，我们按照 $P(x)$ 完全独立随机地从 $\mathcal{X}$ 中选取字母作为码字的符号。从 $x^n(1)$ 的第 1 个码符号开始一直到 $x^n(M)$ 的最后一个码符号，从而得到全部 2^{nR} 个码字，它们组成一个码（书）。

$$\mathcal{C} = \begin{bmatrix} x^n(1) \\ \vdots \\ x^n(M) \end{bmatrix} = \begin{bmatrix} x_1(1) & x_2(1) & \cdots & x_n(1) \\ \vdots & & & \\ x_1(M) & x_2(M) & \cdots & x_n(M) \end{bmatrix} \longrightarrow \begin{array}{l} \text{一个码字,} \\ \text{可能存在重复码字} \\ \text{只是概率很小} \end{array}$$

  Each entry in this matrix is generated i.i.d. according to $P(x)$. The probability that we generate a particular code is

$$P(\mathcal{C}) = \prod_{w=1}^{M} \prod_{i=1}^{n} P\left(x_i(w)\right)$$

- The total number of possible codebooks is $\left(|\mathcal{X}|^n\right)^M = |\mathcal{X}|^{n2^{nR}}$. (所有可能产生的码的总数)
  (When $|\mathcal{X}| = 2, n = 16, M = 2^8$, 总数$\approx 10^{1233}$)
- Shannon 不是计算某一特定好码的性能，而是计算这一码集的平均性能。

- All messages are equiprobable：$P(W = w) = 2^{-nR}$, for $w = 1, \ldots, 2^{nR}$

(2) Encoding：
Depend on the message $W = w$, transmit the codeword $x^n(w)$ through n usages of the channel.

(3) Decoding：*typical set decoding*

- For a given y^n, if there exists unique $x^n(w) \in \mathcal{C}$ such that $(x^n(w), y^n)$ is jointly typical (i.e., $(x^n(w), y^n) \in T_\epsilon^{(n)}$), then decode $\hat{W} = w$.

$$\text{Note:} \quad Y^n \sim P\left(y^n | x^n(w)\right) = \prod_{i=1}^{n} P(y_i | x_i(w))$$

- If no such w exists or if there is more than one such, then an error is declared.

(4) Calculate the probability of error averaged over all codewords in the codebook and averaged over all codebooks (i.e., over the ensemble of the codes). If 平均错误概率 is small, then there exists one code with smaller prob. of error.

$$\begin{aligned} P_e &= \sum_{\mathcal{C}} P(\mathcal{C}) P_e(\mathcal{C}) \\ &= \sum_{\mathcal{C}} P(\mathcal{C}) \cdot \sum_{w} P\left(x^n(w)\right) P\left(\text{error} | x^n(w)\right) \\ &= \sum_{\mathcal{C}} P(\mathcal{C}) \cdot \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} P_{e|w}(\mathcal{C}) \\ &= \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \sum_{\mathcal{C}} \underbrace{P(\mathcal{C}) P_{e|w}(\mathcal{C})}_{\text{在码上平均}} \end{aligned}$$

$$\left(P_{e|w}(\mathcal{C}) = P\left(\text{error} | \mathcal{C}, W = w\right)\right)$$

- Random coding creates symmetry. So the average prob. of error averaged over all codes does not depend on the particular index that was sent.
- So we can assume the message $W = 1$ was sent. Thus

$$P_e = \sum_{\mathcal{C}} P(\mathcal{C}) P_{e|1}(\mathcal{C}) = P_{e|1} = P(\text{error}|w = 1)$$

(5) Using Joint AEP

- Define the event

$$E_i = \left\{ (x^n(i), y^n) \in T_\epsilon^{(n)} \right\}, \ i \in \{1, 2, \ldots, 2^{nR}\}$$
$$E_i^c = \left\{ (x^n(i), y^n) \notin T_\epsilon^{(n)} \right\}$$

- Then

$$\begin{aligned} P(\text{error}|w = 1) &= P(E_1^c \cup E_2 \cup \cdots \cup E_M) \\ &\leq P(E_1^c) + \sum_{i=2}^{M} P(E_i) \end{aligned}$$

- In computing the Prob. of E_i, what the distribution of X^n, Y^n we should use?
  - For E_1, X^n, Y^n are drawn from the joint distribution of $P(x^n, y^n)$.
  - For $E_i, i \geq 2$, $X^n, Y^n \sim P(x^n)P(y^n)$.
    ($X^n(1)$ and $X^n(i)$ are independent, so are Y^n and $X^n(i),\ i \neq 1$)
- For large enough n, $P(E_1^c) \leq \epsilon$; for $i \geq 2$, $P(E_i) \leq 2^{-n[I(X;Y)-3\epsilon]}$. Therefore,

$$\begin{aligned} P_e = P(\text{error}|w = 1) &\leq \epsilon + \sum_{i=2}^{M} 2^{-n[I(X;Y)-3\epsilon]} \\ &= \epsilon + (2^{nR} - 1) \cdot 2^{-n[I(X;Y)-3\epsilon]} \\ &\leq \epsilon + 2^{nR} \cdot 2^{-n[I(X;Y)-3\epsilon]} \\ &= \epsilon + 2^{-n[I(X;Y)-R-3\epsilon]} \end{aligned}$$

- If $R < I(X;Y) - 3\epsilon$, we can choose n large enough such that $2^{-n[I(X;Y)-R-3\epsilon]} \leq \epsilon$. Then we have $P_e \leq 2\epsilon \triangleq \delta$.

(6) Choose $P(x)$ to be $P^*(x)$, the distribution on X that achieves capacity, then $R < I(X;Y) \longrightarrow R < C$. As long as $R < C = \max I(X;Y)$, the rate is achievable.

(7) The average performance of the random code is "good", so that exists a "good" code.

□

Remaining questions:

1) Long codewords are good, but how long?
What is the performance for finite codeword length?

2) Random codes seems good, but how do we decode?

- Is there any better structured code that can do as well?
- The randomness in coding is a computational trick or a fundamental requirement to achieve capacity?

6.10 Converse of coding theorem

- Recall Fano's inequality:
  For any r.v.'s X and Y, we try to guess X by $\hat{X} = g(Y)$. The error probability $P_e = P(\hat{X} \neq X)$ satisfies

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|Y)$$

- Converse of coding theorem:
  Now consider trying to guess the message W based on the observation Y^n. Let $P_e^{(n)} = P(\hat{W} \neq W)$. Define

$$E = \begin{cases} 1, & \text{if } \hat{W} \neq W \\ 0, & \text{if } \hat{W} = W \end{cases}$$

  Using similar derivation and noting that $\log_2 |\mathcal{W}| = \log_2 |\mathcal{X}| = nR$, we have

$$H(W|Y^n) \leq 1 + P_e^{(n)} nR$$

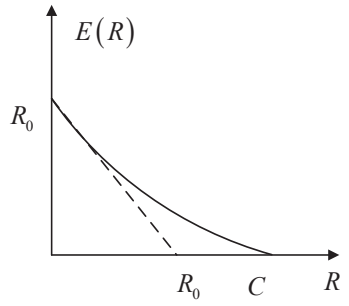
$$\begin{aligned} nR = H(W) &= H(W|Y^n) + I(W; Y^n) \\ &\leq H(W|Y^n) + I(X^n(W); Y^n) \\ &\leq 1 + P_e^{(n)} nR + I(X^n; Y^n) \\ &\leq 1 + P_e^{(n)} nR + nC \end{aligned}$$

  Rearrange, $P_e^{(n)} \geq 1 - \frac{C}{R} - \frac{1}{nR}$.
  If $R > C$, for n large enough, $P_e^{(n)}$ is bounded away from 0.

- Strong (converse to the) coding theorem — Gallager bound
  Encoding: Random coding
  Decoding: ML

$$P_e^{(n)} \leq 2^{-NE(R)}$$

  其中 $E(R) \triangleq \max_{0<\rho<1} \max_{Q(x)} [E_0(\rho, Q) - \rho R]$ 称为随机编码错误指数。



6.11 Joint source-channel coding

6.11.1 Compare source and channel coding

- Notation

- An i.i.d. source sequence V^n.
- Source coder $\mathcal{V}^n \to \{0,1\}^*$, maps source sequence into binary sequence U^*.
- Channel coder $\{0,1\}^* \to \mathcal{X}^m$, maps binary sequence U^* into transmitted sequence X^m.
- Receiver receives Y^m.
- Channel decoder recovers $\hat{U}^*$.
- Source decoder guess $\hat{V}^n$.

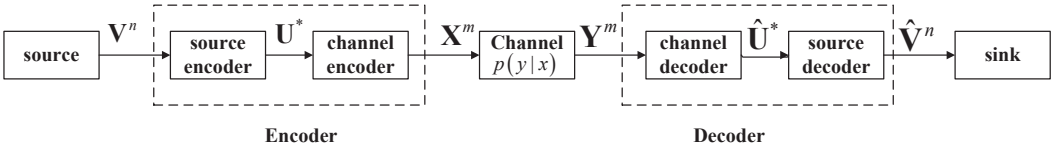


Figure 6.11: 系统框图

- Source coding: use less bits to represent every possible source sequence – *sphere covering* in $\mathcal{V}^n$.
- Channel coding: use m transmitted symbol to represent as many as possible bits, while having small P_e – *sphere packing* in $\mathcal{X}^m$.
- Source coding reduces redundancy: $|\mathcal{V}|^n \to 2^{nH(V)}$.
  Example: Bern(p) sequence: n bits $\to nH(p)$.
- Channel coding adds redundancy: $2^{mI(X;Y)} \to 2^{mH(X)}$.
  Example: BSC: $m\,(1 - H(p)) \to m$ bits.

6.11.2 Joint source-channel coding (The source-channel separation theorem)

- Overall goal: transmit i.i.d. source V^n (that satisfies the AEP) over a DMC $P(y|x)$ reliably:

$$V^n \to X^m \to Y^m \to \hat{V}^n$$
$$P_e^{(n)} = P(V^n \neq \hat{V}^n) \to 0$$

- **Source-channel coding theorem:** This can be done whenever $H(V) < C$.

  Proof. We will encode only $V^n \in T_\epsilon$; all other sequences will result in an error.
  可达性:

$$P_e^{(n)} \leq \underbrace{P\left(V^n \notin T_\epsilon^{(n)}\right)}_{V^n \text{ is not correctly encoded}} + \underbrace{P\left(\hat{V}^n \neq V^n | V^n \in T_\epsilon^{(n)}\right)}_{U^{nR} \text{ is not correctly recoverd}}$$
$$\leq \epsilon + \epsilon = 2\epsilon$$

$$(\text{if } n \to \infty, \text{and } H(V) < R < C)\ (H(V) + \epsilon = R < C)$$

□

- The two-stage process/method is as good as one stage process. So source and channel coding can be done separately.

- Converse of source-channel coding theorem:
  Joint source-channel encoder and decoder:

$$\begin{cases} X^n(V^n) : \mathcal{V}^n \to \mathcal{X}^n & \text{(take } m = n\text{)} \\ \hat{V}^n(Y^n) : \mathcal{Y}^n \to \mathcal{V}^n \end{cases}$$

  Question: Is it possible to reliably transmit a source with $H(V) > C$?
  (We will show that $P_e^{(n)} \to 0$ implies $H(V) \le C$)

  Proof. By Fano's inequality, $H(V^n|\hat{V}^n) \le 1 + P_e^{(n)} \underbrace{n \log |\mathcal{V}|}_{\log |\mathcal{V}|^n}$

  Now

$$\begin{aligned} H(V) &= \frac{1}{n} H(V^n) \\ &= \frac{1}{n} H(V^n|\hat{V}^n) + \frac{1}{n} I(V^n; \hat{V}^n) \\ &\le \frac{1}{n} \left(1 + P_e^{(n)} n \log |\mathcal{V}|\right) + \frac{1}{n} I(V^n; \hat{V}^n) \qquad (\because \text{Markov chain}) \\ &\le \frac{1}{n} + P_e^{(n)} \log |\mathcal{V}| + C \qquad \text{(memoryless)} \end{aligned}$$

  So $P_e^{(n)} \to 0 \Longrightarrow H(V) \le C$. $(n \to \infty)$ □

Theorem 6.11.1 (Source-channel coding theorem)**.** *For an i.i.d source V^n and the DMC $p(y|x)$, as long as $H(V) < C$, there exists a source channel code with $P_e^{(n)} \to 0$. Conversely, if $H(V) > C$, the probability of error is bounded away from 0. It is not possible to send the source over the channel with arbitrarily low prob. of error.*

- Applies for general cases:
  - source with memory
  - channel with memory
- It is not always true :

  (1) Channel is correlated with the source
  (2) Multiple access with correlated sources
  (3) Broadcasting channels (non-degraded)

  } separation theorem does not hold

6.12 Channel with feedback

- Does feedback help us to transmit higher data rate?
- Perfect feedback: Assume all previously received signals $(Y_1, \ldots, Y_{i-1})$ is available at the transmitter at time i.
  – Why do we ignore the possible errors in the feedback link?

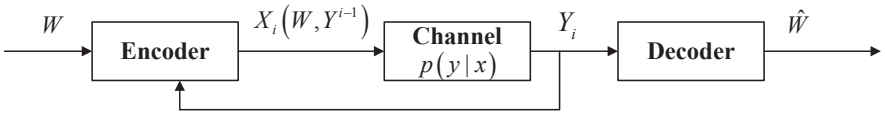


Figure 6.12: DMC with feedback

6.12.1 Feedback capacity

- Feedback code: A $(2^{nR}, n)$ feedback code is a sequences of mapping

$$X_i = X_i(W, Y^{i-1}), \quad i = 1, \ldots, n$$

  A decoder recover W as $\hat{W} = g(Y^n)$.

- DMC with feedback:

- A rate is available if there exists a sequences of $(2^{nR}, n)$ codes such that the probability of error $P_e^{(n)} = P(\hat{W} \neq W) \to 0$.

- The capacity with feedback, C_{FB}, of a DMC is the supremum of all rates achievable with feedback codes.
  Obviously, $C_{\text{FB}} \geq C$. ($\because$ a non-feedback code is a special case of a feedback code)

Theorem 6.12.1. *For a DMC,*

$$C_{\text{FB}} = C = \max_{P(x)} I(X;Y)$$

—Feedback does not improve the capacity at all.

Proof. Repeat the converse of the coding theorem without feedback:

$$\begin{aligned} nR = H(W) &= H(W|Y^n) + I(W;Y^n) \\ &\leq H(W|Y^n) + I(X^n;Y^n) \\ &\leq 1 + P_e^{(n)} nR + I(X^n;Y^n) \\ &\leq 1 + P_e^{(n)} nR + nC \end{aligned}$$

What is wrong?

$$\begin{aligned} I(X^n;Y^n) &= H(Y^n) - H(Y^n|X^n) \\ &\leq H(Y^n) + \sum_{i=1}^{n} H(Y_i|Y_1^{i-1}, X^n) \\ &\quad \downarrow \quad \text{with feedback, this does not hold since } Y_i \text{ depends on the future } X's \\ &\leq H(Y^n) + \sum_{i=1}^{n} H(Y_i|X_i) \end{aligned}$$

correct:

$$\begin{aligned}
I(W;Y^n) &= H(Y^n) - H(Y^n|W) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|Y_1^{i-1}, W) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|Y_1^{i-1}, W, X_i) \quad (\text{since } X_i \text{ 是} Y_1^{i-1} \text{和} W \text{的函数}) \\
&= H(Y^n) - \sum_{i=1}^{n} H(Y_i|X_i) \\
&\le \sum_{i=1}^{n} H(Y_i) - \sum_{i=1}^{n} H(Y_i|X_i) \\
&\le nC
\end{aligned}$$

□

Example 6.12.1. *BEC with feedback (ARQ)*

- Capacity of BEC: $C = 1 - \alpha$
- Consider a BEC with perfect feedback. Feedback reveals if a transmission is successful.
- Design a feedback code as:
  - transmit a new bit if the last transmission is successful. (收到ACK)
  - retransmit if it failed. (NAK)
- The average (total) number of channel uses is

$$1 + \alpha + \alpha^2 + \cdots = \frac{1}{1-\alpha}$$

- Average number of bits transmitted per channel use is

$$\frac{1}{\frac{1}{1-\alpha}} = 1 - \alpha$$

- The capacity is not improved at all, but *the code is greatly simplified.*
- Discussions: Is this the only reason that feedback is used?
  - DMC does not benefit from feedback
  - General channels with memory
    $\left\{ \begin{array}{l} \text{Wireless channels} \\ \text{network protocols} \end{array} \right.$

6.13 小结

(1) Source coding

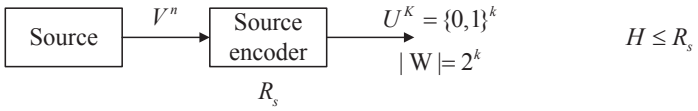


Figure 6.13: Source coding

(2) Channel coding

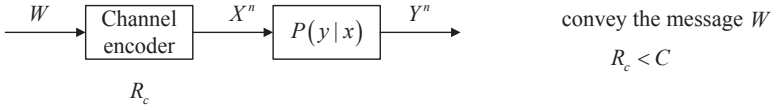


Figure 6.14: Channel coding

(3) Separation of source and channel coding: convey the inform source V^n

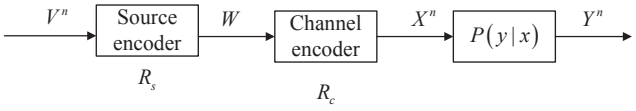


Figure 6.15: Separation of source and channel coding

This is possible 只要源码字个数 $2^{nR_s} \leqslant$ 信道码字个数 2^{nR_c}.
能可靠地convey信源 V^n: if $nH \leqslant nR_s \leqslant nR_c \leqslant nC$，即 $H \leqslant R_s \leqslant R_c \leqslant C$.

(4) Joint source-channel coding

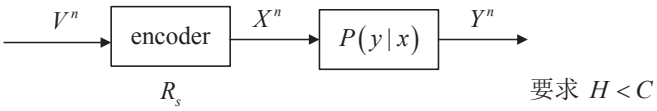


Figure 6.16: Joint source-channel coding

所以，信源、信道可分开编码 without loss of optimality.

Note: 尽管信源、信道分开编码可达到渐近最佳性，但是对于有限码长，联合编码能取得reduced prob. of error.

Chapter 7

The Gaussian Channel

- Discrete-time memoryless channel

$$\begin{array}{ccc} p(\mathbf{y}|\mathbf{x}) & = & \prod_{i=1}^{n} p(y_i|x_i) \\ \uparrow & & \uparrow \\ \text{pdf} & & \text{pdf} \end{array}$$

  例：$x_i \in \mathcal{X} \subset \mathbb{R}, y_i \in \mathcal{Y} = \mathbb{R}$

- For memoryless channels:

$$I(X^n; Y^n) \leq \sum_{i=1}^{n} I(X_i; Y_i) \leq nC$$

$$C \triangleq \max_{P(x)} I(X;Y) = \max_{P(x)} [h(Y) - h(Y|X)]$$

7.1 Additive White Gaussian Noise (AWGN) Channel

- Consider the channel

$$Y = X + Z$$

  with power constraint $\mathbb{E}[X^2] \leq P$, and $Z \sim \mathcal{N}(0, \sigma^2)$.

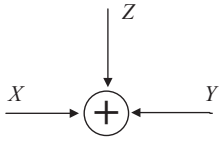


Figure 7.1:

Definition 7.1.1. *The capacity of the Gaussian channel with power constraint P is defined as*

$$C = \max_{P(x):\mathbb{E}[X^2] \leq P} I(X;Y)$$

- Calculation:

$$\left.\begin{aligned} I(X;Y) &= h(Y) - h(Y|X) \\ &= h(Y) - h(X+Z|X) \\ &= h(Y) - h(Z) \end{aligned}\right\} \text{since } Z \text{ is independent of } X.$$
$$= h(Y) - \frac{1}{2}\log 2\pi e\sigma^2$$

Since

$$\begin{aligned}\mathbb{E}[Y^2] = \mathbb{E}[(X+Z)^2] &= \mathbb{E}[X^2] + 2\mathbb{E}[X]\mathbb{E}[Z] + \mathbb{E}[Z^2] \\ &= \sigma_x^2 + \sigma^2 \leq P + \sigma^2\end{aligned}$$

and recalling that the normal maximizes the entropy for a given variance, we have

$$h(Y) \leq \frac{1}{2}\log_2 2\pi e(P+\sigma^2), \text{ with equality if } Y \sim \mathcal{N}(0, P+\sigma^2)$$

Thus,

$$\begin{aligned} I(X;Y) &\leq \frac{1}{2}\log_2 2\pi e(P+\sigma^2) - \frac{1}{2}\log_2 2\pi e\sigma^2 \\ &= \frac{1}{2}\log_2\left(1 + \frac{P}{\sigma^2}\right)\end{aligned}$$

Hence,

$$\begin{aligned} C &= \max_{\mathbb{E}[X^2]\leq P} I(X;Y) \\ &= \frac{1}{2}\log_2\left(1+\frac{P}{\sigma^2}\right) \\ &= \frac{1}{2}\log_2(1+\text{SNR}) \qquad \text{bits/symbol or bits/channel use}\end{aligned}$$

and the maximum is attained when $X \sim \mathcal{N}(0, P)$.

- Note: Maximize $h(Y)$:

$$\left.\begin{aligned} &Y \sim \mathcal{N}(0, P+\sigma^2) \\ &Z \sim \mathcal{N}(0, \sigma^2) \\ &Y = X + Z \end{aligned}\right\} \Rightarrow X \sim \mathcal{N}(0, P)$$

7.2 Capacity as an Estimation Problem

Consider

$$\begin{aligned} I(X;Y) &= h(X) - h(X|Y) \\ &= h(X) - h\left(X - g(Y)|Y\right)\end{aligned}$$

for any function $g(\cdot)$.

- Choose X to be $\mathcal{N}(0, \sigma_x^2)$ distributed.

- Choose $g(\cdot)$ to be the Linear Minimum Mean-Square Error estimate of X. (L-MMSE)
  In the Gaussian case:

$$g(Y) = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2} Y$$

$$\text{Var}\left[X - g(Y)\right] = \frac{\sigma_x^2 \sigma_z^2}{\sigma_x^2 + \sigma_z^2}$$

  and $\underbrace{X - g(Y) \text{ is independent of } Y.}_{\text{正交性原理}}$

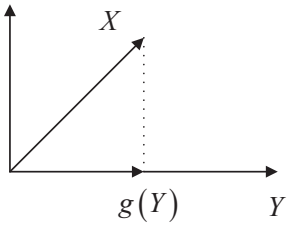


Figure 7.2:

  Now

$$\begin{aligned} I(X;Y) &= \frac{1}{2}\log\left(2\pi e \sigma_x^2\right) - \frac{1}{2}\log\left(2\pi e \frac{\sigma_z^2 \sigma_x^2}{\sigma_z^2 + \sigma_x^2}\right) \\ &= \frac{1}{2}\log\left(1 + \frac{\sigma_x^2}{\sigma_z^2}\right) \end{aligned}$$

- Discussions:
  - Denote $\hat{x} = g(Y)$, we call $\hat{x}$ a sufficient statistics if

$$\begin{cases} X \longrightarrow Y \longrightarrow \hat{X} \\ X \longrightarrow \hat{X} \longrightarrow Y \end{cases}$$

  - In Gaussian estimation problems, the L-MMSE estimate $\hat{X}$ satisfies this.

$$I(X;Y) = I(X;\hat{X})$$

L-MMSE:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}$$

$$\hat{x} = \mu_x + \mathbf{K}_{xy}\mathbf{K}_y^{-1}(y - \mu_y)$$

7.3 Non-Gaussian noise channel

- For general distributions of Z with $\mathbb{E}[Z] = 0$, $\text{Var}[Z] = \sigma^2$, consider the channel $Y = X + Z$.

$$I(X;Y) = h(Y) - h(Z)$$

$$h(Y) \leq \frac{1}{2}\log 2\pi e(P + \sigma^2)$$

$$h(Z) = \frac{1}{2}\log 2\pi e\bar{\sigma}^2 \quad (\text{熵功率})$$

Therefore,

$$\begin{aligned} I(X;Y) &\leq \frac{1}{2}\log 2\pi e(P+\sigma^2) - \frac{1}{2}\log 2\pi e\bar{\sigma}^2 \\ &= \frac{1}{2}\log_2\left(\frac{P+\sigma^2}{\bar{\sigma}^2}\right) \end{aligned}$$

$$h(Z) \leq \frac{1}{2}\log 2\pi e\sigma^2 \Rightarrow C \geq \frac{1}{2}\log\left(1+\frac{P}{\sigma^2}\right)$$

Equality holds only for the Gaussian noise: **AWGN is the worst noise.**

- A mutual information game:
  - The transmitter tries to maximize $I(X;Y)$ by choosing $P(X)$, subject to a power constraint $\mathbb{E}[X^2] = \sigma_x^2 \leq \rho$.
  - The channel (jammer) tries to minimize $I(X;Y)$ by choosing a noise $P(Z)$, subject to $\mathbb{E}[Z^2] = \sigma_z^2$.
  - **Saddle point:** the optimal input is Gaussian; the worst noise is also Gaussian.

7.4 Coding theorem for Gaussian channels

- More realistic: Consider the channel

$$Y_i = X_i + Z_i$$

where Z_i is i.i.d. $\mathcal{N}(0,\sigma_z^2)$, and the input has power constraint

$$\frac{1}{n}\sum_{i=1}^{n} x_i^2 \leq \sigma_x^2$$

Theorem 7.4.1. $C = \frac{1}{2}\log\left(1+\frac{\sigma_x^2}{\sigma_z^2}\right)$ *is the maximum achievable rate.*

Proof. (1) Generate random codebook with 2^{nR} codewords, each of length n, with iid $\mathcal{N}(0,\sigma_x^2-\epsilon)$ entries.

(2) Encoding: $w \longrightarrow X^n(w)$

(3) Jointly typical decoding.

(4) Compute the error probability:
Without loss of generality, assume that the 1st codeword $\mathbf{X}_1 = X^n(1)$ is transmitted.

- If the generated codeword violates the power the power constraint, declare an error.

$$E_0 = \left\{\frac{1}{n}\sum_{i=1}^{n} X_i^2(1) \geq \sigma_x^2\right\}$$

- Define $E_i = \{(X^n(i), Y^n) \text{ is jointly typical}\}$
  Then

$$P(E_1) \to 1$$
$$P(E_i) \approx 2^{-nI(X;Y)} \quad \text{for } i \neq 1$$

$$\begin{aligned} P_e^{(n)} = P(e|w=1) &= P\left(E_0 \cup E_1^c \cup E_2 \cdots \cup E_{2^{nR}}\right) \\ &\leq P\left(E_0\right) + P\left(E_1^c\right) + \sum_{i=2}^{2^{nR}} P\left(E_i\right) \\ &\leq \epsilon + \epsilon + \sum_{i=2}^{2^{nR}} 2^{-n[I(X;Y)-3\epsilon]} \\ &= 2\epsilon + \left(2^{nR}-1\right) 2^{-n[I(X;Y)-3\epsilon]} \\ &\leq 2\epsilon + 2^{nR} \cdot 2^{-n[I(X;Y)-3\epsilon]} \\ &= 2\epsilon + 2^{-n[I(X;Y)-R-3\epsilon]} \\ &\leq 3\epsilon \end{aligned}$$

  for n sufficiently large and $R < I(X;Y) - 3\epsilon$.

□

- **Converse to the coding theorem**: (If $P_e \to 0$, then $R \leq C$)

$$\begin{aligned} nR = H(W) &= I(W;Y^n) + H(W|Y^n) \\ &\leq I(W;Y^n) + 1 + nRP_e^{(n)} \\ &\leq I(X^n;Y^n) + 1 + nRP_e^{(n)} \\ &\leq \sum_{i=1}^{n} I(X_i;Y_i) + 1 + nRP_e^{(n)} \end{aligned}$$

  To drive $P_e^{(n)} \to 0$, need

$$R \leq \frac{1}{n}\sum_{i=1}^{n} I(X_i;Y_i)$$

  – **Key:** Individual power constraint v.s. average power constraint
    Let $P_i = \frac{1}{2^{nR}} \sum_w x_i^2(w)$, and $\frac{1}{n}\sum_i P_i \leqslant P = \sigma_x^2$. Then

$$\begin{aligned} R &\leq \frac{1}{n}\sum_{i=1}^{n} I(X_i;Y_i) \\ &\leq \frac{1}{n}\sum_i \frac{1}{2}\log\left(1+\frac{P_i}{\sigma_z^2}\right) \\ &\leq \frac{1}{2}\log\left(1+\frac{1}{n}\frac{\sum_i P_i}{\sigma_z^2}\right) \text{ (By Jensen's 不等式, since } f(x) = \frac{1}{2}\log(1+x) \\ &\qquad\qquad \text{is a concave function of } x) \\ &\leq \frac{1}{2}\log\left(1+\frac{P}{\sigma_z^2}\right) \end{aligned}$$

- **Sphere packing in the Gaussian channel**
  Sphere packing — Useful geometric approach to channel capacity, an optimal input distribution.
  - For a transmitted codeword x^n, the set of y^n that are jointly typical with x^n is a sphere centered at x^n with radius $\sqrt{n\sigma_z^2}$.
  - The volume of an n-dim sphere of radius r

$$\text{Vol}\,(n,r) = \frac{\left(\pi r^2\right)^{n/2}}{(n/2)!} = A_n r^n$$

  - The maximum data rate is

$$R = \frac{1}{n}\log\frac{A_n\left(n\sigma_y^2\right)^{\frac{n}{2}}}{A_n\left(n\sigma_z^2\right)^{\frac{n}{2}}} = \frac{1}{2}\log\left(\frac{\sigma_y^2}{\sigma_z^2}\right)$$

    Need Y^n to be iid Gaussian.
  - For x^n and y^n independently chosen according to the marginal distribution, the probability that they are jointly typical is

$$\frac{A_n\left(n\sigma_Z^2\right)^{\frac{n}{2}}}{A_n\left(n\sigma_Y^2\right)^{\frac{n}{2}}} \approx 2^{-nI(X;Y)}$$

  - For a codebook of $x^n(w), w = 1,\ldots,2^{nR}$, and an independently chosen y^n, the probability that there exists w, such that $(x^n(w), y^n)$ are jointly typical

$$2^{nR}\frac{A_n\left(n\sigma_Z^2\right)^{\frac{n}{2}}}{A_n\left(n\sigma_Y^2\right)^{\frac{n}{2}}} \approx 2^{-n[I(X;Y)-R]} \longrightarrow 0$$

- **A Greedy way to spend power (Rate-splitting):**
  Consider the AWGN channel, with power constraint

$$\frac{1}{n}\sum_i x_i^2 \le P$$

  The transmitter
  - Split the power into $P = P_1 + P_2$, and the data stream into $R = R_1 + R_2$, and assign to two virtual users.
  - Each user has a random codebook, generate codewords X_1^n and X_2^n. The superposition $X^n = X_1^n + X_2^n$ is transmitted.
    The receiver receives

$$\mathbf{Y} = \mathbf{X}_1 + \mathbf{X}_2 + \mathbf{Z}$$

  - First treat $\mathbf{X}_2 + \mathbf{Z}$ as the noise, can support

$$nR_1 \le I(\mathbf{X}_1;\mathbf{Y}) = \frac{n}{2}\log\left(1 + \frac{P_1}{P_2+\sigma_z^2}\right)$$

  - Upon decoding $\mathbf{X}_1$, subtract $\mathbf{X}_1$ from $\mathbf{Y}$, then decode $\mathbf{X}_2$. Can support rate

$$nR_2 \le I(\mathbf{X}_2;\mathbf{Y}|\mathbf{X}_1) = \frac{n}{2}\log\left(1 + \frac{P_2}{\sigma_z^2}\right)$$

  - The overall data rate is $R_1 + R_2$, the channel capacity is achieved.

- Discussions:

  - To generalize, we can have many virtual users.
    Useful in multiple access channels.
  - Highly depends on the fact that the sum of independent Gaussian r.v.'s is Gaussian.

7.5 Parallel Gaussian Channels

- Assume that we have a set of Gaussian channels in parallel,

$$Y_i = X_i + Z_i, \quad i = 1, 2, \ldots, K$$

  where $Z_i \sim \mathcal{N}(0, \sigma_i^2)$ are independent of each other.

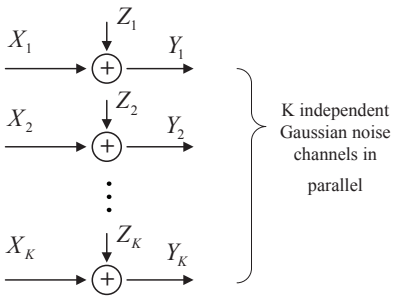


Figure 7.3:

- The power constraint:

$$\sum_{i=1}^{K} P_i = \mathbb{E}\left(\sum_{i=1}^{K} X_i^2\right) \le P$$

- Consider how to distribute the power among the various channels so as to maximize the total capacity.

$$I(X_1, \ldots, X_K; Y_1, \ldots, Y_K) \le \sum_{i=1}^{K} I(X_i; Y_i) \le \sum_i \frac{1}{2} \log\left(1 + \frac{P_i}{\sigma_i^2}\right)$$

  Equality is achieved by independent $\{X_i\}$, $X_i \sim \mathcal{N}(0, P_i)$

$$C = \max_{P(\mathbf{x}):\sum \mathbb{E}[X_i^2] \le P} I(X_1, \ldots, X_K; Y_1, \ldots, Y_K) = \frac{1}{2} \sum_{i=1}^{K} \log_2\left(1 + \frac{P_i}{\sigma_i^2}\right)$$

- Now the problem is reduced to find the optimal power allocation:

$$\begin{cases} \text{Maximize} & \frac{1}{2}\sum_i \log_2\left(1 + \frac{P_i}{\sigma_i^2}\right) \\ \text{s.t.} & \sum_i P_i \le P \end{cases}$$

- Using Lagrange multiplier λ, the object function is

$$J(P_1, \ldots, P_K) = \frac{1}{2} \sum_i \log\left(1 + \frac{P_i}{\sigma_i^2}\right) + \lambda \left(\sum_i P_i\right)$$

$$\frac{\partial J}{\partial P_i} = \frac{1}{2} \cdot \frac{1}{P_i + \sigma_i^2} + \lambda = 0 \Rightarrow P_i + \sigma_i^2 = -\frac{1}{2\lambda} \triangleq \gamma(\text{常数})$$

So $P_i = \gamma - \sigma_i^2$.
Note that the P_i's must be non-negative, we choose

$$P_i = (\gamma - \sigma_i^2)^+ = \begin{cases} \gamma - \sigma_i^2, & \text{if } \gamma \geq \sigma_i^2 \\ 0, & \text{otherwise} \end{cases}$$

where constant γ is solved by that

$$\sum_i (\gamma - \sigma_i)^+ = P$$

Therefore, the capacity of this channel is given by

$$C = \sum_{i=1}^{K} \frac{1}{2} \log\left[1 + \frac{(\gamma - \sigma_i^2)^+}{\sigma_i^2}\right]$$

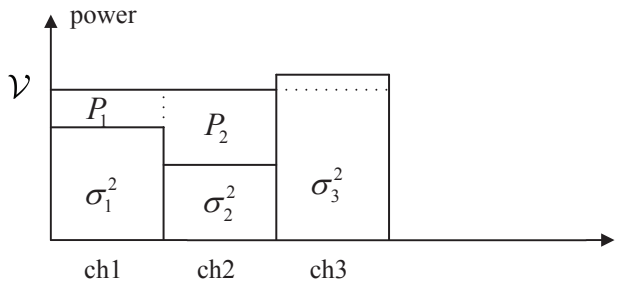


Figure 7.4: Water-pouring for parallel channels

7.6 Waveform channel (Bandlimited AWGN channels)

- **Channel model:** $y(t) = x(t) + z(t)$
where $z(t)$ is a Gaussian random process. Any set of samples are iid Gaussian distributed with zero mean and variance $\mathbb{E}[z(t)^2] = \frac{N_0}{2}$.

- Front end of the receiver: Provide sufficient statistics for bandlimited transmitted signals.

  - Assume that the $x(t)$ has non-zero frequency components only in $[-W, W]$. Then $x(t)$ and $y(t)$ can be sampled at $T_s = \frac{1}{2W}$.
  - Can be thought as a bank of matched filter matched to $\text{sinc}\left(t - \frac{n}{2W}\right)$, where $\text{sinc}(t) = \frac{\sin(2\pi W t)}{2\pi W t}$.

- **Discrete-time channel model:**

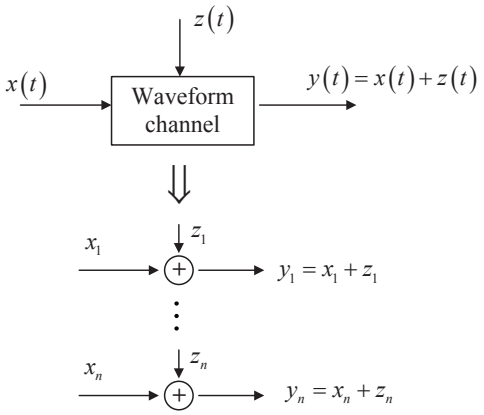


Figure 7.5:

- $2W$ samples per second.
- The noise power in each sample is $\sigma^2 = \frac{N_0}{2} \quad \left(= \frac{N_0 W}{2W}\right)$
- Average signal power (energy) per symbol: $P/2W$

We obtain $Y_i = X_i + Z_i$.

- The overall capacity in [bits/sec] is

$$C = 2W \cdot \frac{1}{2} \log\left(1 + \frac{P/2W}{N_0/2}\right) = W \log_2\left(1 + \frac{P}{N_0 W}\right) \quad (P\text{:Walt, } N_0\text{:Walt/Hz})$$

- As $W \to \infty$,

$$\begin{aligned} C_\infty &= \lim_{W\to\infty} W \log_2\left(1 + \frac{P}{N_0 W}\right) \\ &= \lim_{W\to\infty} \frac{P}{N_0}\left[\frac{N_0 W}{P} \log_2\left(1 + \frac{P}{N_0 W}\right)\right] \\ &= \lim_{W\to\infty} \frac{P}{N_0} \log\left(1+x\right)^{\frac{1}{x}} \qquad \ln\left(1+x\right)^{\frac{1}{x}} \overset{x\to 0}{\longrightarrow} 1 \\ &= \frac{P}{N_0} \log_2 e \text{ [bits/s]} \qquad \text{(capacity grows linearly with the power. Power-efficient regime.)} \end{aligned}$$

- The energy required to send 1 bit is $E_b = \frac{P}{C}$ (Walts × 1 sec = 焦耳)
发送 1 bit 所需最小 SNR:

$$\frac{E_b}{N_0} = \frac{1}{\log_2 e} = 0.693 \quad (-1.59 \text{ dB})$$

- 更一般地，in order to be capable of transmitting at rate R bits/s with an arbitrarily small error probability, we must have $R < C$; i.e.,

$$R < W \log_2\left(1 + \frac{P}{N_0 W}\right)$$

Define the spectral efficiency (bandwidth efficiency) as $\eta = R/W$ (bits/s/Hz). Then

$$\eta < \log_2\left(1 + \frac{P}{N_0 W}\right) = \log_2(1 + \text{SNR})$$

This result is called the *Shannon limit*. An equivalent statement is $\text{SNR} > 2^\eta - 1$.

- Using the relation $E_b = PT_b = \frac{P}{R}$, we obtain

$$\frac{E_b}{N_0} = \frac{P}{\eta W N_0} = \frac{\text{SNR}}{\eta}$$

Here, E_b denotes the average (received) energy per information bit. The Shannon limit in terms of E_b/N_0 may be expressed as

$$\frac{E_b}{N_0} > \frac{2^\eta - 1}{\eta}$$

As $\eta \to 0$, it approaches the *ultimate Shannon limit*: $E_b/N_0 > \ln 2$ (-1.59dB).

- Note that when the SNR is high, the capacity increases logarithmically with P: $C \approx W \log_2(\frac{P}{N_0 W})$. Increasing W yields a rapid increase in capacity. The system is in the *bandwidth-limited regime*.

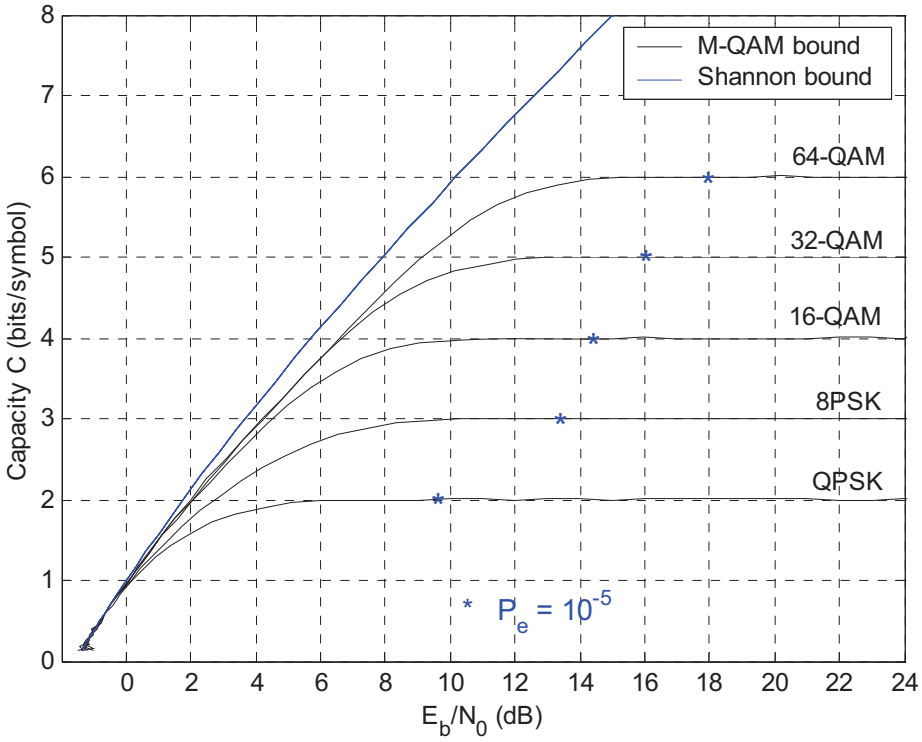


Figure 7.6: Shannon 容量限

- **Example:** 有线电话信道: $W = 3$ kHz, SNR $= 30$ dB, $C \approx 33$ kps. (Voice modem)

- Discussions:

  – Waveform channels are simply $2WT$ parallel channels.

  – White Gaussian noise allows us to look at the signal space with any chosen basis.

7.7 Gaussian channel with colored noise

Chapter 8

Rate Distortion Theory

8.1 Introduction

- Representation problem of a continuous r.v.
- **Quantization**
  X – r.v., $\hat{X}(X)$ – representation of X

  If we are given R bits to represent X, then $\hat{X}$ can take on 2^R values. Then problem is to find

  $$\begin{cases}\text{the optimum sets of values for } \hat{X}\\ \text{the regions associated with each value } \hat{X}\end{cases}$$

  For example, $X \sim \mathcal{N}(0, \sigma^2)$, minimize $\mathbb{E}\big[(X-\hat{X})^2\big]$
- 若传输的信息允许有一定失真，则信息率可以降低。

8.2 Distortion measure and the rate-distortion function

- Consider a source sequence $x^n = (x_1, x_2, \ldots, x_n) \sim$ i.i.d. $P(x)$ (or $u^n = (u_1, u_2, \ldots, x_n)$) and a reproduction sequence $\hat{x}^n = (\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_n)$ (or $v^n = (v_1, v_2, \ldots, v_n)$)
- 若 U, V 为离散随机变量，且字符集为 $\mathcal{U} = \mathcal{V} = \{a_1, a_2, \ldots, a_k\}$，则

  Definition 8.2.1. *A (single-letter) distortion function (measure) (非负函数) is a mapping*

  $$d : \mathcal{X} \times \hat{\mathcal{X}} \to \mathbb{R}^+$$
  $$(d : \mathcal{U} \times \mathcal{V} \to \mathbb{R}^+)$$

  失真函数：$d_{ij} = d(u = a_i, v = a_j)$ or $d(x_i, y_j)$, $d(u_i, v_j)$

  失真矩阵：

  $$[D] = \begin{bmatrix} d(u_1, v_1) & d(u_1, v_2) & \ldots & d(u_1, v_m) \\ d(u_2, v_1) & d(u_2, v_2) & \ldots & d(u_2, v_m) \\ \vdots & \vdots & \ddots & \vdots \\ d(u_n, v_1) & d(u_n, v_2) & \ldots & d(u_n, v_m) \end{bmatrix}$$

- 平均失真：

$$\begin{aligned}\bar{d} = \mathbb{E}\big[D(u,v)\big] &= \sum_u \sum_v P(u,v)d(u,v) \\ &= \begin{cases} \sum_i \sum_j Q_i P_{ji} d_{ij}, & P_{ji} = \Pr\{v = a_j | u = a_i\} \\ \int_u \int_v Q(u)P(v|u)d(u,v)dudv \end{cases}\end{aligned}$$

- 矢量间的平均失真：

$$\begin{aligned}\bar{d}_L &= \mathbb{E}\big[d(u^L, v^L)\big] \\ &= \frac{1}{L}\sum_{l=1}^{L} \mathbb{E}\big[d(u_l, v_l)\big] \\ &= \frac{1}{L}\sum_{l=1}^{L} \bar{d}_l\end{aligned}$$

- Hamming 失真 measure:

$$d(x,\hat{x}) = \begin{cases} 0, \text{ if } x = \hat{x} \\ 1, \text{ if } x \neq \hat{x} \end{cases}$$

$$\text{or} \quad d(u,v) = \begin{cases} 0, \text{ if } u = v \\ 1, \text{ if } u \neq v \end{cases}$$

$$d_{ij} = \begin{cases} 0, \text{ if } i = j \\ 1, \text{ if } i \neq j \end{cases}$$

Hamming 失真矩阵:

$$[D] = \begin{bmatrix} 0 & 1 & 1 & \dots & 1 \\ 1 & 0 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 0 \end{bmatrix}$$

- 平均 Hamming 失真:

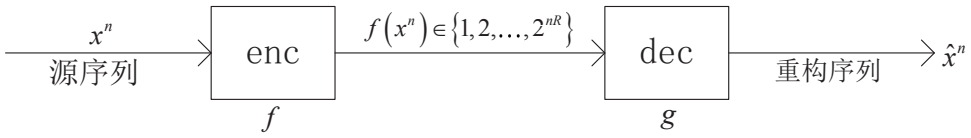
$$\begin{aligned}\mathbb{E}\big[d(x,\hat{x})\big] &= \sum_x \sum_{\hat{x}} P(x,\hat{x})d(x,\hat{x}) \\ &= P(x = \hat{x}) \cdot 0 + P(x \neq \hat{x}) \cdot 1 \\ &= P(x \neq \hat{x}) \triangleq P_e\end{aligned}$$

- Squared error distortion: $d(x,\hat{x}) = (x - \hat{x})^2$
  平均失真度量 $\rightarrow$ mean-square error

8.2.1 率失真编码

Definition 8.2.2. *A $(2^{nR}, n)$ rate-distortion code consists of an encoding function $f : \mathcal{X}^n \rightarrow \{1, 2, \dots, 2^{nR}\}$ and a decoding function $g : \{1, 2, \dots, 2^{nR}\} \rightarrow \hat{\mathcal{X}}^n$.*

$\hat{\mathcal{X}}^n$ 中的重构序列 $g\big(f(1)\big), g\big(f(2)\big), \dots, g\big(f(M)\big)$ 称为码字。

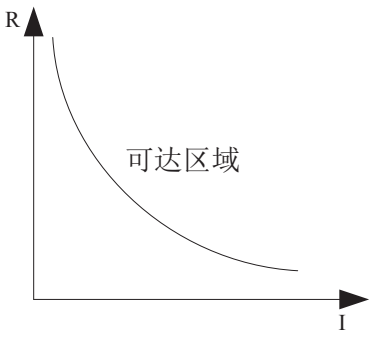


- The associated distortion is:

$$\begin{aligned} D &= \mathbb{E}\Big[d\Big(x^n, g\big(f(x^n)\big)\Big)\Big] \\ &= \sum_{x^n} P(x^n) d(x^n, \hat{x}^n) \end{aligned}$$

- 定义：A rate-distortion pair (R, D) is said to be achievable if there exists a sequence of $(2^{nR}, n)$ rate-distortion codes with $\lim\limits_{n\to\infty} \mathbb{E}\Big[d\Big(x^n, g\big(f(x^n)\big)\Big)\Big] \le D$.

- Rate-distortion region: closure of the set of achievable $R - D$ pairs. (凸的)



- The *rate-distortion function* $R(D)$ is the minimum of all rates R for a given distortion D such that (R, D) is achievable.

- The *distortion-rate function* $D(R)$ is the minimum of all distortions for a given rate R such that (R, D) is achievable.

- 可以证明，$R(D)$ is non-negative and non-increasing in D.

Definition 8.2.3. *The information rate-distortion function is defined by*

$$\begin{aligned} R^{(I)}(D) &= \min_{\hat{x}:\mathbb{E}\left[d(x,\hat{x})\right] \le D} I(X;\hat{X}) \\ &= \min_{P(\hat{x}|x):\sum\limits_{x,\hat{x}} P(x)P(\hat{x}|x)d(x,\hat{x}) \le D} I(X;\hat{X}) \qquad (\because P(x) \text{ is given}) \\ &= \min_{P(\hat{x}|x)} \left\{ \sum_x \sum_{\hat{x}} P(x)P(\hat{x}|x) \log \frac{P(\hat{x}|x)}{\sum\limits_x P(x)P(\hat{x}|x)} \right\} \end{aligned}$$

Theorem 8.2.1. *For an i.i.d. source,* $R(D) = R^{(I)}(D)$.

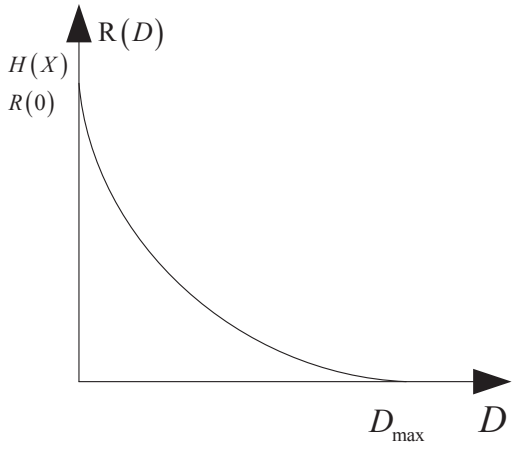
It states that the minimum coding rate at distortion D is $R^{(I)}(D)$.

- 率失真函数的性质:

- $R(D)$ 是 D 的非增函数
- $R(D)$ 是凸-U的：

$$R(D_\theta) \le \theta R(D_1) + (1-\theta)R(D_2)$$
$$\text{其中 } D_\theta = \theta D_1 + (1-\theta)D_2$$

- $0 \le R(D) < H(U)$
- $\lim\limits_{D\to 0} R(D) = H(U)$ (连续r.v. 的情况下，$R(D) \to \infty$)
- 对于 $D \ge D_{max}, R(D) = 0$



- 令最小平均失真为 D_{min}，则

$$D_{min} = \sum_x P(x) \min_{\hat{x}} d(x, \hat{x})$$

相当于在失真矩阵的每一行找出一个最小的 d，然后对这些最小的 d 求平均。

- $D_{min} = 0$ 时, 有 $R(0) = H(X)$。
- D_{max} 是 $R(D) = 0$ 时 D 的最小值，即

$$\begin{aligned} D_{max} &= \min_{p_{ji}\in\mathcal{P}} E\big[d(u,v)\big] \\ &\qquad \text{(其中 } \mathcal{P} \text{ 是使 } I(p_{ji}) = 0 \text{ 的全体 } P(v|u) \text{ 集合)} \\ \because I(p_{ji}) = 0 &\Leftrightarrow U \text{ 与 } V \text{ 独立} \to P(v|u) = P(v) \\ \therefore D_{max} &= \min_{P(v)} \sum_v P(v) \sum_u Q(u) d(u,v) \\ &= \min_{v\in V} \sum_u Q(u) d(u,v) \qquad \text{即，使 } P(v^*) = 1 \\ &= E\big[d(U, v^*)\big] \qquad v^* \text{ 是使 } D_{max} \text{ 最小的 } v \end{aligned}$$

- D_{max} 的另一定义:
  令 $\hat{x}^* = \arg\min\limits_{\hat{x}\in\hat{\mathcal{X}}} \mathbb{E}\big[d(X,\hat{x})\big] = \arg\min\limits_{\hat{x}} \sum\limits_x P(x) d(x,\hat{x})$
  定义 $D_{max} = \mathbb{E}\big[d(X,\hat{x}^*)\big]$
  - $\hat{x}^*$ 表示不管信源产生什么符号，若都把它恢复成某一个确定的再生值 $\hat{x}$，当这个再生值取 $\hat{x}^*$ 时，平均失真为最小，等于 D_{max}。

– 所以，D_{max} 也表示无需对信源作任何描述时，可以达到的最小失真，即 D_{max} 是 $R(D)=0$ 时的最小平均失真。

- 计算举例：
  $\mathcal{X}=\hat{\mathcal{X}}=\{0,1,2,3\}$ 且 $P(x=i)=\frac{1}{4}, i\in\mathcal{X}$. 失真矩阵 $d(x,\hat{x})$ 为

$$[d_{ij}]=\begin{bmatrix}0&1&1&1\\1&0&1&1\\1&1&0&1\\1&1&1&0\end{bmatrix}$$

求 $D_{max}, R(D_{max}), R(D)$.

解:

$$D_{min}=\sum_i P_i \min_j d_{ij}=0, \qquad \text{此时 } R(D_{min})=\log_2 4=2 \text{ bits}$$

$$D_{max}=\min_j \sum_i P_i d_{ij}=\frac{3}{4}, \qquad R(D_{max})=0$$

$$\because H(X|\hat{X})\le H(P_e)+P_e\log_2(|\mathcal{X}|-1)=H_2(D)+D\log_2 3$$
$$\therefore I(X;\hat{X})=H(X)-H(X|\hat{X})\ge 2-H(D)-D\log_2 3$$

$$R(D)=\min I(X;\hat{X})=\begin{cases}2-H_2(D)-D\log_2 3, & 0<D<\frac{3}{4}\\ 0, & D\ge\frac{3}{4}\end{cases}$$

注: $R^{(I)}(D)$ 的计算与容量 C 的计算有相似之处：

– $R^{(I)}(D)$ 是在信源分布 $P(X)$ 已知的条件下，在受失真限制的转移概率 $P(\hat{x}|x)$ 上互信息的最小值。（在矩阵变量上求）

– 而信道容量是在转移概率已知条件下，对不同的输入分布求互信息的最大值。（在概率矢量上求）

8.3 率失真函数的计算

1. Bernoulli source with Hamming distortion:

$$R(D)=\begin{cases}H(p)-H(D), & 0\leqslant D<\min(p,1-p)\\ 0, & D\geqslant\min(p,1-p)\end{cases}$$

Bernoulli源是一个二元无记忆源，其中输出‘1’的概率为 p，即 $P(X=1)=p$.

Proof: $X\sim Bernoulli(p), \quad \mathcal{X}=\hat{\mathcal{X}}=\{0,1\}, \quad p<\frac{1}{2}$

(思路：先找一个下限，再证明其可达，于是它就是所求的最小值)

$$\begin{aligned}I(X;\hat{X})&=H(X)-H(X|\hat{X})\\&=H(p)-H(X\oplus\hat{X}|\hat{X})\\&\ge H(p)-H(X\oplus\hat{X}) \quad ①\\&\ge H(p)-H(D) \quad ②\end{aligned}$$

Thus, $R(D) \geq H(P) - H(D)$.

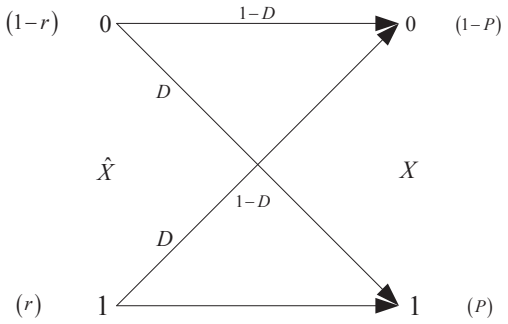
从①到②的过程是：$\because H(x \oplus \hat{x}) = H\big(P(x \neq \hat{x})\big)$, 而 $P(x \neq \hat{x}) = E\big[d(x, \hat{x})\big] \leq D$

$\therefore$ 当 $D \leq \frac{1}{2}$ 时, $H\big(P(x \neq \hat{x})\big) \leq H(D), \quad 0 \leq D \leq p \leq \frac{1}{2}$

下面证明上式中等号可成立（即下界可达）：

$\because I(X; \hat{X}) = I(\hat{X}; X)$,

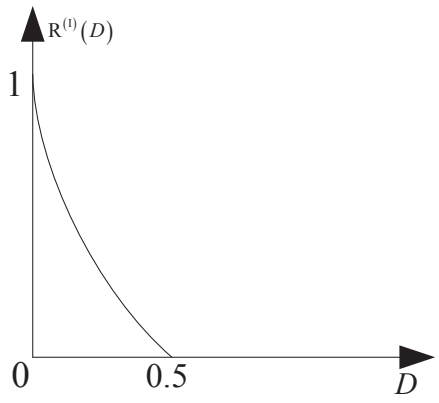
$\therefore$ 可研究反向信道。



Let $r = P(\hat{x} = 1)$, which is chosen such that $r(1 - D) + (1 - r)D = p$ or $r = \frac{p-D}{1-2D}, \quad p \geq D$

Then $H(X|\hat{X} = 0) = H(X|\hat{X} = 1) = H_2(D)$

$$\Rightarrow I(X; \hat{X}) = H(X) - H(X|\hat{X})$$
$$= H_2(p) - H_2(D)$$



(注：使等号成立的分布 $P(\hat{x}|x)$ 可利用Bayes公式由 $P(x|\hat{x}), P(\hat{x}), P(x)$ 获取)

If $D \geq p$, then we can achieve $R(D) = 0$ by setting $\hat{X} = 0$ with probability 1. In this case, $D = p$; Similarly, if $D \geq 1 - p$, we can achieve $D = 1 - p$ by letting $\hat{X} = 1$ with probability 1. (无需进行任何编码，$R = 0$)

Hence，for a binary source

$$R(D) = \begin{cases} H_2(p) - H_2(D), & 0 \leqslant \quad D < \min\{p, 1-p\} \\ 0, & \quad D \geqslant \min\{p, 1-p\} \end{cases}$$

2. Gaussian source with squared error distortion:

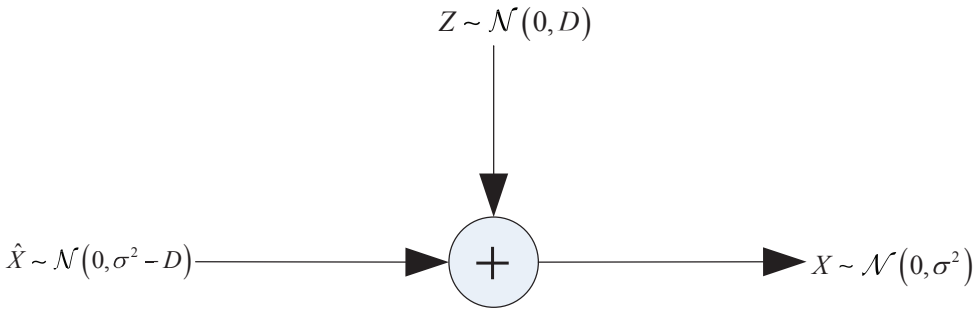
$$R(D) = \begin{cases} \frac{1}{2}\log\frac{\sigma^2}{D}, & 0 \leq \quad D \leq \sigma^2 \\ 0, & \quad D > \sigma^2 \end{cases} = \big(\frac{1}{2}\log\frac{\sigma^2}{D}\big)^+ = \max\left\{\frac{1}{2}\log\frac{\sigma^2}{D}, 0\right\}$$

Proof: Let $X \sim \mathcal{N}(0, \sigma^2)$. Then

$$
\begin{aligned}
R(D) &= \min_{f(\hat{X}|X):E\left[(\hat{X}-X)^2\right]\le D} I(X;\hat{X}) \\
I(X;\hat{X}) &= h(X) - h(X|\hat{X}) \\
&= \frac{1}{2}\log(2\pi e)\sigma^2 - h(X-\hat{X}|\hat{X}) \\
&\ge \frac{1}{2}\log 2\pi e\sigma^2 - h(X-\hat{X}) \\
&\ge \frac{1}{2}\log 2\pi e\sigma^2 - \frac{1}{2}\log 2\pi e E(X-\hat{X})^2 \\
&\ge \frac{1}{2}\log 2\pi e\sigma^2 - \frac{1}{2}\log 2\pi e D \\
&= \frac{1}{2}\log\frac{\sigma^2}{D} \\
\text{Hence, } R(D) &\ge \frac{1}{2}\log 2\pi e\frac{\sigma^2}{D} \qquad (*)
\end{aligned}
$$

Next, we want to find the density $f(\hat{x}|x)$ that achieves this lower bound.

(根据对称性) We choose the joint distribution as follows:



Clearly, $I(X;\hat{X}) = \frac{1}{2}\log\frac{\sigma^2}{D}$ and $E\left[(X-\hat{X})^2\right] = D$, thus achieving the bound in $(*)$. If $D > \sigma^2$, we choose $\hat{X} = 0$ with probability 1 , achieving $R(D) = 0$. (此时平均误差不会超过 D)

达到 $R - D$函数的概率分布$f(\hat{x}|x)$ 可由Bayes公式求出。

➢ We can rewrite $R(D)$ as

$$D(R) = \sigma^2 2^{-2R}$$

Each bit of description can reduce the expected distortion by a factor of $\frac{1}{4}$.

3. Gaussian vector source:

➢ Consider representing m independent r.v.'s $X_1, X_2, \ldots, X_m, X_i \sim \mathcal{N}(0, \sigma_i^2)$

Problem: Given R bits, how we should allot these bits to the different components to minimize the total distortion ?

or Given the total distortion D, what's the minimum R?

$$
\begin{aligned}
R(D) &= \min_{f(\hat{x}^m|x^m):E\left[d(x^m,\hat{x}^m)\right]} I(x^m;\hat{x}^m) \\
&\text{where } d(x^m,\hat{x}^m) = \sum_i (x_i - \hat{x}_i)^2
\end{aligned}
$$

➢ Solution:

$$\begin{aligned} I(x^m;\hat{x}^m) &= h(x^m) - h(x^m|\hat{x}^m) \\ &= \sum_i h(x_i) - \sum_i h(x_i|x^{i-1},\hat{x}^m) \\ &\geq \sum_i h(x_i) - \sum_i h(x_i|\hat{x}_i) \\ &= \sum_i I(x_i;\hat{x}_i) \\ &\geq \sum_i R(D_i) \quad \text{(率失真定义)} \\ &= \sum_i \max\{\frac{1}{2}\log\frac{\sigma_i^2}{D_i}, 0\} \end{aligned}$$

We have the optimization problem:

$$R(D) = \min_{\sum_i D_i = D} \sum_i \max\left\{\frac{1}{2}\log\frac{\sigma_i^2}{D_i}, 0\right\}$$

➢ Using Lagrange multipliers, we have

$$\begin{aligned} J(D) &= \sum_i \frac{1}{2}\log\frac{\sigma_i^2}{D_i} + \lambda\sum_i D_i \\ \frac{\partial J}{\partial D_i} &= -\frac{1}{2}\frac{1}{D_i} + \lambda = 0 \quad \Rightarrow D_i = \lambda' = \frac{D}{m} \qquad \text{(度分量等失真)} \end{aligned}$$

但这仅当 $\frac{D}{m} \leq \min\{\sigma_i^2\}$ 时才可行。因此必须利用Kuhn-Tucker条件，即选 λ 使

$$\frac{\partial J}{\partial D_i} = \begin{cases} 0, & \text{if } D_i < \sigma_i^2 \\ \leq 0, & \text{if } D_i \geq \sigma_i^2 \end{cases}$$

➢ Hence,

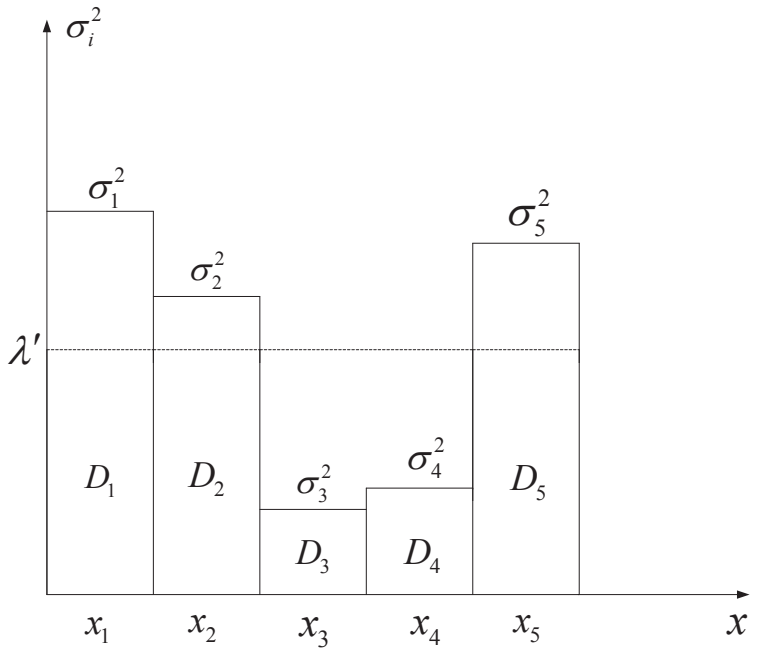
$$\begin{aligned} R(D) &= \sum_{i=1}^m \frac{1}{2}\log\frac{\sigma_i^2}{D_i} \\ \text{where} D_i &= \begin{cases} \lambda', & \text{if } \lambda' < \sigma_i^2 \\ \sigma_i^2, & \text{if } \lambda' \geq \sigma_i^2 \end{cases} \\ \text{and} \ \ \lambda' &\text{ is chosen so that } \sum_i D_i = D \end{aligned}$$

只描述方差 $> \lambda'$ 的r.v.'s，而方差 $< \lambda'$ 的不分配编码比特。

➢ 率失真函数的性质:

- $R(D_{min}) \leq H(X)$
- $R(D)$ 为单调递减的下凸(convex)函数。

  凸性证明：考虑两个率失真对 (R_1, D_1) and (R_2, D_2), 令达到这两个率失真对的分布分别为：



$$P_1(x,\hat{x}) = P(x)P_1(\hat{x}|x)$$
$$P_2(x,\hat{x}) = P(x)P_2(\hat{x}|x)$$

构造 $P_\lambda = \lambda P_1 + (1-\lambda)P_2$, 则有 $D(P_\lambda) = \lambda D_1 + (1-\lambda)D_2$。

从而

$$\begin{aligned} R(D(P_\lambda)) &\le I_{P_\lambda}(X;\hat{X}) \\ &\le \lambda I_{P_1}(X;\hat{X}) + (1-\lambda) I_{P_2}(X;\hat{X}) \\ &\qquad \text{（由互信息是条件分布的凹函数得）} \\ &= \lambda R(D_1) + (1-\lambda) R(D_2) \end{aligned}$$

所以，$R(D)$ 为 D 的 convex-U 函数。

8.4 率失真编码定理（The rate-distortion theorem）

$$R(D) = R^{(I)}(D) = \min_{P(\hat{x}|x)} I(X;\hat{X})$$

➢ Converse: For any $(2^{nR}, n)R - D$ code with distortion $\le D$, R must satisfy $R \ge R^{(I)}(D)$. Thus $R(D) \ge R^{(I)}(D)$.

Proof: Consider any $(2^{nR}, n)$ $R-D$ code. Let $\hat{X}^n = g_n\big(f_n(x^n)\big)$.

$$
\begin{aligned}
nR &\geq H(\hat{X}^n) \qquad (\because \hat{X}^n \text{ 序列至多有 } 2^{nR} \text{ 个}) \\
&\geq H(\hat{X}^n) - H(\hat{X}^n|X^n) \\
&= I(\hat{X}^n; X^n) \\
&= H(X^n) - H(X^n|\hat{X}^n) \\
&= \sum_i H(X_i) - \sum_i H(X_i|\hat{X}^n, X_{i-1}, \ldots, X_1) \\
&\geq \sum_i H(X_i) - \sum_i H(X_i|\hat{X}_i) \\
&= \sum_i I(X_i; \hat{X}_i) \\
&\geq \sum_i R^{(I)}(E[d(X_i; \hat{X}_i)]) \qquad (\text{由 } R^{(I)}(D) \text{ 的定义}) \\
&= n \sum_i \frac{1}{n} R^{(I)}(E[d(X_i; \hat{X}_i)]) \\
&\geq nR^{(I)}\left(\frac{1}{n}\sum_i E\big[d(X_i; \hat{X}_i)\big]\right) \qquad (\text{由 } R^{(I)}(D) \text{ 的下凸性}) \\
&= nR^{(I)}\Big(E\big[d(X^n, \hat{X}^n)\big]\Big) \\
&= nR^{(I)}(D)
\end{aligned}
$$

➢ Achievability:

- $(x^n, \hat{x}^n) \in A_{d,\varepsilon}^{(n)} \subset A_\varepsilon^{(n)}$: distortion typical set

$$
\begin{aligned}
&\left| -\frac{1}{n}\log P(X^n) - H(X) \right| < \varepsilon \\
&\left| -\frac{1}{n}\log P(\hat{X}^n) - H(\hat{X}) \right| < \varepsilon \\
&\left| -\frac{1}{n}\log P(X^n, \hat{X}^n) - H(X, \hat{X}) \right| < \varepsilon \\
&\left| d(X^n, \hat{X}^n) - E\big[d(X, \hat{X})\big] \right| < \varepsilon
\end{aligned}
$$

- $d(X^n, \hat{X}^n) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{d(X_i, \hat{X}_i)}_{\text{r.v.}}$
- If $(X_i, \hat{X}_i)$ be drawn i.i.d. $\sim P(x, \hat{x})$. Then $P\big(A_{d,\varepsilon}^{(n)}\big) \to 1$
- $R^{(I)}(D)$ 的可达性: $R^{(I)}(D) \geq R(D)$.

  证明只要 $R > R^{(I)}(D)$, 总存在速率为 R, 失真 $\leq D + \delta$ 的 $R-D$ 码，由于 δ 的任意性，所以 $\big(R^{(I)}(D), D\big)$ 是可达的，从而 $R^{(I)}(D) \geq R(D)$。

Chapter 9

复习

➢ 熵与互信息：

性质，链式法则，不等式

➢ 编码定理：

无损压缩，有噪信道，限失真，信源信道分离性

➢ 无损信源编码：

Huffman, Shannon-Fano, 算术, LZ, 码率（平均码长）, 效率

➢ 容量:

对称，准对称信道，最佳输入分布，C 的计算

$C = I(X = k; Y), \qquad Q_k > 0$ for all k

信道的组合，高斯信道，并列高斯信道，注水定理，波形信道

➢ FEC:

G, H, d_{min} 计算, $d \geq 2t + 1$

系统与非系统码: $\hat{C} \to \hat{U} : \hat{C} = \hat{U}G$, Syndrome, 标准阵陪集首, Pe 计算,

译码准则: MAP, ML, MD

➢ 率失真: